KEMIAN LAITOS
JYVÄSKYLÄN YLIOPISTO

Evaluation of the impact of coffee quality on acrylamide levels using analytical techniques



Master's Thesis
University of Jyväskylä
Department of Chemistry
4.11.2025
Tia Grahn

Abstract

Acrylamide is formed in coffee during roasting and degrades during storage. The coffee quality and roasting style affect the chemical profile of coffee. This work studies the physicochemical differences between specialty coffees and commercial coffees, thermal stability under oxygen atmosphere, and the differences observed in acrylamide levels between coffee quality and roast levels. In addition, the dynamics of acrylamide formation and degradation were evaluated in a production-scale roasting process, as well as the effects of storage time on the degradation rate of acrylamide. The aim of the study was to produce results that reflect real industrial processing conditions.

A fit-for-purpose high-performance liquid chromatography-tandem mass spectrometry (HPLC-MS/MS) method was developed using an isotope-labeled internal standard (AA-d_3). The method was validated according to the SANTE/11813/2021 guideline for linearity, sensitivity, repeatability, accuracy, precision, recovery and trueness. The calibration showed excellent linearity ($R^2 > 0.9999$). The limit of detection and quantification were 3 and 10 µg/L, respectively. The sample extraction was performed with methanol, and the recovery was within the range of 72–99%. Roast color (L*) and moisture (%) were measured for each sample. Analysis of covariance (ANCOVA) was used to compare the different quality groups, with L* as a covariate. In addition, thermogravimetric analysis (TGA) was used to characterize compositional differences.

Tracking the acrylamide (AA) levels through an industrial roasting process, the concentration reached a maximum after 557 seconds of roasting, at a temperature of 181 °C at 557 seconds, after which AA levels start to decrease. A 12–38% decrease in AA content was observed during 28–63 days of storage in different coffees. The average moisture content of commercial coffees was 3.46%, whereas specialty coffees had 2.20%.

TGA showed systematic compositional differences between specialty and commercial coffees within all roast levels. Commercial coffees lost more mass at the lowest temperatures, which are indicative of moisture and volatile compounds. For specialty coffees, mass loss was more concentrated at higher temperatures, where lipid and protein fractions are degraded.

Tiivistelmä

Akryyliamidia (AA) muodostuu kahvissa paahtoprosessissa ja se hajoaa varastoinnin aikana. Kahvin laatu ja paahtotyyli vaikuttavat kahvin kemialliseen profiiliin. Tässä työssä tutkittiin erikoiskahvien ja kaupallisten kahvien välisiä fysikokemiallisia eroja, termistä stabiilisuutta happi atmosfäärissä sekä millaisia eroja havaittiin akryyliamidipitoisuuksissa kahvilaatujen sekä paahtoasteiden välillä. Lisäksi arvioitiin myös akryyliamidin muodostumis- ja hajoamisdynamiikkaa tuotantomittavaan paahtoprosessissa, sekä varastointiajan vaikutuksia akryyliamidin hajoamisnopeuteen. Tutkimuksen tavoitteena oli tuottaa tuloksia, jotka heijastavat todellisia prosessiolosuhteita.

Työssä kehitettiin tarkoitukseen sopiva korkean erotuskyvyn nestekromatografia-tandemmassaspektrometria (HPLC-MS/MS) -menetelmä, jossa käytetiin isotooppi leimattua sisäistä standardia (AA-d_3). Menetelmä validoitiin SANTE/11813/2021-ohjeen mukaisesti lineaarisuuden, herkkyyden, toistettavuuden, tarkkuuden, oikeellisuuden, saannon ja totuudenmukaisuuden osalta. Kalibrointi osoitti erinomaista lineaarisuutta ($R^2 > 0,9999$) ja havaitsemis-/kvantifiointirajat olivat 3 ja 10 µg/L. Näytteen uutto suoritettiin metanolilla ja talteenotto oli 72–99%. Jokaisesta näytteestä mitattiin paahtoväri (L*) ja kosteus (%). Lisäksi termogravimetrisella analyysilla (TGA) karakterisoitiin koostumuseroja. Eri laaturyhmien vertailuun käytettiin kovarianssianalyysia (ANCOVA), jossa L* toimi kovariaattina.

Teollisen paahtoprosessin aikana akryyliamidi (AA) konsentraatio saavutti paahtomaksimin 181 °C asteessa 557 sekunnin kohdalla, minkä jälkeen sen konsentraatio alkoi vähenemään. AA-pitoisuudessa havaittiin 12–38% lasku 28–63 päivän varastoinnin aikana eri kahveissa. Kaupallisissa kahveissa oli keskimäärin 3,46% kosteutta, kun erikoiskahveissa oli 2,20%. Akryyliamidia oli kaupallisissa kahveissa keskimäärin 27% enemmän kuin erikoiskahvissa (n=17).

TGA osoitti systemaattisia koostumuseroja erikois- ja kaupallisten kahvien välillä jokaisella paahtoasteella. Kaupalliset kahvit menettivät enemmän massaa matalimmissa lämpötiloissa, jotka viittaavat kosteuteen ja haihtuviin yhdisteisiin. Erikoiskahvien osalta taas massahäviö painottui korkeampiin lämpötiloihin, joissa tapahtuu lipidi- ja proteiinifraktioiden hajoamista.

Preface

This Master's thesis was done at the Department of Chemistry in University of Jyväskylä. The experimental part of the work was carried out in the spring 2025, and thesis was completed in November 2025. The literature was searched through the Web of Science, ScienceDirect and Google Scholar databases. The literature review was limited to supporting the research questions and the results of the experimental work. Artificial intelligence tools have not been used in this thesis.

The work was supervised by Senior Lecturer Dr. Elina Kalenius and Dr. Anniina Kiesilä. Throughout this process, I have learned a lot and gained confidence in working independently in the laboratory. I am deeply grateful to my supervisors for their expert guidance and continuous support. I would also like to thank Associate Professor Manu Lahtinen for the opportunity to include thermal analysis as part of this work. Finally, I wish to express my biggest gratitude to my husband for his motivation and moral support during this academic journey.

Helsinki 4.11.2025
Tia Grahn

Table of Content

Abstract ..... **iii**
Tiivistelmä ..... **iv**
Preface ..... **v**
Table of Content ..... **vi**
Abbreviations and Terms ..... **viii**
1 Introduction **2**
2 Acrylamide **4**
2.1 Acrylamide Regulations in Coffee 5
2.2 Chemical Pathways of Acrylamide Formation 5
2.3 Integrated Risk Assessment of Acrylamide and Antioxidants 7
3 Impact of Coffee Quality on Precursors of Acrylamide **9**
3.1 Preharvesting Factors Effect on Acrylamide Precursors 9
3.2 Effect of Post-harvesting on Green Coffee 11
4 Roasting and Post-Roasting Dynamics **13**
4.1 Physicochemical Changes During Roasting 13
4.2 Degree of Roast 14
4.3 Roasting Styles (HTST vs. LTLT) 15
4.4 Degassing and Secondary Shelf Life 16
4.5 Acrylamide Stability During Storage 17
5 Analytical Chemistry in Coffee Research **19**
5.1 Sample Preparation Methods 19
5.1.1 QuEChERS 20
5.1.2 SPE 21
5.1.3 Challenges 21
5.2 Chromatographic Techniques 22
5.2.1 High-Performance Liquid Chromatography 23
5.2.2 Other Relevant Chromatographic Systems 24
5.2.3 Chromatographic Performance Parameters 25
5.3 Mass Spectrometry 27
5.3.1 Electrospray Ionization (ESI) 28
5.3.2 Triple Quadrupole (QQQ) 29
5.4 Method Validation 30
5.5 Physico-chemical Characterisation of Coffee 33

6 Materials and Methods 35
6.1 Materials 35
6.1.1 Coffee Samples 36
6.2 Methods 37
6.2.1 HPLC-MS/MS 37
6.2.2 Moisture Analysis 38
6.2.3 Colorimetric analysis 38
6.2.4 Thermogravimetric Analysis (TGA) 39
6.3 Sample Preparation 39
6.3.1 Calibration Standards 39
6.3.2 Coffee Samples 40
7 Method Development and Validation 42
7.1 Chromatographic Method Development 42
7.1.1 Column Screening 42
7.1.2 Chromatographic Parameters 46
7.1.3 Matrix Interferences 48
7.2 Optimizing the Ion Source Parameters 49
7.3 Sample Preparation 50
7.3.1 Liquid Samples 50
7.3.2 Solid-Liquid Extraction 51
7.4 Method Validation 52
7.4.1 Linearity 52
7.4.2 Sensitivity (LOD and LOQ) 53
7.4.3 Precision 53
7.4.4 Trueness 54
7.4.5 Recovery 55
7.4.6 Acrylamide Stability 56
7.4.7 Summary 57
8 Results and Discussion 59
8.1 Physicochemical Differences between Specialty and Commercial Coffee 59
8.2 Acrylamide Analysis 61
8.3 Acrylamide Behavior During Roasting 66
8.4 Chemical Stability in Commercial and Specialty Coffee 68
8.5 Thermogravimetric Analysis 70
9 Conclusion 74
References 77
List of Appendix 87

Abbreviations and Terms

AA	Acrylamide
C18	Octadecyl
CRM	Certified reference material
DSC	Differential scanning calorimetry
dSPE	Dispersive solid-phase extraction
ESI	Electrospray ionization
HMF	5-hydroxymethylfurfural
HPLC	High-performance liquid chromatography
HTST	High-temperature short time
LOD	Limit of detection
LOQ	Limit of quantification
LTLT	Low-temperature long-time
MRM	Multiple reaction monitoring
MS/MS	Tandem mass spectrometry
QC	Quality control
QQQ	Triple quadrupole
QuEChERS	Quick Easy Cheap Effective Rugged Safe
RT	Retention time
S/N	Signal-to-noise ratio
SPE	Solid-phase extraction
TGA	Thermogravimetric analysis
%RSD	Relative standard deviation

1 Introduction

Coffee belongs to the *genus Coffea*, which has two commercially important varieties: *Coffea Arabica* (Arabica) and *Coffea Canephora* (Robusta).[1] Arabica has higher quality and it accounts for 75% of the world's coffee production, with Brazil being the largest exporter. Coffee is one of the most consumed beverages in the world and the average person drinks nearly 1500 cups of coffee annually in Finland.[2] Coffee is valued for its taste and for the stimulating effect. It contains bioactive compounds, which have health-promoting benefits due to antioxidant activity.[3] However, roasting coffee also leads to the formation of potentially harmful compounds, including acrylamide, furans, and 5-hydroxymethylfurfural (HMF).[4,5] Given the high per-capita coffee consumption in the Nordics, coffee has become the primary source of dietary acrylamide, which is a cumulative neurotoxin and potential carcinogen.[6]

The coffee roasting process is a complex series of thermal reactions in which the coffee's moisture content, internal pressure, and temperature control the formation and degradation of chemical compounds.[7] Even small changes in these variables can alter the chemical composition and sensory profile of the coffee. Because of this complex matrix, analytical studies often require a multivariate approach.[3] From an analytical chemistry perspective, the reliability of results is based on two key premises: 1) the development and validation of a robust method, and 2) a thorough understanding of the matrix that guides appropriate sampling. Although many studies have successfully developed methods to investigate acrylamide in coffee, some study designs have not fully accounted for key variables such as country of origin and processing method, or the roasting setups have been unrealistic, which do not reflect the industrial roasting process.[8] As a result, some reported findings may not fully capture the complexity of the process, and in certain cases, the conclusions may have been oversimplified.

One of the goals in this thesis is to narrow the gap between scientific research and practical roastery work. Therefore, the study focuses on the formation and degradation of acrylamide in a real-world roasting environment making the results adaptable in practice. Another focus is the behavior of acrylamide during storage. The compound is water-soluble and reactive, and its concentration can change over time. Post-roast quenching can affect the moisture content of the bean and thereby the rate of acrylamide degradation.[9] This work investigates how acrylamide levels change during storage for different coffee types.

In addition, the differences between specialty coffees and commercial coffees at different roasting degrees are studied. Specialty coffees are often roasted more slowly and at lower temperatures, while industrial roasting processes are faster and conducted at higher temperatures. These differences may significantly affect the formation and degradation of acrylamide. The quality and handling of green coffees beans may also differ. Specialty coffees are produced from hand-picked ripe cherries, and they are processed carefully, while the origin of commercial blends are often less transparent. The study focuses on 100% Arabica coffees. To minimize variables, all specialty coffees originate from Brazil, while commercial blends contain coffee from South, Central and Latin America. In Finland, a section on acrylamide was added to the in-house control plan of the roasteries a few years ago, but the mechanisms of acrylamide formation and key variables are still unclear to many roasters. This highlights the relevance and practical importance of the study.

This thesis consists of a literature review and an experimental part. The literature review includes both chemical and process-related factors that influence acrylamide formation and degradation, especially considering variables that have often been overlooked in previous studies. It also summarizes the current knowledge on the formation and degradation of acrylamide in coffee and identifies the analytical challenges related to its determination. The experimental part includes the development and validation of an HPLC-MS/MS method for acrylamide analysis, the preparation of coffee samples and the determination of acrylamide concentrations in coffee samples. In addition, complementary analyses such as thermogravimetric analysis (TGA), moisture content and color measurements are performed to understand better the relationship between roasting conditions and acrylamide concentration.

2 Acrylamide

Acrylamide, also known as 2-propenamide, is a white, odorless solid with a molecular weight of 71.08 g/mol and the formula C_3H_5NO. It is highly soluble in water (2.04 kg/L at 25 °C) and dissolves in several organic solvents such as methanol and acetonitrile. It melts at 82–86 °C and boils at 252.5 °C.[10] Structure of the molecule is presented in Figure 1, and its physical and chemical properties are shown in Table 1.

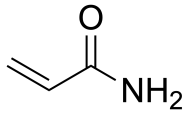


Figure 1. Structure of acrylamide (AA).

Table 1. Physical and chemical properties of acrylamide[10]

Characteristic	Value
IUPAC name	2-propenamide
Molecular formula	C_3H_5NO
Molecular weight	71.08 g/mol
Exact mass	71.037113783
Solubility in water	2.04 kg/L in 25 ºC
Melting point	82–86 ºC
Boiling point	252.5 ºC

In 2002, a Swedish team discovered that several foods contain acrylamide, which raised the interest of many researchers.[11] Acrylamide is carcinogenic, cyto- and genotoxic compound, which is formed mostly during Maillard reaction and its formation mechanisms are presented in Chapter 2.2.[12] Most everyday foods contain acrylamide if they have been fried, baked, or roasted in temperatures above 120 °C. The most common sources of acrylamide in the Finnish diet are starchy casseroles, rye bread, and coffee. Although coffee does not contain the maximum level of acrylamide among the sources mentioned, it is the most significant single source of acrylamide for Finnish adults due to the high consumption.[13] Acrylamide is classified as Group 2A carcinogen by the International Agency for Research on Cancer (IARC). Group 2A stands for *'probably carcinogenic to humans',* which means that acrylamide has shown carcinogenicity in animal studies but there's limited evidence in humans.[14]

2.1 Acrylamide Regulations in Coffee

In 2015, the European Food Safety Authority[15] (EFSA) published a scientific opinion, which stated that acrylamide formed in food may increase the risk of cancer for consumers of all age groups. After this, the European Union confirmed that acrylamide is a chemical hazard in the food chain, as defined in Council Regulation (EEC) No 315/93. Based on EFSA's assessment, the European Commission established mitigation measures and set benchmark levels for acrylamide, including 850 µg/kg for instant coffee and 400 µg/kg for roasted coffee, in Commission Regulation (EU) 2017/2158.

Instant coffee is often reported to contain more acrylamide per unit of mass than coffee beans or grounds.[16] However, the differences in extraction ratios are often overlooked when instant coffee and filter coffee are compared. Less instant coffee is required to achieve a similar total dissolved solids (TDS) level with filter coffee.[17,18] Most of the coffee grounds' acrylamide is extracted into the coffee brew within a 4–5-minute brewing time, and therefore the exposure can ultimately be very similar between the two types of coffee.[19] The amount of acrylamide in coffee is most affected by the time and degree of roasting, the variety of coffee, physical properties, and the storage condition and time.

2.2 Chemical Pathways of Acrylamide Formation

Coffee roasting is typically divided into three phases: the drying phase, the Maillard reaction phase, and the development phase.[20] During the Maillard phase, several chemical reactions occur between reducing sugars and amino acids, generally known as the Maillard reaction. The formation of acrylamide has been proposed as a possible side pathway within this reaction.[21]
The Maillard reaction is often described as the primary pathway for acrylamide formation.[22] In the reaction, free asparagine reacts with reducing sugars such as glucose or fructose to form a Schiff base, which is then deaminated to produce 3-aminopropionamide (3-APA), which is then deaminated to acrylamide.[23] The mechanism is shown in Figure 2.

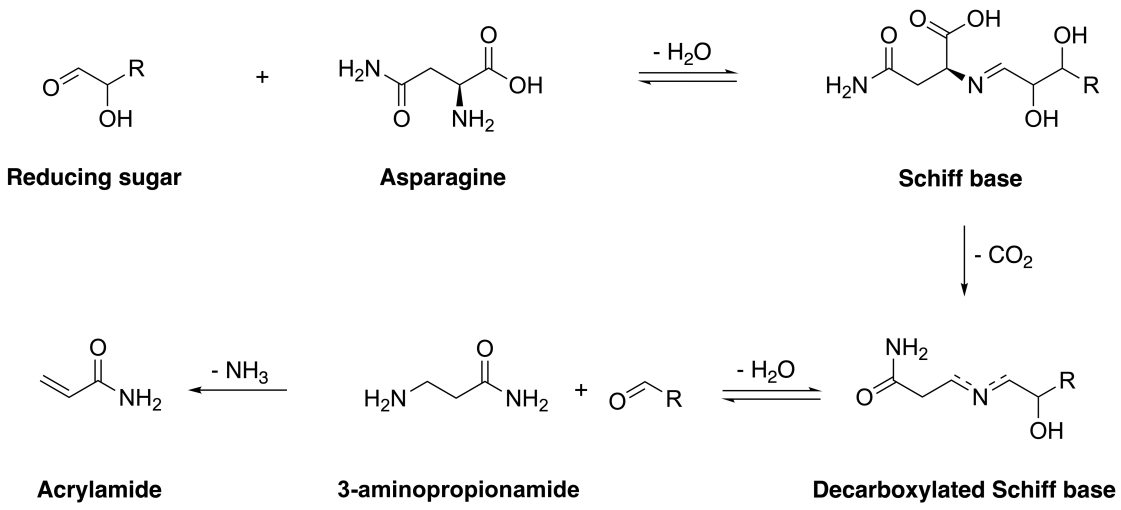


Figure 2. Traditional Maillard reaction -pathway.

The reaction has been demonstrated in various model systems, where reactions are studied under controlled conditions using pure reagents.[21] The laboratory environment allows precise control of individual variables, but the reaction pathway has not been confirmed in the coffee matrix. Therefore, extrapolation to the complex chemical environment of coffee must be approached with caution.

Hamzalıoğlu and Gökmen[24] (2020) developed a kinetic model that describes how different compounds react with each other during coffee roasting. Based on their results, the previously discussed Maillard reaction may not be the most important pathway for acrylamide formation in coffee. During roasting, sucrose breaks down into glucose and a highly reactive intermediate called fructofuranans cation (FFC). This compound can further degrade into 5-hydroxymethylfurfural (HMF), which is considered one of the main routes for HMF formation under coffee roasting conditions. HMF can further react with asparagine to form acrylamide, making this an important alternative pathway for acrylamide formation in coffee.[24] This reaction pathway is illustrated in Figure 3.

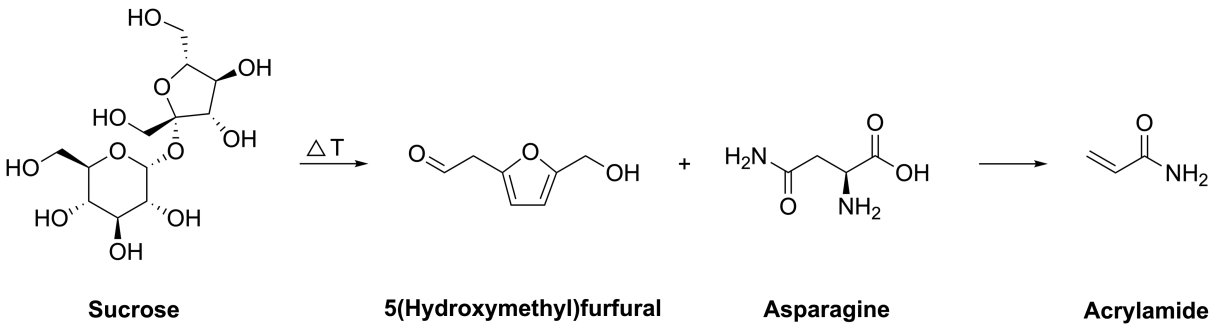


Figure 3. HMF as a precursor in acrylamide formation.

Other acrylamide formation mechanisms have also been proposed to occur in coffee. Green coffee beans contain 8–18% lipids, and linoleic acid is one of the main fatty acids. During roasting, linoleic acid undergoes oxidation, producing reactive carbonyl compounds that can later react with asparagine, ultimately leading to the formation of acrylamide.[25]

Based on the mechanisms presented, it is justified to examine the factors that influence the precursors of acrylamide. This provides a foundation for understanding and controlling acrylamide formation.

2.3 Integrated Risk Assessment of Acrylamide and Antioxidants

Acrylamide is absorbed following oral intake and is metabolized in vivo via glutathione conjugation and CYP2E1-mediated epoxidation to form the reactive intermediate glycidamide (GA). AA has been shown to act as both a neurotoxin and a genotoxin, inducing oxidative stress and the formation of deoxyribonucleic acid (DNA) adducts. In vivo animal studies have demonstrated that GA has greater genotoxic potency than AA and is likely the primary driver of acrylamide's carcinogenicity. At the cellular level, AA also triggers excessive production of reactive oxygen species (ROS), thereby exacerbating oxidative damage.[12]

However, coffee contains a range of bioactive compounds that show antioxidant, anti-inflammatory and antigenotoxic activities. These can protect cells against oxidative damage, which is induced by AA.[26] Bioactive compounds in coffee include, among others, chlorogenic acids (CGAs) and their degradation products, Maillard reaction products formed during roasting such as melanoidins, and alkaloids such as caffeine and trigonelline. Antioxidant activity can be assessed by measuring the concentrations of these compounds or by determining

the total phenolic content.[27,28] Therefore, it may be misleading to isolate and evaluate a single compound from the complex food matrix and make conclusions about its overall safety.[29]

3 Impact of Coffee Quality on Precursors of Acrylamide

The International Coffee Organization (ICO) and the Specialty Coffee Association have established a protocol for evaluating and scoring coffee using trained sensory panelists.[30,31] Based on the score, coffee can be classified either specialty or non-specialty coffee, which is often referred as commercial coffee.[32] Some chemical quality indicators have been shown to align with high-quality coffee based on sensory evaluation.[33,34]

Determining overall coffee quality requires considering multiple variables. From an analytical perspective, these variables are not only descriptors of sensory quality but also sources of systematic variability in acrylamide analysis. For example, asparagine levels may correlate with acrylamide in one coffee but not in another, which demonstrates the importance of matrix-matched calibration and internal standards.[35] A comprehensive assessment should consider chemical, physical, and sensory properties, as well as health-related compounds, such as potential carcinogens.[36] Before roasting, the quality of coffee is affected by the growing environment, genotype, cultivation practices, maturity of cherries, harvesting technique, processing, and drying conditions. These pre-harvest and post-harvest factors influence the concentration of acrylamide precursors and the overall matrix composition of the sample.

Although coffee quality is typically defined by evaluating its sensory attributes[30], all qualitative differences are linked to measurable chemical and physical properties, such as amino acid and sugar content, moisture and color. Everything from the elemental composition of the soil to the storage conditions of the coffee affects on its quality and acrylamide levels.[37,38]

3.1 Preharvesting Factors Effect on Acrylamide Precursors

Preharvest factors explain approximately 40% of coffee quality.[39] These include the cultivated coffee variety, geographical location, climate, soil composition, agronomic practices, and the ripeness of the cherries at the time of harvest.[3] Coffee grows on a plant that produces cherries, which typically contain two seeds known as coffee beans. The maturity of the coffee cherry can be divided into different stages based on their color and firmness.[40,41] Different compounds undergo biosynthetic processes during different maturity stages. Levels of acrylamide precursors vary throughout these stages.[41]

The differences in the maturity stages of coffee cherries impact on the behavior of green coffee throughout processing, drying, storage, and roasting. The relationship between water content and equilibrium relative humidity differs across fruit layers and maturity stages.[41] Unripe cherries require a longer drying phase in natural-processed coffee, which induces a state of stress in the beans. In response to this stress, the beans produce and accumulate asparagine, to maintain metabolic balance. Stress conditions also slow down protein synthesis, resulting in an increased accumulation of asparagine, which is the primary amino acid in unripe cherries. Drying takes longer when the cherry is unripe leading to higher asparagine content because of the stress response.[42] The ripening process is asynchronous, meaning that cherries won't reach optimal ripeness at the same time.[41] There is also variance between coffee cultivars, which are affected by genetic and environmental factors. Some coffee variety reach optimal ripeness faster than others, which leads to varying chemical composition, including sucrose.[41,43] Therefore, results from one region cannot be generalized, for example, by making absolute statements such as *'Brazilian coffees have a certain chemical profile'.* This highlights the need for a multivariate approach when determining the chemical composition of coffee. Both genetic and environmental factors must be considered when designing the sampling strategy.

Cultivating coffee trees in shaded environment reduces their exposure to solar radiation and lowers the temperature around. Maturing takes longer but it leads to more even physical quality of the bean, including a consistent size beans, which affect to further practices.[30] Shade-grown coffee can yield 80% higher sugar content of fructose, glucose, and sucrose than one grown in direct sun light. Shade-grown also complimented content of chlorogenic acids by 43%, which had strong correlation ($r = 0.83$) with antioxidant capacity.[1] Chemical composition of coffee is also affected by water stress. When coffee tree doesn't get enough water, it has shown to increase sucrose levels, while controlled irrigation increases content of antioxidants.[44] Another study found no influence of irrigation on levels of reducing sugars, suggesting that temperature is more significant factor. Higher temperatures elevated the levels of both reducing sugars and free amino acids.[33]

Elements in the soil have been shown to influence the bioactive compounds in coffee.[44] Concentration of reducing sugars reached the highest value in ripe cherries and lowest in unripe cherries. Fertilizing soil with nitrogen significantly increased reducing sugars only in ripe cherries. The content of total free amino acids is greatest in green and lowest in dried cherries, but increased linearly as nitrogen supply was risen. One study has shown that asparagine responds differently to soil nutrient ratios than other free amino acids.[45,46] To our best

knowledge, no studies have yet examined a direct link between fertilization and the formation of free asparagine in coffee, but similar relationships have been investigated in other crops.

In tea cultivation, researchers have explored how different forms of potassium-, magnesium- and sulfur-containing fertilizers affect the leaves' free amino acids. The decisive factor proved to be the fertilizer's anion: potassium sulfate (K_2SO_4) increased asparagine content, while potassium chloride (KCl) lowered it. The amount of free asparagine depends on how efficiently the plant reduces nitrate and stores nitrogen in amide form. A high overall nitrogen supply generally increases asparagine, whereas carbonate and oxide forms (K_2CO_3, MgO) raise soil pH, impair K and Mg uptake, and thus decrease asparagine accumulation. Study showed that using magnesium or potassium sulfate will increase overall free amino acid content.[45] Although it will increase asparagine, it should be noted that amino acids are important for coffee quality[42] and the present of other free amino acids will lead to lower acrylamide concentration after roasting by possible promoting competing reactions.[47]

As climate change rises temperatures, quality and biochemistry of coffee are negatively affected.[48] It can be concluded that (i) more acrylamide precursors are formed in areas where the plant is unshaded and therefore exposed to higher temperatures, and (ii) in the future, under the current production model, rising temperatures may increase precursor formation and thus acrylamide levels in coffee, unless the current production factors are managed.

3.2 Effect of Post-harvesting on Green Coffee

Post-harvest stages include harvesting, processing, drying, and roasting. Coffee cherries can be harvested either mechanically or manually by hand. After harvesting, the cherries are often processed using natural or washed processing method. Coffee drying occurs on patios or using mechanical dryers. Roasting is a key part in this study and therefore, it is discussed in a separate chapter.

The structural differences over maturity stages lead to uneven drying. Some green coffee have low water activity, while others absorb excess moisture.[41] In specialty coffee production, only optimally ripe cherries are selectively hand-picked, which often requires multiple harvesting rounds, resulting in higher costs. In contrast, manual picking is less common within commercial coffee production, and cherries are often harvested in bulk, which increases the likelihood of including both unripe and overripe fruits.[40] This type of heterogeneity in physical quality leads

to an uneven distribution of density and porosity, which influences reaction kinetics during roasting and leads to an inconsistent chemical profile for the final product.[41]

In natural processing, the whole coffee cherry is laid out on patios and dried in the sun for 3–9 days, which is why it's sometimes referred as sun-dried. The cherries are turned regularly until they are completely dry. Afterward, the parchment skin is mechanically removed.[49] The process increases the levels of glucose and fructose.[49,50] In high-quality lots, only fully ripe cherries are used, which promotes early germination of the bean, activates enzymes, and helps metabolize polysaccharides. Unevenly dried cherries lead to poor-quality coffee.[51]

The first stage of washed coffee processing involves depulping, in which the outer skin of the coffee cherry is mechanically removed, and then the cherries are placed in a clean water tank. The mesocarp layer surrounding the bean is composed mostly of water (84%), along with proteins, sugars, and pectins. During processing, fermentation takes place, during which microbes break down pectins and sugars into organic acids, alcohols, and other metabolic byproducts. The fermentation process depends on many factors such as temperature and pH. To achieve high-quality coffee, the level of fermentation should be kept as consistent as possible. Finally, the cherries are washed and dried either mechanically or on patios. Green coffee is dried to moisture level of 8–12% and it has neutral pH.[24]

Sucrose is the major low-molecular-weight sugar in green coffee beans, and its concentration is hardly affected by processing. However, sucrose breaks down during roasting into fructose and glucose, which are precursors of acrylamide.[52,53] It was found that, 8% of dry mass (DM) of Brazilian green coffees is sucrose, which accounts for more than 90% of the total sugars, while glucose and fructose are roughly 20 times less abundant. In Brazilian coffees, natural-processed beans contained over 6.6 times more fructose and more than 4.4 times glucose compared to washed-processed beans. A similar trend was observed in Tanzanian samples, where natural-processed coffees had approximately 13 times more fructose and 7.5 times more glucose than their washed-processed counterparts.[52]

However, it is possible that pre-harvest factors have a major effect on the chemical profile of coffee. The study analyzed fructose and glucose from several washed and natural Arabica green coffees available on the market, which demonstrated significant variation.[52]

4 Roasting and Post-Roasting Dynamics

The physical quality of the bean influences internal temperature, water activity, pressure, and mass transfer phenomena, which in turn control the kinetics of chemical reactions during the roasting process.[54] There are three main variable controlling the roasting process: temperature, time, and air flow. The roasting process includes a drying phase, Maillard reaction phase, development phase, and a cooling phase.[20,25] Roasting process does not only determine the sensory properties, but also strongly influence the formation and degradation of acrylamide. Understanding the dynamics of roasting is therefore essential for the interpretation of acrylamide concentrations and for ensuring comparability of analytical results between different coffees and roasting conditions.

4.1 Physicochemical Changes During Roasting

Roasting can be divided into endothermic and exothermic phases. The endothermic phase occurs at the beginning of roasting, when the process requires a lot of energy to evaporate the water, create pressure inside the bean and start chemical reactions.[54] When the internal temperature of the coffee bean reaches 180–200 °C, coffee start to crack and produces a popping sound, known as the first crack. Roasting can be continued beyond this point, eventually leading to a later event called the second crack. These acoustic phenomena are used to monitor the progress of roasting.[55]

One of the most important factors in the kinetics of chemical and physical changes in coffee is the total water content and water activity at different stages of roasting. The initial moisture content of coffee affects the reaction kinetics.[7] Beside the initial moisture content of the green coffee, a great amount of water is formed during the roasting as a result of chemical reactions, which eventually evaporates. If the moisture content is too low at a certain point in the roasting, the occurrence of chemical reactions is decreased.[54]

The volume changes, dehydration and chemical reactions that occur during roasting change the microstructure of the bean, which affects the shelf life of the coffee. High temperature short time (HTST) refers to high temperature roasting conditions, while low temperature long time (LTLT) refers to low temperature roasting condition.[54]

The density of coffee changes during roasting, resulting in a more porous structure, when coffee is roasted as darker. The internal pressure of the bean creates volatile pressure, which determines the geometric development of the pore microstructure and cracks formed. This further affects the shelf life of coffee.[56] Another important factor that affect shelf life is the choice of cooling process after roasting. Thermal reactions must be stopped rapidly using effective cooling. In the specialty coffee industry, cooling is done with air flow and in larger industrial roasters using water. However, it should be noted that the final moisture content of the beans is affected by post-roasting cooling, as the beans can absorb water if they are sprayed with water during the cooling phase.[9,54]

4.2 Degree of Roast

There is no single absolute way for determining the degree of roast. It is often determined during roasting using color, aroma, roasting parameters and kinetics. The first crack and second crack are widely used in practice as an indicator for monitoring roasting. However, in the literature, the degree of roast is often determined based on color of the coffee.[57,58]

Green coffee beans exhibit a green-grayish color, which changes to brown or even black as roasting progresses.[30] One of the most common methods for determining coffee color is using the CIE L*a*b* color system, which converts the reflected or transmitted light into numerical scale, where L*=100 corresponds to white and 0 to black. In this system, a* represent the green-red axis, with a* = -50 indicating green and a* = 50 indicating red. The blue-yellow component is described by the variable b*, where b* = -50 correspond to blue and b* = 50 to yellow. Light roast coffee usually has an L* of around 30, medium roast coffee around 25, and dark roast coffee closer to 20.[54,57]

Schouten *et al.*[58] (2021) studied acrylamide formation and antioxidant activity on different roasting profiles. The study provided much useful information on the relationship between the physical properties of coffee and acrylamide formation and antioxidant activity. Different degree of roast was achieved by applying identical temperature-time parameters. Roasting was stopped when the coffee sample reached the desired L* value. Thus, different roasts were achieved by changing only the total roasting time. In their study light roast was achieved at 147 °C at 6 minutes, medium roast at 172 °C at 11 minutes and dark roast 191 °C at 15 minutes. Acrylamide concentrations were found to be approximately 775 µg/kg, 250 µg/kg and 110 µg/kg at different roasting degrees. CGAs levels decrease 49% from light to medium roast

and 81% from light to dark roast.[58] The design of roasting condition differs from industrial roasting practices, and therefore, the results may not be directly comparable to those obtained under real production conditions.

In turn, Acquaticci *et al.*[49] (2023) studied how different roasting profiles of washed and naturally processed Brazilian Arabica coffees affect the formation of acrylamide and furans. The roasting was carried out in a controlled laboratory environment. Five roasting profiles were applied individually to each coffee sample, regardless of their processing method. Natural processed light, medium, and dark roasts contained 326 µg/kg, 259 µg/kg, and 274 µg/kg of acrylamide, respectively. Washed coffees had AA levels of 248 µg/kg in both light and medium roasts, and 184 µg/kg in dark roast. However, the reduction in AA between light and dark roasts was only 16–26%, which is approximately 3–6 times lower than the decrease reported by Schouten *et al.*[49,58] Although roasting coffee to a darker level decreases the acrylamide levels, it also increases the formation of 5-hydroxymethylfurfural (5-HMF), furfural alcohol and furan.[5] Roasting coffee darker may also compromise the flavor. A strong negative correlation has been found between acrylamide levels and perceived off-flavors ($r = -0.85$).[59]

L* color values have showed variation between processing methods, suggesting that roasting kinetics proceed differently depending on whether the coffee was washed or naturally processed.[49] This is not unexpected, considering that the physical quality and chemical profile of green coffee vary significantly depending on the processing method. It is unlikely that applying the same roasting parameters to all coffees would yield sensorially pleasant results, since coffees differ in their processing backgrounds and initial chemical profile.[60] Coffees behave differently during roasting depending on the processing method.[49] As a result the roasting profiles are optimized in industry for each individual coffee to maximize the organoleptic properties.

4.3 Roasting Styles (HTST vs. LTLT)

HTST and LTLT indicate the reality of roasting conditions between commercial and specialty coffee. HTST aligns more closely with commercial coffee roasting profiles, while LTLT is typically applied to specialty coffee roasting profiles.

The time–temperature condition during roasting influences the physicochemical properties of coffee. Quantitative measurements of these properties provide valuable insight into the thermal

and chemical transformations occurring both during and after roasting, as they reliably reflect the reaction kinetics. The formation and degradation of various compounds are linked to these kinetics and changes in the internal bean temperature and moisture content.[61,62] Physical parameters, such as bean color, density, volume, moisture, and mass change throughout the roasting process, which reflect complex chemical reactions, including Maillard reactions and thermal degradation processes.[54,61]

Different time–temperature profiles create different physical and chemical development pathways. Within dark roasted coffee HTST profile has shown a more rapid decrease in moisture content and density, which in turn influences the formation and degradation of aroma compounds. For instance, some desired flavor compounds break down under these conditions, while unpleasant compounds continue to increase.[61] Also, acrylamide levels in coffee vary greatly depending on the roasting time and temperature, when the degree of roast is the same. AA was formed at 200 µg/kg with a 300-second roast, 285 µg/kg with a 150-second roast, and 350 µg/kg with only a 90-second roast.[19]

When green coffee is treated with steam, it reaches the target color faster, resulting in a shorter roasting time. This also induces chemical changes: the content of monosaccharides, such as fructose and glucose, increases and a reduction in free amino acid content occurs. These compounds have shown that the rapid roasting process is associated with an increase in acrylamide content.[62] Energy is transferred more intensely within the bean when the air velocity of roasting is increased, which result in a higher concentration of acrylamide. Conversely, if the humidity is increased during roasting, acrylamide formation is suppressed. However, this also reduces the antioxidant capacity.[63]

4.4 Degassing and Secondary Shelf Life

After roasting, the internal pressure of a coffee bean is primarily caused by carbon dioxide (CO_2) generated during the roasting process. This CO_2 then diffuses out through the bean's cell walls.[9] In the coffee industry, this phenomenon is generally referred to as degassing.[64] After roasting, CO_2 desorption is at its highest rate during the first hours and then gradually settles. The desorption rate is influenced on the cell structure, surface area, and the water activity of roasted coffee bean. Some studies have shown correlation between certain volatile compounds and CO_2 release during roasting suggesting that desorption and diffusion have a main role in the post-roasting reduction of aroma compounds.[9,64]

Aromas are released faster when the moisture content of the bean is higher.[65] In the specialty coffee industry, coffees are roasted in small batches, where cooling is done by air cooling without water contact. However, on a larger scale, it is common to use water as part of the cooling by applying spray or film quenching. In spray quenching, water is applied to the beans as a fine mist, leaving a moisture content 11% higher than air cooling. Another method is film quenching, where the beans pass through a thin layer of water, resulting in a final moisture content of 120% higher compared to air cooling.[9]

Although dark roast coffee has lower moisture content than lighter roasts, it still loses aromatic compounds faster, which imply that moisture alone does not account for this phenomenon. In general, darker roasts experience more physical expansion and partial collapse of their cell structure during roasting, creating micro-cracks, which generates new routes for gas to escape.[56] Secondary shelf life (SSL) describes how long coffee quality is maintained in opened coffee packages. Lower temperature and water activity significantly extend the SSL of coffee.[43] Anese *et al.*[65] (2006) investigated how water activity is related to the quantitative loss of coffee aromatic compounds. The study showed that, depending on water activity, ground coffee loses 60% of its original aroma within 13–23 days when stored at 30 °C and opened briefly once daily, which stimulates SLL under normal conditions.

4.5 Acrylamide Stability During Storage

The effects of storage on AA have been studied at different storage conditions across several food matrices. The reduction of acrylamide during storage is mostly influenced by temperature, time, storage conditions, food matrix, water activity, and pH.

Hoenicke and Gatermann[37] (2005) studied the stability of acrylamide during three-month storage at 10–12 °C. They found that concentration in coffee beans decreased 30%, from 285 to 200 µg/kg, and in ground coffee 31%, from 305 to 210 µg/kg, respectively. However, no information was provided to confirm whether these different sample types had similar roasting and production date, which is unlikely, considering that the roasted coffee beans initially had a lower acrylamide concentration than the ground coffee. Lantz *et al.*[19] (2006) also studied how storage temperature and time affect to acrylamide concentration in ground coffee. They found that approximately 15% of acrylamide is lost within two week and around 40% within three months of storage at room temperature. This indicates that reduction rate of

acrylamide is not linear, which highlights the importance of reporting roasting or production date to ensure comparability of analytical results.

The water activity affects to the kinetics of acrylamide and 5-hydroxymethylfurfural (HMF). AA decrease during storage, while HMF levels increase. The authors found a negative correlation (r = -0.66) between water activity and AA reduction across 90 different food samples.[16]

Acrylamide reacts with sulfur-containing amino acids (e.g. cysteine and SH groups), free amino and hydroxyl groups during storage. Acrylamide has an α,β-unsaturated amide structure, whose double bond behaves as Lewis acid and gets attacked by strong nucleophiles, such as sulfur-containing (SH) groups. In this process, the nucleophile adds to the β-carbon of acrylamide via Michael addition. This breaks down the structure of acrylamide creating new products, which is observed as a decrease in its concentration.[37] Nucleophilic attack on acrylamide is presented in Figure 4.

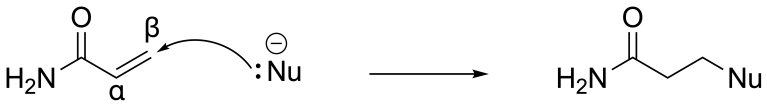


Figure 4. Nucleophilic attack on acrylamide.

Higher water content shortens the shelf life of coffee aromas. Dimethyl sulfide breaks down more rapidly in coffees with higher moisture. In ground coffee, dimethyl sulfide content can decrease by up to 80% within the first 24 hours after opening the package. Other aroma compounds follow a similar trend: an exponential decrease during the first days, followed by either moderate aroma loss or stabilization.[66] Degassing phase occurs right after roasting, which might explain this rapid decrease and stabilizing. It is possible that similar pattern is occurring with acrylamide. Same factors might affect to reduction rate of acrylamide.

5 Analytical Chemistry in Coffee Research

In recent decades, the development of measurement technology and automation has shifted the focus of analytical chemistry from routine measurements towards method validation, data processing and deeper interpretation of results.[67] Analytical chemistry covers the entire research process from formulating a research question to selecting and developing a method and evaluating the measurement results.[68] The goal is to ensure that the chosen method is fit for purpose, meaning it produce reliable information about the analyte.[69]

Analytical chemistry provides tools for identifying compounds and determining their concentrations across various sample matrices. The analytical process has several steps: sampling, sample preparation, separation, detection and evaluation. Each step contributes to the reliability of the final result.[68] It is noteworthy that sampling can be a major source of error in analytical chemistry. Poorly executed sampling can cause up to 10 times greater error than other analytical steps. This highlights the importance of a representative sample, especially in complex matrices like coffee.[70] Coffee contains hundreds of compounds, including polyphenols, Maillard reaction products, and lipids, which can interfere with the quantification of target analytes, especially acrylamide, which occurs at ppb levels.[34,71,72]

Acrylamide is particularly challenging compound due to its chemical nature. It is a small, polar and unstable compound, requiring sensitive and selective analysis for its accurate determination.[73] Chromatographic techniques (HPLC, GC, SFC) combined with mass spectrometry are often used minimize matrix effects and detect the compound at low concentrations.[74]

The study of physicochemical properties, such as moisture content, color and thermogravimetric behavior (TGA), can provide important additional information on the processing history of the sample and the state of the matrix.[57] Understanding these parameters supports the assessment of acrylamide formation and stability.

5.1 Sample Preparation Methods

Sample preparation is a vital step in the determination of acrylamide, as it often exists in low concentrations in a complex matrix. Acrylamide is found widely in different food matrices, which is why sample preparation methods also vary broadly in the literature.[75] Acrylamide may

be unevenly distributed within the sample matrix, which makes homogenizing sample essential to ensure the reliability and repeatability of the analysis.[76]

Sample preparation process begins with the extraction of the homogenized solid sample using water or an organic solvent.[72,77] The next step may include spiking an isotopically labeled analyte (e.g. AA-d_3) at this stage, which allows to estimate the overall losses during sample preparation.[76] Extraction is usually enhanced by mixing or by sonication.[72,78] After extraction, the solid and liquid phases are typically separated by centrifugation, and the supernatant is collected.[77]

The pre-clean-up stage includes removing proteins with Carrez reagents, while lipids are often removed using hexane. In both cases, the sample is mixed and centrifuged before proceeding to the next step.[79] However, it has been found that lipid purification does not improve the chromatography baseline for coffee, and is therefore overlooked in many studies.[80]

The three most commonly used extraction techniques for acrylamide are Dispersive Liquid–Liquid Microextraction (DLLME), Solid Phase Extraction (SPE), and the QuEChERS method, where QuEChERS stands for "Quick, Easy, Cheap, Effective, Rugged, and Safe".[75] SPE and QuEChERS are the most commonly applied methods for coffee and therefore, theory of other techniques has been excluded from this work.

5.1.1 QuEChERS

The QuEChERS method has four main steps, starting with the extraction of a homogenized sample using acetonitrile (ACN). After extraction, a mixture of salts, typically magnesium sulfate ($MgSO_4$) and sodium chloride (NaCl), are added to separate phases by removing water from the organic phase and thus improving extraction efficiency. The sample is then centrifuged, and the resulting organic layer (ACN) is collected. In the final stage, dispersive solid-phase extraction (d-SPE) is performed by adding specific adsorbents. These typically include PSA (primary-secondary amine) to remove organic acids and sugars, C18 (octadecyl) to eliminate lipids and fats, and additional $MgSO_4$ to remove residual moisture. The mixture is vortexed, centrifuged once more, and the supernatant is collected for analysis or further clean-up.[72]

5.1.2 SPE

Solid Phase Extraction (SPE) is a widely used sample clean-up technique. A typical SPE protocol has four steps. First, the sorbent is activated and equilibrated, usually with methanol and water. The sample is then loaded onto the sorbent**,** followed by a washing step to remove interfering matrix compounds**.** Finally, the target analyte is eluted.[68,75] The process is shown in the Figure 5.

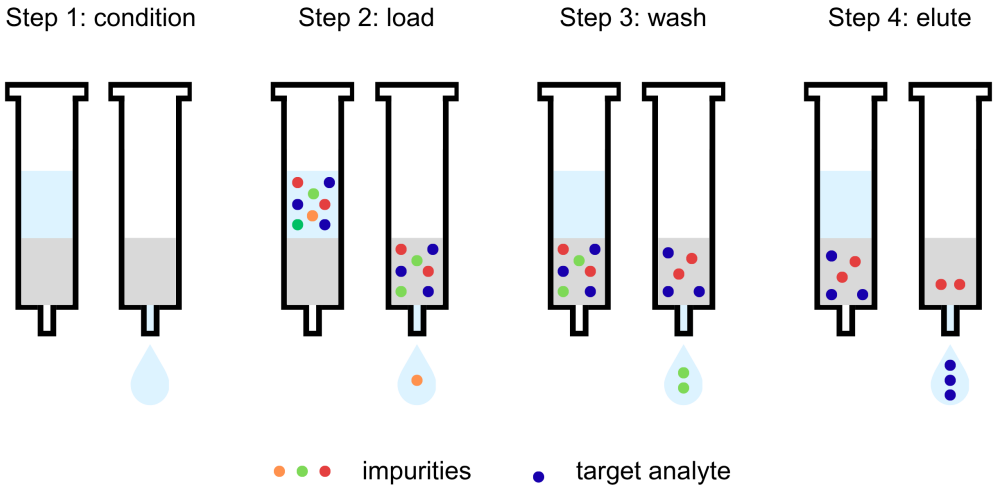


Figure 5. Solid phase extraction steps illustrated.

Commonly, the analyte binds selectively to the sorbent and is eluted in the final step. However, in some SPE formats, the target analyte passes through the sorbent during the loading step, while the interfering compounds are retained in the sorbent.[68,75]

5.1.3 Challenges

Although QuEChERS and SPE methods are effective cleanup options, their application in challenging matrices exist practical challenges. Coffee is known as a challenging matrix in acrylamide analysis due to its high content of polyphenols, caffeine, and other interfering compounds. Acrylamide is soluble in water and in polar organic solvents such as methanol and acetonitrile. Even though all these solvents have been used extracting acrylamide from coffee matrices, their efficiency has not been systematically compared, for example, by using the same coffee sample with different sample preparation protocol.

Water is the most commonly used solvent for extraction.[76] Several studies have noted that aqueous extraction in coffee increases the need for careful sample clean-up before the analysis.

For example, Şenyuva & Gökmen[81] (2005) found that water extracted coffee contained high levels of matrix interferences and even two consecutive SPE steps were insufficient. This resulted in poor separation and notable signal suppression due to matrix effects.

Water based extraction methods can be highly complex. In the method developed by Bertuzzi *et al.*[79] (2017) water extraction was combined with Carrez reagents and hexane, followed by QuEChERS partitioning using acetonitrile, and a two-step clean-up. First, the sample was treated with dispersive SPE (dSPE) using basic alumina, and then with an OASIS hydrophilic-lipophilic balance (HLB) column. This approach reduced matrix effects and improved repeatability without compromising analyte recovery, but it increases sample preparation time.

Bortolomeazzi *et al.*[78] (2012) also extracted acrylamide with water using ultrasound-assisted extraction. The samples were then cleaned using solid-phase extraction (SPE) with a mixed-bed sorbent, which include three different materials: reversed-phase C18, strong anion exchange resin (SAX), and strong cation exchange resin (SCX). This combination enabled effective retention of both hydrophobic and ionic compounds, which improved clean-up efficiency for analysis. Wang *et al.*[82] (2022) investigated how different SPE cartridges performed after QuEChERS-type pre-extraction to isolate acrylamide from coffee samples during sample preparation. They found that the samples were not sufficiently purified with individual hydrophilic-lipophilic balance (HLB), mixed-mode strong anion exchange (MAX) and mixed-mode strong cation exchange (MCX) cartridges, and therefore, acrylamide could not be reliably detected due to signal interference from the matrix. On the other hand, their mixed sorbent cartridge provided the best purification efficiency and reproducibility. This one-step SPE cartridge contained graphitized carbon black (Carb), together with SCX and primary-secondary amine (PSA). In conclusion, while several extraction and cleanup methods are available, their applicability depends on the sample matrix and available resources.

5.2 Chromatographic Techniques

Acrylamide is often analyzed using high-performance liquid chromatography coupled with tandem mass spectrometry (HPLC-MS/MS). Although gas chromatography with a mass spectrometer (GC-MS) has been widely used, it has mostly been replaced by HPLC-MS/MS. In turn, supercritical fluid chromatography-tandem mass spectrometry (SFC-MS/MS) has

shown an approximately 11 fold higher sensitivity in the determination of acrylamide in coffee than HPLC-MS/MS.[74]

Acrylamide is often ionized using the triple quadrupole (QQQ), although Orbitrap analyzers have also been applied.[83] The ionization mode is typically electrospray ionization in positive mode (ESI^+), but the use of atmospheric pressure chemical ionization (APCI) has been reported.[76] Time-of-flight (TOF) instruments have shown promising results in resolving acrylamide from interfering compounds, thus improving sensitivity. Although HPLC with other detectors, such as diode array detection (HPLC–DAD), has been used, these methods are generally less sensitive than tandem mass spectrometry (MS/MS).[72,76]

In this thesis the analysis of acrylamide is carried out using HPLC-MS/MS triple quadrupole in ESI+ mode. Relevant theoretical background is discussed, and the basic principles of gas chromatography and supercritical fluid chromatography (SFC) are mentioned. The SFC–MS/MS has shown exceptional sensitivity and is therefore covered here even though it has not yet been widely used.

5.2.1 High-Performance Liquid Chromatography

The HPLC instrument consists of five main components: the solvent delivery system, the sample injection system, the column, the detector, and the control and data-acquisition system. Instrumentation of HPLC is shown in Figure 6. The solvent delivery system includes a pump and a degasser, which removes dissolved gases from the mobile phase before reaching the pump. The mobile phase is typically a mixture of water and an organic solvent.[68]

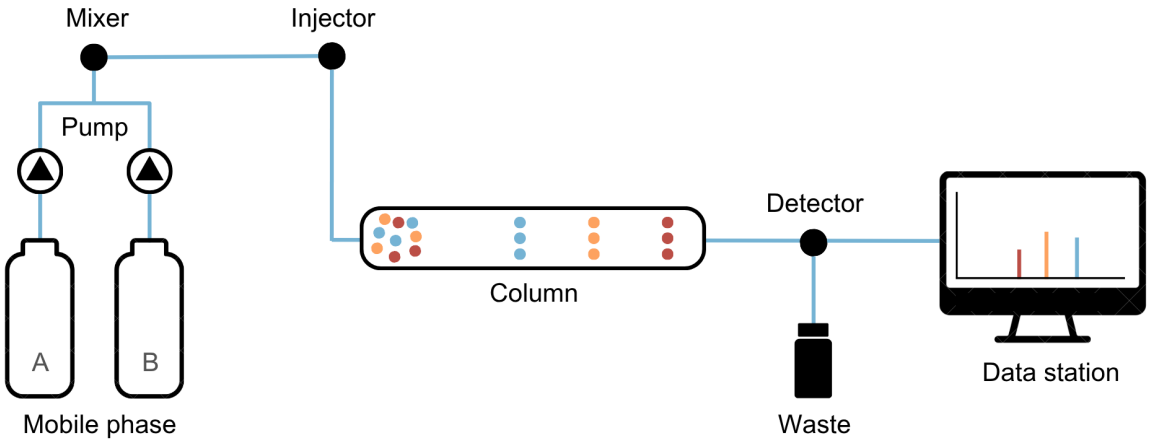


Figure 6. Simplified flow diagram of LC-system.

Sample injection is performed by an autosampler, where an injector withdraws the sample from a vial and introduces it into the column. The column is normally a stainless-steel or PEEK tube packed with silica particles coated with a stationary phase (commonly C18-material). Smaller particle sizes improve separation efficiency but increase backpressure. A guard column is placed to protect the analytical column to filter out particulates and strongly retained impurities.[68]

Compounds in the sample separate in the column based on their chemical interactions with the stationary phase. The stronger a compound interacts, the longer it is retained. Eluting analytes pass through the detector (e.g. MS), which converts their concentration into an electrical signal. The resulting chromatographic data are collected and processed by the data-acquisition software, producing a digital chromatogram on the computer. In HPLC analysis, the peak area is directly proportional to the analyte concentration. [68]

The choice of column and mobile phase has potentially the most significant effect on the retention of acrylamide and the reproducibility of the analysis. Nature of acrylamide might cause some challenges when using reversed-phase columns due to its polarity and small size. It often elutes too close to void time, and it's suggested that retention time should be at least twice the length of the void time to ensure sufficient retention.[77] This often requires a nearly 100% aqueous phase, which may result in retention loss, poor reproducibility, and peak distortion when using a column packed with densely bonded C18 silica particles. This phenomenon is caused by water dewetting from hydrophobic mesopores, which reduces analyte interaction with the stationary phase.[84–86]

Hydrophilic interaction chromatography HILIC columns have also been used for acrylamide determination, but reversed-phase columns operating in 100% aqueous mobile phase are still preferred.[77] HILIC columns do not retain acrylamide sufficiently, causing the analyte signal to co-elute with matrix components and resulting in reduced sensitivity and quantification accuracy.[87]

5.2.2 Other Relevant Chromatographic Systems

The basic principle of supercritical fluid chromatography (SFC) is similar to HPLC. This system uses supercritical carbon dioxide ($SCCO_2$) as the mobile phase, which has low viscosity and polarity. This allows to use higher flow velocity leading to faster analysis time of wider

range of different compounds. It is possible to use polar organic solvent as modifier of mobile phase with $SCCO_2$ making it easier to separate acrylamide from complex matrix and retain selectivity.[74]

In Gas Chromatography (GC) the sample is injected into the column using a heated injector, which rapidly vaporize the sample. In split mode, most of the sample is vented out, and only a small portion enters the column, producing narrow peaks. In spitless mode, the entire sample enters the column, increasing sensitivity. High-quality inert carrier gas is used to carry the sample through the capillary column that is coated with a stationary phase. This stationary phase is usually a nonvolatile liquid that is covalently bonded to the inner wall of a fused-silica tube. The internal diameter and length of the column affect its separation efficiency, as well as on the thickness of the stationary phase film. GC run programs/methods allow the adjustment of various parameters: including oven temperature program, which control the heating rate and maximum temperature, carrier gas flow rates and detector settings.[68]

Using GC-MS to determine acrylamide often requires sample derivatization, which is time-consuming. If any of the precursors of acrylamide are still present, they might react and form more acrylamide during analysis because the process is carried out at a high temperature. On the other hand, the high temperature during analysis could also degrade acrylamide into other compounds. Both scenarios would lead to an unreliable absolute concentration of acrylamide. In addition, the run time is rather long.[74]

5.2.3 Chromatographic Performance Parameters

The separation performance of a chromatographic method can be evaluated by calculating the column efficiency, retention factor, selectivity, and resolution.

Column efficiency is determined for different columns to compare their performance. It is expressed as the theoretical plate number (N), which can be calculated using the following Equation 1:

$$N = 16 \left(\frac{t_r}{W}\right)^2, \quad (1)$$

where N is the theoretical plate number, t_r is the retention time and W is the peak width at the baseline.

The retention factor (k) describes the strength of interaction between the analyte and the stationary phase. It is defined as the ratio of the time the analyte spends in the stationary phase to the time it would take to elute if it had no interaction, which is called the dead time. The retention factor is calculated using the following Equation 2:

$$k = \frac{t_r - t_m}{t_m}, \tag{2}$$

where, t_r is the retention time of the analyte and t_m is the dead time, meaning the time required for an unretained compound to pass through the column. These are demonstrated in Figure 7. A retention factor ≥ 2 has been suggested as the minimum threshold for reliable quantification of acrylamide.[77]

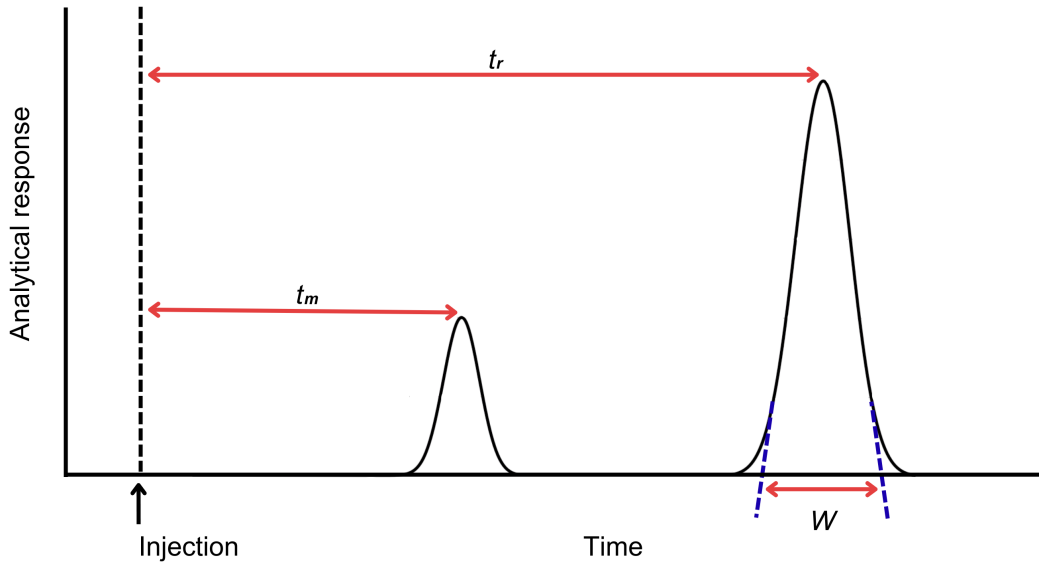


Figure 7. Illustrated chromatogram of retention time (t_r), dead time (t_m) and the peak width (W).

The separation factor (α) describes how well two analytes are separated based on their retention time. It is defined as the ratio of the retention factors (k) of two peaks, where the second peak has the higher retention time. It can be calculted using the following Equation 3:

$$\alpha = \frac{k_2}{k_1} \tag{3}$$

If the separation factor is 1, it means that the two compounds co-elute, while values greater than 1 indicate separation. Selectivity is mostly influenced by the chemical nature of the mobile and stationary phases, and it can be improved by changing either phase or adjusting temperature.

Chromatographic performance can also be assessed by peak symmetry, which is often done by determining the asymmetry factor (A_s). It is calculated from the peak width at 10% of the peak height using the following formula:

$$A_s = \frac{a}{b} \ , \tag{4}$$

where b is the width of the back of the peak and a is the width of the front. Ideally, the peak is Gaussian and A_s = 1.0. However, good symmetry is indicated by values of 0.9–1.2, while higher values indicate tailing and lower values indicate fronting. Peak symmetry is important for resolution, since tailing or fronting may compromise the accuracy of quantification.

Resolution (R_s) describes how well two peaks are separated, and it depends on column efficiency, analyte selectivity, and retention. It is calculated by using the variables defined in the previously presented equations as follows:

$$R_s = \frac{\sqrt{N}}{4} \ \frac{(\alpha-1)}{\alpha} \ \frac{k}{k+1} \tag{5}$$

Together, these parameters provide a comprehensive overview of the method's performance and robustness.

5.3 Mass Spectrometry

Mass spectrometry (MS) is an analytical technique used to determine the mass-to-charge ratios (*m/z*) of ions derived from compounds. The method is based on converting the molecules under study into ions, which are separated based on their *m/z* values, and their abundance is measured. This creates a mass spectrum that provides information about a structure of the compound. The main components of a mass spectrometer are an ion source, a mass analyzer, and a detector.

Ionization techniques are generally categorized based on whether ion formation occurs under atmospheric pressure or in a vacuum. In liquid chromatography (LC), ionization typically takes place at atmospheric pressure using methods such as electrospray ionization (ESI) or atmospheric pressure chemical ionization (APCI). In contrast, gas chromatography (GC) uses vacuum-based ionization techniques, such as electron ionization (EI) and chemical ionization

CI. The most common mass analyzers are quadrupole, triple quadrupole (QQQ), time-of-flight (TOF), and Orbitrap. The most frequently used detector is the microchannel plate (MCP), which detects the arrival of ions and converts the signal into an electrical response. In this work, the theoretical focus is limited to tandem mass spectrometry (MS/MS) using ESI as the ionization method and a triple quadrupole (QQQ) mass analyzer for ion selection and fragmentation analysis.

5.3.1 Electrospray Ionization (ESI)

Electrospray ionization (ESI) is a soft ionization technique, meaning it mainly produces intact molecular ions without breaking the analyte into fragments. In the process, a diluted sample solution is introduced through a metal capillary held at high voltage (typically 2–5 kV), which produces a Taylor cone at the tip. From this cone, a fine spray of charged droplets is emitted. The solvent is evaporated under a flow of heated nitrogen gas, leading to progressive droplet fission due to Coulombic repulsion. This results in smaller progeny droplets and finally the release of gas-phase ions.[88] The mechanism of ESI is illustrated in Figure 8.

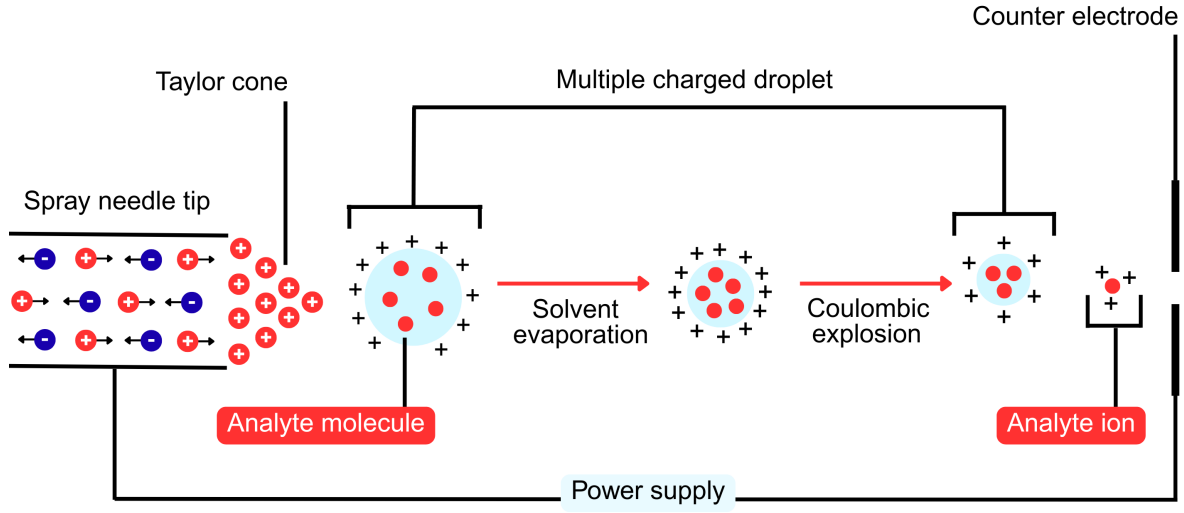


Figure 8. Diagram showing the electrospray ionization mechanism.

Two models have been proposed to explain ion formation: 1) the ion evaporation model (IEM), where small ions are emitted directly from the droplet surface, and 2) the charge residue model (CRM), which assumes the droplet evaporates completely, leaving a single ion behind.[88]
ESI can be operated in positive or negative ion mode, depending on the analyte's chemical properties. In the case of acrylamide, positive ion mode (ESI+) is typically applied, generating $[M+H]^+$ ions. However, when electrospray ionization (ESI), is used to quantify acrylamide in

coffee, studies have shown that isobaric interferences can occur, leading to overestimation of acrylamide levels.[89]

5.3.2 Triple Quadrupole (QQQ)

The quantification of acrylamide requires high analytical sensitivity and selectivity, which is effectively achieved using tandem mass spectrometry (MS/MS) coupled with a triple quadrupole (QQQ) mass analyzer. A triple quadrupole system consists of three quadrupoles: Q1 acts as a mass filter selecting the precursor ion, Q2 serves as a collision cell for fragmentation, and Q3 detects the resulting product ions.[90]

Multiple reaction monitoring (MRM) is the most employed acquisition mode in QQQ systems for acrylamide analysis.[90] It allows monitoring of specific transitions from precursor to product ions, which enhances selectivity in complex food matrices. For acrylamide, typical MRM transitions include *m/z* 72 → 55 and 72 → 27. Other ions may also form, and the proposed fragmentation pattern is shown in Figure 9.

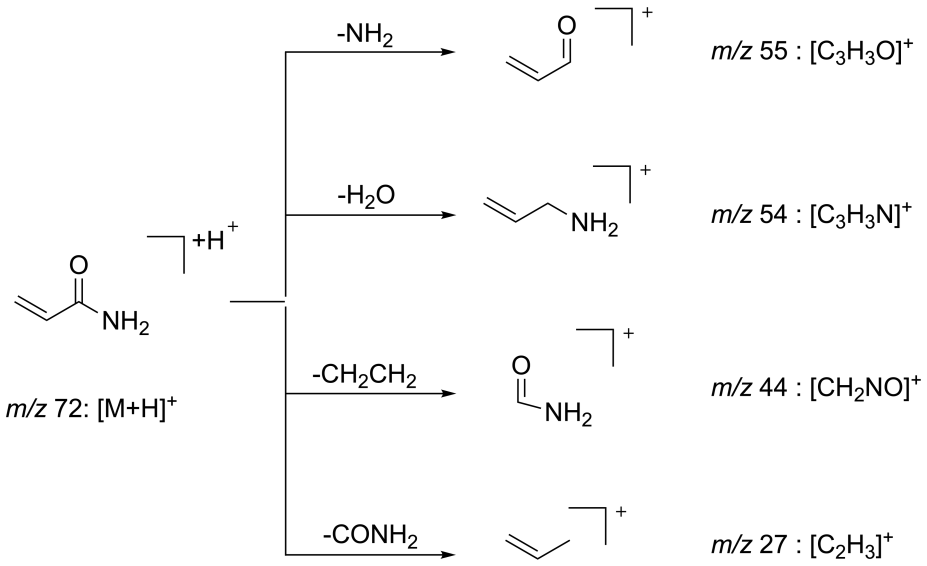


Figure 9. Fragmentation pattern of acrylamide in triple quadrupole MS using ESI+.

One of the main challenges in acrylamide analysis is the presence of isobaric interferences, where co-eluting matrix compounds produce precursor or product ions with the same *m/z* values as acrylamide. It has been reported that compounds such as N-acetyl-β-alanine, 3-aminopropionamide, and lactamide can go through in-source fragmentation, generating *m/z* 72 ions. These ions may further produce fragments identical to those of acrylamide, which

might lead to a false positive signal or overestimation of acrylamide concentrations in MRM mode.[89,91]

These interferences can be minimized by optimizing chromatographic separation and mobile phase. For example, adjusting the mobile phase pH or using an appropriate LC column can prevent co-elution of interfering compounds. This ensures that only acrylamide is detected within the MRM transitions, which improves the accuracy and reliability of the method.[89]

5.4 Method Validation

Validation is the process of evaluating the performance of an analytical method and providing objective evidence that it is fit for its intended purpose by meeting the specified requirements. In other words, it involves performing various tests and procedures to assess the reliability of the analytical method. Validation can be carried out using either an interlaboratory comparison or a single-laboratory approach. Typically, validation includes the assessment of selectivity, LOD & LOQ linearity, repeatability, accuracy, and trueness of the method.[68,92]

Single-laboratory validation is used, for example, when a laboratory develops a new method or needs to validate a modified method for its own use. This approach is particularly relevant if all the equipment or materials required for a published standard are not available.[92] Interlaboratory validation means that the same samples are analyzed and validated in multiple laboratories using the same method. This approach provides information on the method's reproducibility across different laboratories and increases the reliability of the validation. For example, in one study in 2009, a method validation by collaborative trial was carried out where coffee samples were analyzed in 11 different laboratories across Europe.[93] In this thesis, a new method was developed because the specific analytical setup used had not been previously applied to this type of analysis. Therefore, single-laboratory validation was carried out to demonstrate that the method performs reliably under the specific conditions of this study.

In analytical chemistry, specificity refers to the ability of a method to truly identify and quantify an analyte without any interference from other components. Selectivity, on the other hands, describes the ability of a method to reliably determine the analyte within a complex matrix, without interferences from other matrix components that could introduce bias into the measurement signal.[69] The determination of acrylamide in coffee is typically described in terms of selectivity, as the method must differentiate the analyte from complex matrix components

that may interfere with the measurement.[77] For example, the presence of N-Acetyl-β-alanine in the coffee matrix may cause significant overestimation of acrylamide levels, as mentioned earlier.[89]

The range of analytical method is determined by the interval between the lowest and the highest concentration of analyte that the instrument can reliably detect. The range can be validated by using linear calibration model. Linearity is typically evaluated by preparing independently multiple calibration standards covering the working range of the method, which include several concentration levels evenly distributed across the range. The number of concentration levels varies depending on the guidelines; for example, ICH Q2/R2[94] recommends at least five levels, while Eurachem[69] suggest using six to ten levels to ensure a reliable assessment.

The calibration curve is formed by plotting the analyte concentration (x-axis) against the corresponding response (y-axis). The levels should be chosen to ensure that the analyte concentrations in the samples fall within the range of the curve.[68,94] Typically, linearity is assessed by calculating the correlation coefficient (R^2), the y-intercept and the slope of the regression line. It is generally accepted that R^2 should be at least 0.999.[68] However, it is also important to examine the residuals, as they can reveal systematic deviations from linearity that R^2 alone may not show. Residuals are the differences between the measured values and the predicted values.[69,94]

In cases where the analyte is unstable or where matrix effects are likely, calibration can be performed relative to an internal standard (IS). The IS is typically an isotopically labelled form of the analyte, which is chemically very similar and thus compensates for analyte instability and matrix effects. The concentrations in the calibration curve are expressed as the ratio between the analyte and the IS, while the response reflects their respective signal intensities.[69] This approach has been applied in the analysis of acrylamide in coffee.[89]

After the calibration curve is established, one need to determine which are the lowest concentrations that analyte can be detect and quantify. The lowest point that analyte is reliably detected is called Limit of Detection (LOD), while the lowest point that analyte is confidently quantified is called Limit of Quantification (LOQ). Terminology varies and sometimes these are referred as Detection Limit (DL) and Quantitation Limit (QL). LOQ is the smallest value of working range.

LOD and LOQ can be calculated different ways. Those can be determined based on signal-to-noise ratio (S/N) when analysis form baseline noise. A S/N of 3:1 is generally considered the LOD while S/N of 10:1 is considered LOQ.[94] This is widely used in LC-MS validation.[74] It is also common to calculate the LOD and LOQ based on the standard deviation of a linear response and a slope. LOD and LOQ can be expressed as shown in Equations 5 and 6:[94]

$$\text{LOD} = \frac{3.3\,\sigma}{S}\ , \tag{5}$$

$$\text{LOQ} = \frac{10\,\sigma}{S}\ , \tag{6}$$

where σ = standard deviation (SD) of the response and S = the slope of the calibration curve. The slope can be determined from the regression line or calculated using the LINEST function in Excel. The SD can be calculated from replicate measurements of a low-concentration standard.[69]

Accuracy describes how close the measured results are to the true value. This can be demonstrated by (i) using certified reference materials (CRM) with known concentration, (ii) comparing results obtained with two independent analytical methods, (iii) analyzing a blank matrix spiked with a known level of analyte, or (iv) spiking the sample matrix with the analyte at three different levels.[68] Using the CRM allows for the evaluation of the accuracy of the entire analytical process, including extraction efficiency[69]. In contrast, approaches (iii) and (iv) can be used to assess the accuracy of the measurement phase and the recovery during sample preparation. Previously, a FAPAS CRM for coffee was available and used.[95] Currently the only version available is for instant coffee.

Precision, on the other hand, refers to the consistency of results when the same sample is measured multiple times under defined conditions. It includes repeatability, intermediate precision, and reproducibility, and is typically expressed as the standard deviation or relative standard deviation (%RSD). [69] The precision types and their definitions are presented in Table 2.

Table 2. The different types of precision and their corresponding definitions

Precision type	Definition
Repeatability	The closeness of agreement between results obtained under the same conditions.
Intermediate precision	The variation observed within the same laboratory under different conditions e.g. different days or equipment.
Reproducibility	The precision obtained when the same sample is analyzed in different laboratories under varying conditions.
Ruggedness	The degree of reproducibility under minor variations of method parameters e.g. pH or temperature.

Trueness describes how close the mean value of measurements is to the real value. Recovery experiments provide indirect evidence of trueness and of the efficiency of the sample preparation, when suitable CRM is not available.[69] In a typical recovery study of acrylamide, sample is spiked with a known amount of isotopically labeled acrylamide.[72] Recovery (%) is calculated using the following Equation 7:

$$R = \frac{Area_{spiked}}{Area_{ref}} x\ 100, \quad (7)$$

where $Area_{spiked}$ refers to the peak area of a sample spiked at the beginning of sample preparation, whereas $Area_{ref}$ refers to the peak area of the pure analyte in solvent. Recoveries close to 100 % indicate that the analyte is quantitatively extracted and measured. Significant positive or negative deviations point to losses, matrix effects, or systematic bias in the measurement phase.

5.5 Physico-chemical Characterisation of Coffee

Coffee quality, safety, and shelf life are closely linked to its physicochemical properties. These parameters include moisture content, pH, water activity, and color. These define the sensory properties, influence its chemical stability and the formation of thermal products like acrylamide. For example, higher roasting temperatures lead to lower moisture and darker color, which are often associated with reduced acrylamide levels but also possible loss of volatile compounds contributing to flavor. Additionally, water activity and moisture can be used to

support the determination of chemical stability and oxidation rate during storage, which affect shelf life.

In this study, thermogravimetric analysis (TGA), color measurement, and moisture content analysis were performed to support the interpretation of acrylamide results. As color and moisture content have already been discussed in Chapters 3 and 4 in the context of roasting and storage, only TGA is addressed in this section.

A typical TGA system has a microbalance, a programmable furnace, a temperature control unit, and a gas flow system. The sample is placed in a small crucible on the balance inside the furnace. The balance measures mass change as a function of temperature under fully controlled conditions. The atmosphere can be inert (e.g., nitrogen) or oxidative (e.g., air or oxygen), depending on the analysis goal.[96] Thermogravimetric analysis (TGA) has traditionally been used in the analysis of inorganic materials and thermally stable compounds. Its application to the chemical characterization of foods, and in particular coffee, is a relatively new approach.

As mentioned in earlier chapters roasting coffee is a thermochemical process that changes the bean's aroma, moisture content, and chemical composition. The chemical composition, thermal stability, and expected quality and shelf life of a roasted bean can be assessed with thermal analysis methods.[97] TGA is one of the most informative among these techniques. Grzelczyk *et al.*[98] (2022) demonstrated that TGA can also be applied to coffee samples with complex organic matrices. This allows the assessment of thermal degradation and moisture content.

In turn, Brzezinska *et al.*[97] (2021) characterized the oxidative stability and kinetic parameters of coffee using TGA and DSC. They observed that the lipid fraction of light-roasted coffees is more oxidatively stable than dark-roasted coffees. This is an important finding for coffee quality, as coffee lipids diffuse to the surface and protect aromas from oxidation and evaporation. In coffee research, TGA becomes especially powerful when combined with another instrument, such as differential scanning calorimetry (DSC) or mass spectrometry (MS).[99]

6 Materials and Methods

The main goals for the experimental part were i) developing and validating an HPLC-MS/MS method for the quantitative analysis of acrylamide, ii) finding a sample preparation protocol that isolates acrylamide from the coffee matrix, and iii) obtaining representative coffee samples and quantify their acrylamide levels. A secondary goal was to characterize the relevant physicochemical properties of coffee samples. This included measuring the color of roasted coffee, its moisture content, and thermal behavior using thermogravimetric analysis.

Several different clean-up protocols were evaluated on both ground coffee samples and brewed coffee. The method development process included a systematic series of experiments using different columns, mobile phases and instrument parameters. The methods were evaluated using various analyte concentrations during development. The final was selected and optimized based on the quality of the analyte response, retention time, method sensitivity, and repeatability.

6.1 Materials

This section lists the reagents employed and summarizes the key details of the collected coffee samples (three coffee groups). List of used reagents is shown in Table 3.

Table 3. List of used reagents

Reagent	Molecular Formula	Manufacturer	Purity
Acrylamide	C_3H_5NO	Dr. Ehrenson	N/A
Acetonitrile	CH_3CN	Honeywell	99.9%
Acrylamide-2,3,3 D3	$C_3H_2D_3NO$	Dr. Ehrenson	N/A
Formic Acid	CH_2O_2	SIGMA ALDRICH	>98%
Methanol	CH_3OH	VWR	≥99.9%
Potassium hexacyano-ferrate(II) trihydrate	$K_4Fe(CN)_6 \cdot H_2O$	MERCK	> 99%
Water	H_2O	ELGA	Ultrapure
Zinc sulphate heptahydrate	$ZnSO_4 \cdot 7\ H_2O$	VWR	>99%

6.2 Coffee Samples

Group 1: Differences between specialty and commercial coffee

The first group consisted of specialty and commercial coffee. Coffees were either given or bought from a local grocery store. This sample set consists of 19 coffee samples, including 10 specialty coffees (ID = S1–S10) and 9 commercial coffees (ID = C11–C19), where S and C refer to specialty and commercial coffee, respectively. The roast ages of all samples, as well as the end bean temperature and roasting time for the specialty coffees, are presented in Appendix 1.

Group 2: Impact of storage duration on stability of acrylamide

The second coffee sample group consisted of samples to evaluate the reduction of acrylamide concentration in coffee. Three coffees from Group 1 (S5, S8 and C11) were analyzed at different shelf-life periods to compare acrylamide levels. In addition, a third specialty coffee (ID = S11) was selected, which was collected weekly from a specialty coffee roastery. Each batch of coffee S11 was frozen at -18 °C at different shelf-life periods (7–84 days) after roasting.

Group 3: Dynamics of acrylamide concentration during roasting

Coffee batch of 12 kg was roasted using a Giesen W15A roaster. Samples were collected from the drum every 30 seconds with a sample spoon starting when the bean internal temperature reached 120 °C, which is reported in the literature as the onset temperature of the Maillard reaction.[89] The collected samples were immediately placed into pre-cooled aluminum containers, which had been pre-cooled on dry ice. Then, the containers were placed back on dry ice, and the coffee beans were stirred to ensure rapid and uniform quenching of ongoing chemical reactions as efficiently as possible. A total of 17 samples were collected (ID = R1–R17), where R refers to roasting.

6.3 Methods

6.3.1 HPLC-MS/MS

Acrylamide was analyzed using high-performance liquid chromatography coupled to tandem mass spectrometry (HPLC-MS/MS) with an Agilent Technologies 1290 Infinity LC system coupled to an Agilent Technologies 6460 Triple Quadrupole mass spectrometer, operating in positive electrospray ionization mode (ESI+). The HPLC instrument was equipped with a vacuum degasser, binary pump, a temperature-controlled column oven, and an autosampler. Chromatographic separation was performed using a Waters Acquity UPLC HSS T3 column with VanGuard FIT (50 mm × 2.1 mm, 2.7 µm) at a temperature of 30 °C. The mobile phase consisted of water and acetonitrile (99:1, v/v) containing 0.1% of formic acid. The run was isocratic with a flow rate of 0.3 mL/min for 2 minutes. The injection volume was 4 µL.

Each sample was analyzed in triplicate. After the third replicate of each sample, the column was flushed for 3 minutes at a flow rate of 0.4 mL/min with the initial mobile phase (99:1, H_2O:ACN, v/v), followed by a 2.5-minute wash with 50:50 (H_2O:ACN, v/v). The column was then re-equilibrated for 3 minutes with a flow rate of 0.3 mL/min.

The mass spectrometer was equipped with an Agilent Jet Stream electrospray ionization (AJS ESI) source and operated in positive ion mode. The detailed ion source parameters are summarized in Table 4.

Table 4. Ion source parameters

Parameter	Value
Ionization mode	ESI+
Capillary voltage (V)	1000
Nozzle voltage (V)	400
Nebulizer pressure (psi)	60
Drying gas flow rate (L/min)	5
Drying gas temperature (°C)	250
Sheat gas flow rate (L/min)	11
Sheat gas temperature (°C)	350

Quantification was performed using the MRM transitions *m/z* 72.1 → 55.1 and *m/z* 75.1 → 58.1, while MRM transition *m/z* 72.1 → 27.1 was used as a qualifier ion for identification of acrylamide. In addition, isobaric interference ions were monitored under alternative transitions to verify whether they co-eluted with acrylamide. The fragmentor voltage was set to 80 V, and the dwell time was 100 ms, with all MRM transitions and their respective collision energies listed in Table 5.

Table 5. MRM Transition for Acrylamide, Acrylamide-d_3, and Isobaric Compounds Chosen for Selectivity Assays

Compound	Precursor ion	Product Ions	Collision energy (eV)
Acrylamide	72.1	55.1	7
		27.1	15
Acrylamide-d_3	75.1	58.1	11
3-aminopropanamide	89.0	72.1	9
Lactamide	90.1	72.1	9
N-Acetyl-B-alanine	132.0	72.1	9

6.3.2 Moisture Analysis

Moisture determination was performed in triplicate for each coffee sample using a MX-50 moisture analyzer (A&D Company), following the manufacturer's protocol for ground coffee. Samples were prepared by grinding beans with a Comandante C40 grinder (using setting number 15, corresponding to ~500 µm particle size). A sample mass of 5.0 ± 0.5 g was analyzed at 140 °C with the standard-mid program, which maintained heating until the mass change stabilized at ≤ 0.05% per minute.

6.3.3 Colorimetric analysis

Color was determined using an automatic colorimeter Konica Minolta CR-410 model chroma meter, which was calibrated with a white reference plate before measurements. To minimize measurement uncertainty, all coffee beans were ground using the same grinder and setting as mentioned earlier and weighed equally (5.0 ± 0.5 g) on a watch glass. The samples were lightly pressed flat to avoid light scattering from loose particles. The lighting conditions were standardized, and the measurement chamber was completely light-tight. Each sample was measured in triplicate.

6.3.4 Thermogravimetric Analysis (TGA)

Six coffee samples, representing both specialty and commercial categories, were thermally characterized with PerkinElmer STA 6000 simultaneous thermal analyzer, which records thermogravimetric (TG) and differential scanning calorimetric (DSC) signals from the same sample. The instrument was calibrated for a platinum crucible at heating rates of 2.5 °C/min and 10 °C/min.

TGA and DSC analyses were operated under O_2 atmosphere at a flow rate of 40 mL/min, using a platinum crucible. Heating was started at 20 °C, held for 1 minute, and then ramped at 10 °C/min up to 600 °C. The instrument was operated using PerkinElmer Pyris software, version 13. TG curves and first-derivative DTG traces were recorded and processed using the same Pyris software and Microsoft Excel. Each sample was analyzed in triplicate to confirm reproducibility. Sample sizes ranged from 6.1 to 13.3 mg, and the relative standard deviation (%RSD) of weights were $\leq$ 20% within replicates. Samples were ground using the same Comandante C40 grinder and grind size, and measurements were initiated within one minute of grinding to prevent the loss of volatiles.

6.4 Sample Preparation

6.4.1 Calibration Standards

Stock solutions of AA and AA-d_3 were prepared in glass vials at ambient temperature. Each analyte was accurately weighed on a Mettler Toledo analytical microbalance and dissolved in ultrapure water to a final concentration of 1 g/L. The aqueous acrylamide solution exhibited low surface tension, which affected the formation of a clear meniscus and could influence the accuracy of volumetric preparation of calibration solutions. Therefore, the intermediate solutions were prepared gravimetrically at 1 mg/L. Working calibration standards were prepared by serial pipetting the 1 mg/L intermediate solution into clear glass vials, followed by dilution with ultrapure water to give acrylamide concentrations of 1, 3, 10, 50, 100, 150, 200, 500, 700, and 850 µg/L. Each standard contained 100 µg/L of AA-d_3. Before analysis, the standards were filtered through 0.22 µm PTFE syringe filter.

6.4.2 Coffee Samples

Roasted coffee samples are physically and structurally distinct from the green coffee. Samples collected during roasting are much harder and can in general not be ground using grinders meant for roasted coffee. Therefore, samples were divided into two different groups, and the first step of sample preparation was to select the grinding method based on the group.

i) Roasted coffee samples: Samples were ground using a Comandante C40 grinder set to setting number 15.

ii) Samples collected during roasting: Samples were placed in a plastic decanter, followed by enough liquid nitrogen to just cover the beans. The beans were stirred with a metal spoon until the liquid nitrogen had evaporated, and immediately afterward they were poured into a running grinding mill (Fritsch Pulverisette 14).

1 g of each coffee sample was placed in a Kimax glass tube, followed by 5 mL of MeOH, and spiked with 10 µL of AA-d_3 (c = 10 mg/L) as an internal standard. Samples were shaken at high speed on a horizontal shaker for 30 minutes. Then, samples were vortexed for 1 min at 3000 RPM and then centrifuged at 3500 RPM (Heraeus Labofuge 200) for 5 minutes. The supernatants were filtered into new Kimax tubes using a funnel and a micro-filter (Aeropress). 50 µL of Carrez I and II solutions were added to the filtrate and instantly vortexed for 30 seconds at 3000 RPM. Samples were then placed on a horizontal shaker at high speed for 15 minutes. The samples were centrifuged again at 3500 RPM for 10 minutes.

2 mL of supernatant was transferred into a glass tube and 50 µL of water was added. Methanol was evaporated in a 40 °C water bath under a gentle nitrogen flow. Water was included to ensure the sample would not dry out completely. The residue was then reconstituted in 2 mL of water.
A Strata-X 200 mg/3 mL SPE tube was conditioned with 2 mL of MeOH and equilibrated with 2 mL of water at a flow rate of 2 drops per second. The reconstituted sample was then loaded. Elution was performed with 2 mL of water in two portions (2 × 1 mL), the first 5 drops were discarded, and the remaining eluate was collected at a flow rate of 1 drop per second. The collected sample was finally filtered through a 0.22 µm PTFE membrane into an HPLC vial.

In cases where immediate processing was not possible due to scheduling constraints, filtered extracts were stored in the dark at +4 °C for a maximum of 4 days before further processing.

7 Method Development and Validation

The main goal of the experimental part was to develop a quantitative and qualitative HPLC-MS/MS method for the analysis of acrylamide. The secondary goal was to design a fit-for-purpose sample preparation protocol for coffee with the resources available. This chapter discusses the development and validation of the method, while the final analytical protocol, including the sample preparation and the HPLC-MS/MS method is presented in Chapter 6.

The method development process included a series of experiments using different columns and instrument parameters and different analyte concentrations were evaluated for a calibration curve. The parameters were optimized, and their performance was assessed using analyte-response metrics: retention time (RT), sensitivity (S/N), and reproducibility (RT and area %RSD). Two methods were developed and validated to confirm robustness. Method 1 was validated for linearity, sensitivity (LOD/LOQ), precision, trueness, and recovery. Method 2 was validated for linearity, sensitivity (LOD/LOQ), precision, trueness, within-run stability, and accuracy. No statistically significant difference was found between the calibration curves of the two methods at the 95% confidence level.

7.1 Chromatographic Method Development

7.1.1 Column Screening

In practice, method development is often guided by the instruments, columns and materials available in the laboratory. In this work, only one identical column was used, which has been previously reported in the literature; and therefore, other columns were tested. Every test during method development were performed using a quality control sample containing 100 µg/L of AA and IS, unless otherwise is stated.

Kumari *et al.*[100] (2023) compared three different types of reversed-phase columns in acrylamide analysis. Agilent InfinityLab Poroshell 120 EC-C18 (150 mm × 4.6 mm, 4 µm; column 3) showed the best performance. Therefore, this column was initially chosen for the method development. The column showed promising results, but an unknown impurity was observed in the chromatogram. The ion of this impurity was eluted with the same MRM transition as acrylamide, which made separation difficult. This impurity played the major role in method development as the limiting factor. The impurity was co-eluting with acrylamide when the

mobile phase had more than 5% of organic phase. Increasing aqueous mobile phase also improved peak shape and retention, which is supported by previous research.[81]

Other Agilent InfinityLab Poroshell 120 EC-C18 columns were tested to resolve the co-eluting impurity issue. These columns had dimensions of 2.1 x 150 mm, 2.7 µm (column 2) and 4.6 x 100 mm, 2.7 µm (column 1). None of the columns provided satisfactory retention ($k > 1$) and acceptable repeatability for acrylamide, but the evaluation of the same stationary phase under varied chromatographic dimensions provided valuable insights into acrylamide's behavior and the factors that influence it. It was observed that acrylamide elutes faster when column length or internal diameter decrease. In turn, decreasing particle size improved retention factor.

A standard reversed-phase chromatographic C18 stationary phase has long-chain octadecylsilane moieties, which are hydrophobic. Analyzing polar compounds on C18 columns can be challenging because they often require a highly aqueous mobile phase, which leads to poor retention, distorted peak shapes, and co-elution. Therefore, the method development was continued using water-compatible reversed-phase columns. All the column material is presented in Figure 10.

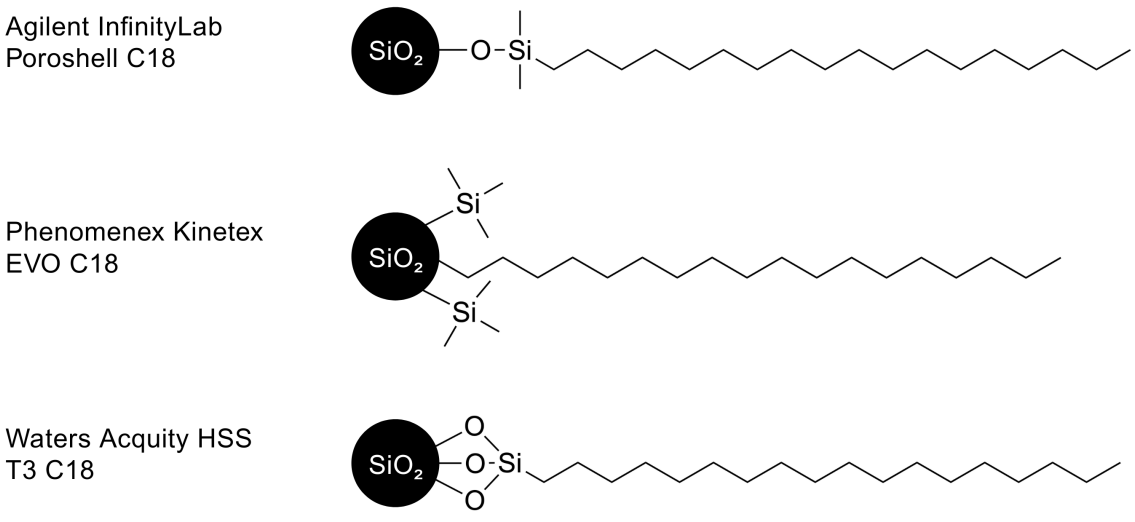


Figure 10. The Stationary Phase Chemistries of the Columns used.[101–103]

According to the literature, reliable quantification of acrylamide in complex matrices requires the use of aqueous stationary phases that can tolerate 100% water as the mobile phase and a retention factor of at least $k \geq 2$.[77] This ensures both decent separation from the solvent front and reliability of the method. The Phenomenex Kinetex EVO C18 is a core–shell C18 column

with enhanced endcapping that reduces silanol activity, improving peak shape and reproducibility for basic analytes.[102] The column is suitability for 100% aquas mobile phase.[102] Its dimensions are 2.1 x 100 mm with a particle size of 2.6 µm (column 4). This column resolved the impurity issue; however, peaks were tailing (A_s = 2.5), and retention was unsatisfactory (k = 1.09). Waters Acquity HSS T3 has been widely used in the determination of acrylamide.[79,82] It is fully porous, high-strength silica column. Its trifunctional bonding and endcapping allow operation with a 100% aqueous mobile phase. The column had dimensions of 50 mm × 2.1 mm with a particle size of 2.7 µm (column 5).

Figure 11 shows the peak shapes of all columns without any smoothing. All peaks are determined with a 72.1 → 55.1 MRM transition from the 100 µg/L standard. The determined S/N values for these peaks are: A) 22.8; B) 80.0; C) 91.0; D) 103.3; E) 241.1; F) 292.3.

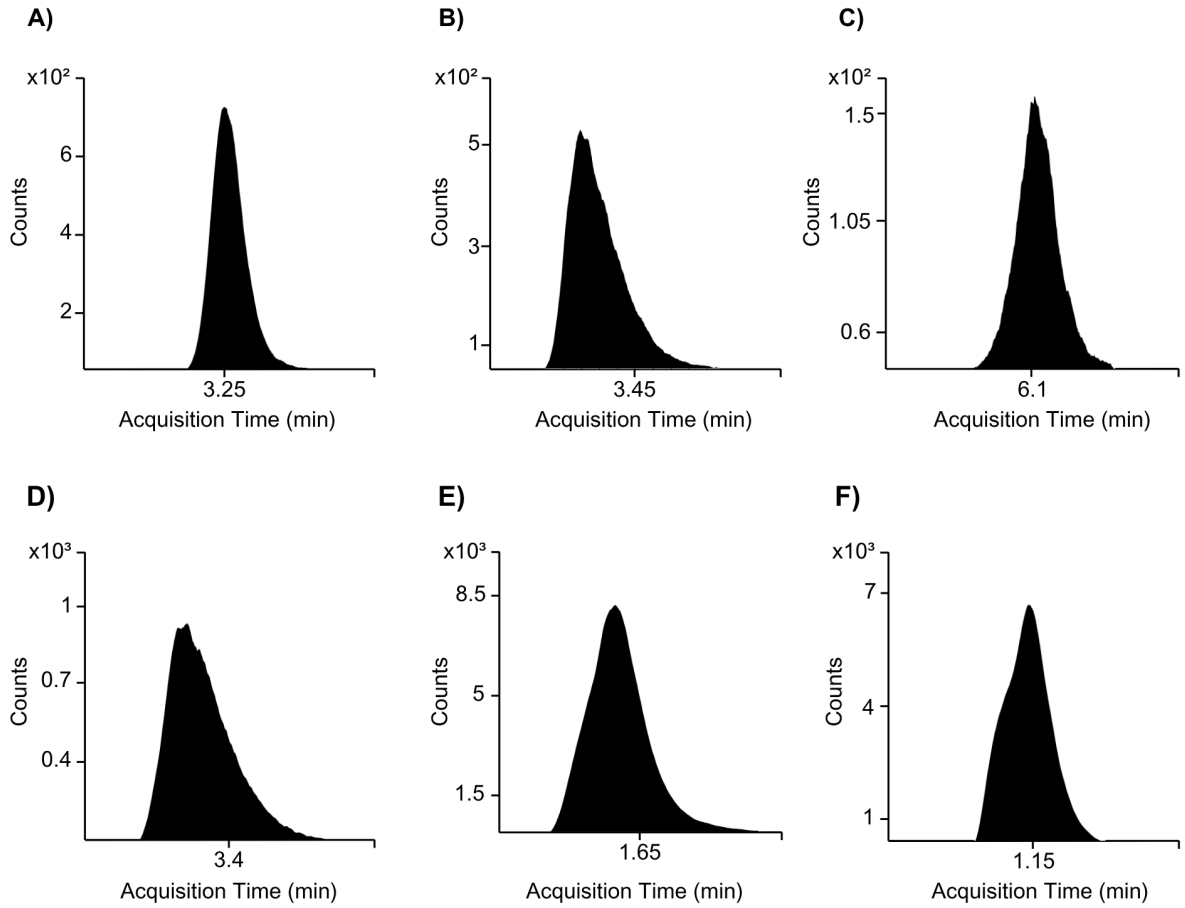


Figure 11. Peak shapes of each column. A–C = columns 1, 2 and 3 (Agilent), D = column 4 (Phenomenex), E = column 5 (Waters) for method 1 and F = column 5 for method 2.

The efficiency of the columns was evaluated using the retention factor (k, see Equation 2). The dead time (t_m) of the columns was calculated theoretically using the Equation:

$$t_m = \frac{V * \varepsilon}{v_f}, \tag{8}$$

where V is the column volume, ε is the column porosity, which is assumed to be 0.68 for the columns used, and v_f is the flow rate (mL/min).

The dimensions and performance of the columns used are presented in Table 6. Columns 1-3 are Agilent InfinityLab Poroshell 120 EC-C18, column 4 is Phenomenex Kinetex EVO C18 and column 5 is Waters Acquity HSS T3. Performance is presented in terms of retention time (t_r); dead time (t_m); full width at half maximum (FWHM); number of theoretical plates (N); retention factor (k), and asymmetry factor (A_s). Two different methods were developed and validated, and therefore, the performance of both are included in the table.

Table 6. Dimensions and performances of columns: t_r *(min)*, t_m *(min)*, FWHM, N, k and A_s

Column	Dimensions ($L \times ID$, d_p)	t_r	t_m	FWHM	N	k	A_s
1	100 mm × 4.6 mm, 2.7 µm	3.21	1.88	0.11	5519	0.70	1.2
2	150 mm × 2.1 mm, 2.7 µm	3.30	1.77	0.15	2747	0.87	2.7
3	150 mm × 4.6 mm, 4.0 µm	6.11	4.84	0.11	17921	0.26	1.1
4	100 mm × 2.1 mm, 2.6 µm	3.28	1.57	0.26	892	1.09	2.5
5_{M1}*	50 mm × 2.1 mm, 2.7 µm	1.58	0.59	0.25	332	1.68	1.1
5_{M2}*	50 mm × 2.1 mm, 2.7 µm	1.13	0.39	0.15	332	1.88	1.2

*M1 refers to method 1 with flowrate of 0.2 mL/min; M2 refers to method 2 with flowrate of 0.3 mL/min

Different columns were tested with slightly different LC-MS parameters during the method development phase, as the conditions were being optimized during the process. Therefore, the results are not directly comparable, but they provide an indication of the resolution, peak shape, and practical applicability of the columns for analyzing acrylamide.

Column efficiencies (N) ranged from a few hundred to nearly 18,000 plates, while retention factors ranged from 0.26 to 1.88. In general, longer and wider columns provided higher efficiencies, while smaller dimensions led to longer retention, but at the expense of peak broadening and lower N. When the tested columns were compared against these criteria,

several showed excellent efficiencies or peak shapes, but could not withstand high water content without pore dewetting. Increasing the amount of organic phase in the mobile phase led to a lower retention factor, reduced S/N and repeatability. Therefore, the selection was finally narrowed down to columns 4 (EVO) and 5 (HSS T3), both of which were compatible with highly aqueous conditions. Column 5 was selected based on peak shape, retention, and sensitivity.

7.1.2 Chromatographic Parameters

The goal was to retain acrylamide as efficiently as possible, but also to distinguish it from other potential interferences. In the early stages of method development, different ratios of organic phase and water were tested on various reversed-phase columns using both isocratic and gradient programs. However, acrylamide eluted too early ($k < 1$) when the proportion of methanol (MeOH) or acetonitrile (ACN) was greater than 5%. Columns 1-3 were not designed to tolerate >95% aqueous phase, which resulted in pore dewetting. In contrast, column 4 was water-tolerant, but produced asymmetric peaks. In addition, acrylamide eluted very quickly even with 100% aqueous phase, and even faster when amount of organic phase was increased. From this point onward, all analysis were performed using column 5.

Column 5 was selected and the gradient program developed by Bertuzzi *et al.*[79] (2017) was tested. The method consisted of a mobile phase of water and ACN (both acidified with 0.1% of formic acid, FA), at a flow rate of 0.2 mL/min. The method included isocratic 100% water for 3 minutes, a linear gradient of 15% ACN, which was held isocratic for two minutes and finally the column was equilibrated for 7 minutes. However, the equilibration was insufficient, as the retention started to shift towards the solvent front and the S/N decreased due to the accumulation of ACN. The column would have required approximately 15 minutes for re-equilibration to achieve repeatable results (%RSD 2.8% n=6), which was not energy-efficient.

In previous studies, several methods have used an isocratic method with a flow rate of 0.1-0.3 mL/min and a low organic phase content (1-3%).[72,79,82,104] In the present study, the column temperature was 30°C and the mobile phase was water and acetonitrile (99:1, v/v). The best retention was achieved at a flow rate of 0.1 mL/min ($k = 1.95$), The addition of acid (0.1% FA) doubled the response for both the analyte and the internal standard. The %RSD for ion

ratios with acid was 1.18%, S/N was 22.7, and RT was 3.47 min (n=3). Without acid, the %RSD for ion ratios was 1.26%, S/N was 21.6, and RT was 3.44 min.

During method development, the column pressure fluctuated and sometimes dropped below 10 bar. This phenomenon occurred a few times, but each time a leak in the capillary tube was traced and the problem was resolved by replacing the tube. These pressure drops made the evaluation of the development process challenging and before the leaks were found it was suspected that they were caused by the low flow rate (0.1 mL/min). Therefore, method development was continued at higher flow rates of 0.2 mL/min (Method 1) and 0.3 mL/min (Method 2). Both methods showed robustness under isocratic conditions using water and ACN (99:1, v/v) with 0.1% FA. In the final method, a wash step was performed with a higher ACN concentration, as described in section 6.2.

The peak obtained with Method 1 showed slightly better symmetry (A_s = 1.1) than that of Method 2 (A_s = 1.2), while retention was poorer with Method 1 (k = 1.68) than with Method 2 (k = 1.88). In addition, the Method 2 was highly repeatable (%RSD 2.95 n=6). The overlaid chromatograms of acrylamide and internal standard of Method 1 and Method 2 are presented in Figure 12.

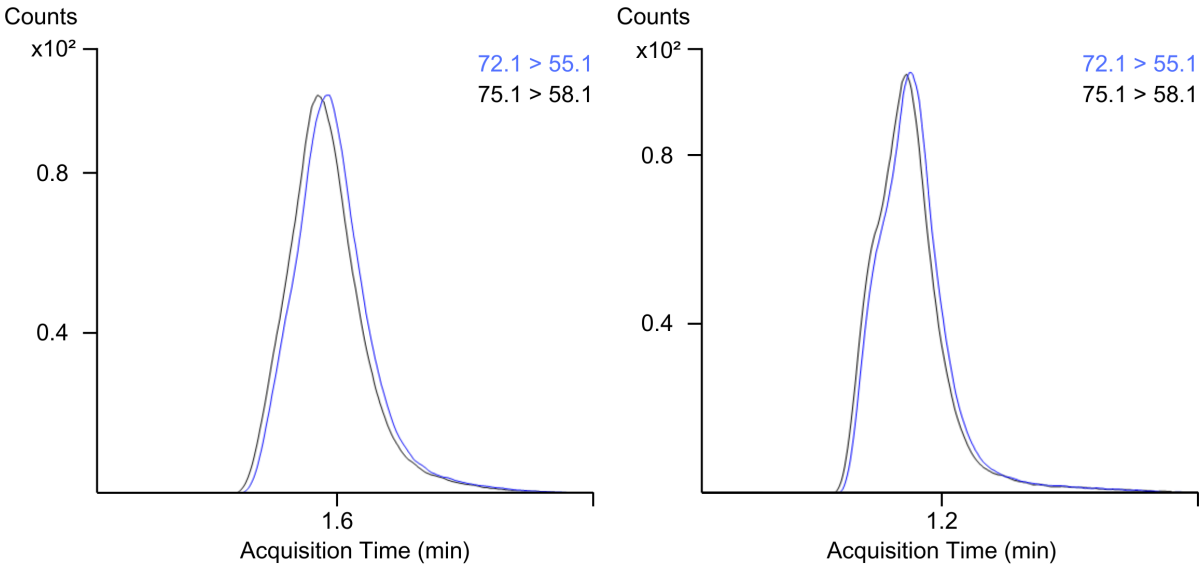


Figure 12. Overlaid chromatograms of AA and IS obtained using Method 1 (on left) and Method 2 (on right).

Acrylamide was analyzed using an isocratic run, and the column was flushed with the organic phase after every three replicate injections to wash out any residues from coffee matrix.

7.1.3 Matrix Interferences

According to the literature, N-acetyl-β-alanine and other isobaric interferences that are present in coffee can produce the same *m/z* 72 fragments as acrylamide during ionization stages and lead to an overestimation of acrylamide unless chromatographic separation is ensured.[89] Retention times of potential isobaric interferences were estimated with secondary transitions MRM 89.0 → 72.1 for 3-aminopropanamide; 132.0 → 72.1 for N-acetyl-β-alanine, and 90.1 → 72.1 for lactamide. Their elution positions relative to acrylamide were evaluated using Method 2. The obtained chromatograms are shown in Figure 13.

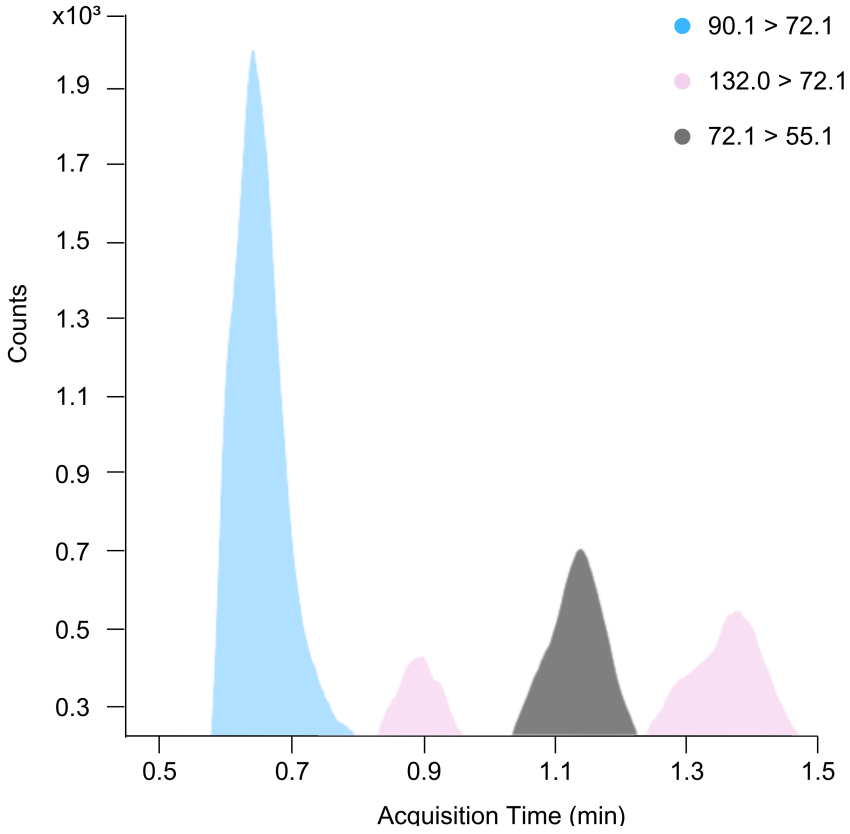


Figure 13. Chromatograms of acrylamide and isobaric compounds overlaid. MRM transition are 90.1 → 72.1 for lactamide (light blue), 132 → 72.1 for N-acetyl-β-alanine (pink) and 72.1 → 55.1 for acrylamide (grey).

No peak was observed at the 89.0 → 72.1 transition. The 90.1 → 72.1 transition gave a strong peak, but the retention time differed from acrylamide. The 132.0 → 72.1 transition produced two peaks in the vicinity of acrylamide, but not at the same retention time. Based on these observations, these compounds do not elute simultaneously with acrylamide, and they were not estimated to influence acrylamide quantitation.

7.2 Optimizing the Ion Source Parameters

The pre-screening of ion source parameters was performed within the same day under the same LC-MS conditions (Method 2), changing one parameter at a time. Therefore, the reported S/N values are comparable. The following optimization rounds, the parameters were fine-tuned stepwise in different runs, so the results are not directly comparable. Changing a single parameter shows trends, but changing multiple parameters simultaneously can cause interaction effects and lead to varied results. The results from pre-screening are presented in Table 7.

Table 7. Effect of MS parameters on signal stability (%RSD) and sensitivity (S/N) during method development

Parameter and value	%RSD	S/N
Fragmentor 135 V	3.45	9.20
Fragmentor 80 V	1.78	107.4
Gas Temperature 250 °C	3.45	9.20
Gas Temperature 350 °C	9.20	8.07
Gas Flow 12 L/min	3.45	9.20
Gas Flow 5 L/min	14.2	101.5
Capillary 2000 V	1.78	107.4
Capillary 2500 V	6.37	87.5
Nebulizer 60 psi	14.2	101.5
Nebulizer 45 psi	2.14	92.2

The fragmentor had the greatest effect on sensitivity. When the fragmentor value was increased from 135 V to 80 V, the S/N increased by 10 times. The same effect was also observed with the gas flow rate. Increasing the flow rate to 12 L min^{-1} collapsed the sensitivity (S/N 9.2). Therefore, 5 L/min was chosen for quantification based on sensitivity. Lower drying gas temperatures gave better results. The %RSD was lower, and the S/N was higher. When the nebulizer value was increased from 45 psi to 60 psi, the sensitivity increased by 10%. Capillary voltage of 2000 V led to lower %RSDs and higher sensitivity (S/N 107.4) than 2500 V (S/N 87.5). In summary, the optimized settings were fragmentor 80 V, gas temperature 250 °C, drying gas 5 L/min, capillary 2000 V, and nebulizer 60 psi.

Finally, the collision energy (CE) for the MRM transitions was optimized based on the S/N to achieve the highest possible sensitivity. The 75 → 58 transition provided a higher S/N at all

tested CE values, so it was optimized so that the ratio of the responses of AA and AA-d_3 is approximately equal to 1 when both concentrations were 100 µg/L. The measured S/N values for the 72 → 55 transition were as follows: CE 7 V: 267.7; CE 8 V: 240; and CE 10 V: 224.7. Therefore, CE = 7 V was selected and for the 75 → 58 transition, 11 V was chosen, giving a ratio of the areas to 1.0 (n=6).

7.3 Sample Preparation

The sample preparation development included three workflows: QuEChERS + dSPE, SPE, and a combination of SPE + QuEChERS. Both coffee beverages and coffee samples extracted with either water or methanol were analyzed.

7.3.1 Liquid Samples

In the QuEChERS method, 9 mL of coffee beverage was transferred to a 50 mL plastic tube. Internal standard was added to the tube (final concentration 100 µg/L). Then, 500 µL of Carrez I and II was added to the sample, followed by vortexing for 1 min and mixing for 10 min. The sample was centrifuged at 4000 rpm for 10 min and 2 mL of supernatant was taken for QuEChERS tube, which contained 200 mg NaCl, 800 mg $MgSO_4$ and 2 mL acetonitrile. The mixture was vortexed for 2 min and centrifuged at 4000 rpm for 20 min. One mL of the acetonitrile layer was collected from the supernatant and transferred to a dSPE tube, which included 50 mg primary secondary amide (PSA) and 150 mg $MgSO_4$. The tube was vortexed for 2 min and in mini-centrifuged at high speed. A 500 µL sample was transferred into a vial and evaporated to dryness under nitrogen. The residue was re-extracted with 500 µl of water and filtered into a vial. The recovery was 31.1%.

Solid Phase Extraction (SPE) was tried to combine with QuEChERS method. A 5 mL coffee drink was taken with an AA-d_3 concentration of 100 µg/L. Then, 250 µL Carrez I and II were added to the sample, after which it was shaken in a horizontal shaker for 15 minutes and centrifuged for 10 minutes at 4000 rpm. Two samples were prepared from this: Strata-X cartridges were conditioned with 3 mL MeOH and 3 mL H_2O, extraction was performed 2 × 1 mL with water, and the fractions were collected in separate vials. The recovery was 64%.

The second sample was repeated by adding 1.5 mL of the extract to a QuEChERS tube containing 250 mg NaCl, 500 mg $MgSO_4$ and 1.5 mL ACN. The mixture was shaken for 20

minutes, centrifuged for 10 minutes at 4000 rpm and collected in a 500 µL vial, the contents of which were dried under nitrogen, redissolved in water and filtered into the vial. The recovery was 61%. The development of the sample preparation was continued using solid sample, so that the same method could be applied to all samples, since samples collected during the roasting stage cannot be used to prepare brewed coffee.

7.3.2 Solid-Liquid Extraction

Two samples were prepared by weighing 1 g of ground coffee and adding 10 mL of water with an AA-d3 concentration of 100 µg/L. The second sample was extracted for 15 minutes by ultrasound without separate heating, and the reference sample was mixed in a vertical shaker for 30 minutes. After this, both were treated identically: Carrez I and II (400 µL) were added, shaken for 20 minutes and centrifuged at 4000 rpm. Strata-X cartridges were conditioned with 3 mL of MeOH and 3 mL of water. Then, 3 mL of sample was loaded into the cartridge and the flow-through was collected to an empty vial, after which the cartridge was extracted with 2 x 1.5 mL of water, which was collected separately. In the case of Ultrasonic Assisted Extraction (UAE), recovery of the flow-through section was 13.7%, and recovery of the eluted extract was 42.0%. For vertical shaker, the flow-through was 23.0% and in the extract 24.7%. UAE can generate heat and more efficient CO_2 degassing occurs, which may change the pH and/or ionic strength, which may affect the retention of acrylamide in the SPE step. Overall, the sample treatment resulted in either strong ion suppression or poor recovery, which is why the methods based on water extraction was not used.

The water-based extraction method resulted in unsatisfactory recoveries and/or ion suppression after purification. Şenyuva & Gökmen[81] (2005) observed a similar problem and their method included methanol extraction and purification with Oasis HLB cartridges. Oasis HLB and StrataX have been reported to be chemically quite similar.[105] StrataX was available in the laboratory and therefore, methanol based extraction method was applied, which is presented in Chapter 6.3.2.

7.4 Method Validation

7.4.1 Linearity

The linearity of the chromatographic methods was determined using a standard curve at different concentrations between 3–850 µg/L. The standard curve was determined relative to the internal standard due to the instability of acrylamide. Although both methods showed excellent linearity in this range (R^2 > 0.9999), considering the satisfactory relative residuals (< 20%), six calibration points were selected for the standard curve between 3–500 µg/L. Results meet the SANTE criteria.[106] The standard curves were determined over a wide range of concentrations, as acrylamide concentrations were expected to increase significantly higher in samples collected during roasting. The calibration curves are shown in Figure 14.

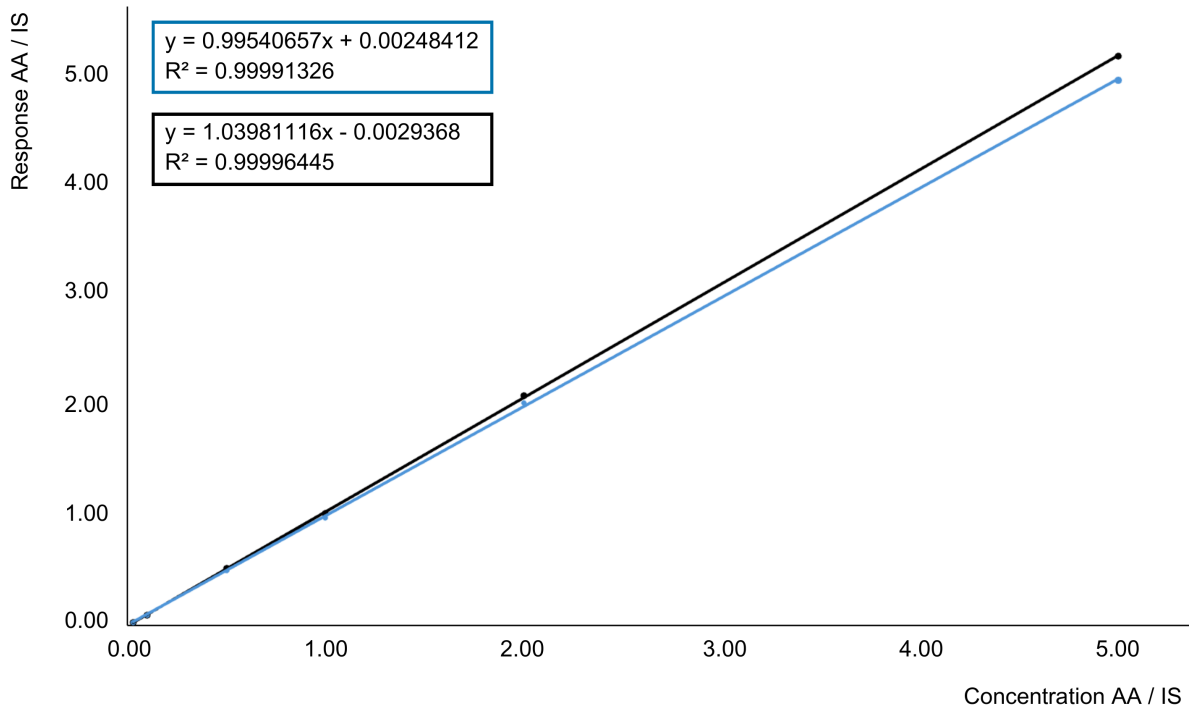


Figure 14. The calibration curve for Method 1 (blue curve) is y=0.99540657x+0.00248412 and for Method 2 (black curve) y=1.03981116x 0.0029368. Linearity has been determined in relation to the internal standard (AA/AA-d_3) over the concentration range of 3–500 µg/L.

It is important to note that between the measurements of Method 1 and Method 2, the standards and sample group 1 were at room temperature for a week. During this period, the samples showed different levels of acrylamide reduction. The results showed that the acrylamide reduction is matrix dependent, which is expected, since the stability of acrylamide is pH

dependent.[107] This may explain the difference in slope and, highlights the need to quantify acrylamide relative to an internal standard. The slope was considered in quantification process.

7.4.2 Sensitivity (LOD and LOQ)

The limits of quantification (LOQ) and detection (LOD) were determined based on signal-to-noise ratios (S/N), corresponding to S/N values of 10 and 3, respectively. LOD and LOQ were 1 µg/L and 3 µg/L for Method 1, and for Method 2 they were 3 µg/L and 10 µg/L, respectively. The difference is explained by the fact that a lower flow rate increases ionization efficiency, which improves sensitivity, reduces the S/N and thus also improves LOD and LOQ.[108]

7.4.3 Precision

Precision was evaluated spiking the same coffee with three acrylamide levels using Method 2. A pooled coffee sample was selected as the base. Three samples were prepared from the base, which were spiked with three concentrations, 10 µg/L, 50 µg/L, and 100 µg/L, of pure acrylamide, and 200 µg/L isotope-labeled acrylamide (AA-d_3). All samples were injected in triplicate and the %RSD was between 3.8–7.9% for acrylamide and 3.1% for all AA-d_3. The raw data is provided in Appendix 2.

Linearity of spiked samples were determined using area of acrylamide and added concentration of acrylamide. R2 was >0.998, which show that recovery is highly repeatable within the same coffee matrix. After subtracting the spiked acrylamide levels, the average native acrylamide concentration in the sample was determined to be 69.3 µg/L and the %RSD was 9.8. This indicates that the method can quantify acrylamide in the coffee matrix consistently

Evaluating precision included validation of repeatability (intra-day precision) and within-lab reproducibility (inter-day precision). Repeatability was evaluated from six QC (200 µg/L of AA-d3) replicates per day for both Method 1 and Method 2. For Method 1, the inter-day precision was determined on three independent days, and the results are presented in Table 8.

Table 8. Intra-day and Inter-day precision of Method 1 (M1) and Method 2 (M2) for 200 µg/L AA-d_3

Precision type	M1-Day 1	M1-Day 2	M1-Day 3	M2-Day 3
Intra-day precision (%RSD)	6.34%	5.93%	1.75%	2.95%
Inter-day precision (%RSD)	5.09%			

The intra-day precision of Method 1 ranged between 1.75–6.34%. The inter-day precision was estimated from the average quality control values for each day, giving an inter-day precision of 5.09%. The repeatability of Method 2 was determined using 6 replicates of same QC, resulting relative standard deviation of 2.95%.

The within-run stability of Method 2 was assessed by injecting three different standard, 3 µg/L 100 µg/L and 500 µg/L, each in triplicate. The standards were re-analyzed after 100 injections and again at the end of the sample sequence. Table 9 presents the responses and their %RSD for each concentration. Responses are expressed as peak areas.

Table 9. The within-run stability of method 2: responses and %RSD at three levels

Level (µg/L)	Start (n=3)	Middle (n=1)	End (n=1)	%RSD (n=5)
3	1984	2005	1964	3.83%
100	60713	60094	65369	3.65%
500	297321	289092	298266	1.52%

The overall variability across the five replicates ranged from 1.52 to 3.83%, which shows excellent within-run stability. This ensured that no matrix effects were present, nor was there any accumulation of impurities on the column, and that data reliability was maintained throughout the analytical sequence. This indicates robust performance of the Method 2 during the run. Overall, the precisions results align with the SANTE guideline (%RSD < 20%).

7.4.4 Trueness

Trueness of quantification was determined as back-calculated accuracy, which was obtained by comparing theoretical and experimental AA/AA-d_3 concentration ratios. Both the slope and the intercept of the regression line were considered. Accuracy was expressed as %bias between calculated and theoretical concentrations, with acceptance limits of ±15 %. All calibration levels met the criteria. The results are shown in Table 10.

Table 10. Trueness of quantification as back-calculated accuracy and %bias for Method 1 and Method 2 across range 3–500 µg/L

c_{theo} (µg/L)	M1-c_{exp} (µg/L)	M2-c_{exp} (µg/L)	M1-%bias	M2-%bias
3	2.97	2.95	1.12%	1.76%
10	9.40	9.26	5.98%	7.40%
50	50.44	50.16	-0.87%	-0.32%
100	98.86	98.14	1.14%	1.86%
200	203.64	201.51	-1.82%	-0.75%
500	499.13	499.28	0.17%	0.14%

Although the slope was higher for Method 2 (1.0398) than for Method 1 (0.9954), this did not produce a statistically significant difference in the results when the results were corrected for slope. The statistical significance between the methods was examined using a paired t-test. No statistically significant difference was observed for Method 1 compared to the theoretical values ($t_{0.05(5)}$ = 0.33), nor was there a statistically significant difference observed for Method 2 compared to the theoretical values ($t_{0.05(5)}$ = -0.62), and finally, no statistically significant difference was observed between Method 1 and Method 2 ($t_{0.05(5)}$ = 1.53). Both methods therefore produce results that do not differ statistically from the theoretical values at the 5% significance level, which indicates that no systematic bias was present.

7.4.5 Recovery

Ground coffee samples were extracted with methanol, treated with Carrez reagents, and finally purified using StrataX SPE cartridges. Recovery was determined using the internal standard acrylamide-d_3 (200 µg/L), which was added to all samples at the first stage of sample preparation. For comparison, a QC reference solution containing 200 µg/L of AA-d_3 was prepared in ultrapure water without a matrix.

Recovery was calculated from a representative subset of three specialty and three commercial coffee samples at different roast degrees (n = 6). The recoveries ranged from 72% to 99%, with an average of 91%. The accuracy test demonstrated consistent recovery within the same sample (%RSD = 3.1%). Results are acceptable according to SANTE guidelines (70–120%).

The recovery was first determined on the same day after sample preparation using Method 1, resulting in an average recovery of 91%. To assess the effect of sample matrix and pH on

acrylamide stability, the recovery was re-evaluated after one week. After one week of storage at room temperature, the apparent recovery decreased to an average of 65%.

7.4.6 Acrylamide Stability

Sample preparation was interrupted and continued after four days. Samples were stored in a refrigerator for this delay, and some were kept at room temperature during the run. The total recovery measured after the delay was clearly lower. Acrylamide may have degraded or passed through the SPE cartridge if storage and delay changed the pH and/or ionic strength, as was also observed in UAE and shaker extracted samples.

After four days, the acrylamide concentration had decreased in different proportions depending on the matrix (standards vs. coffee samples), and after six days even more. Peak areas of AA-d_3 (c=200 µg/L) in coffee sample (matrix) and in quality control sample (QC) are presented in Table 11.

Table 11. Peak areas and recovery (%) for matrix and QC on days 1, 4, and 6 at 200 µg/L

Day	Matrix Area	QC Area	Recovery (%)
1	73718	74460	99%
4	63721	69704	91%
6	65166	97360	67%

This demonstrates the need to calculate the recovery immediately after sample preparation. The chromatograms of the coffee sample from day 1 and day 4 overlaid in Figure 15. Despite the instability of acrylamide, no retention drift occurs, which indicates the stability of the LC conditions and the column.

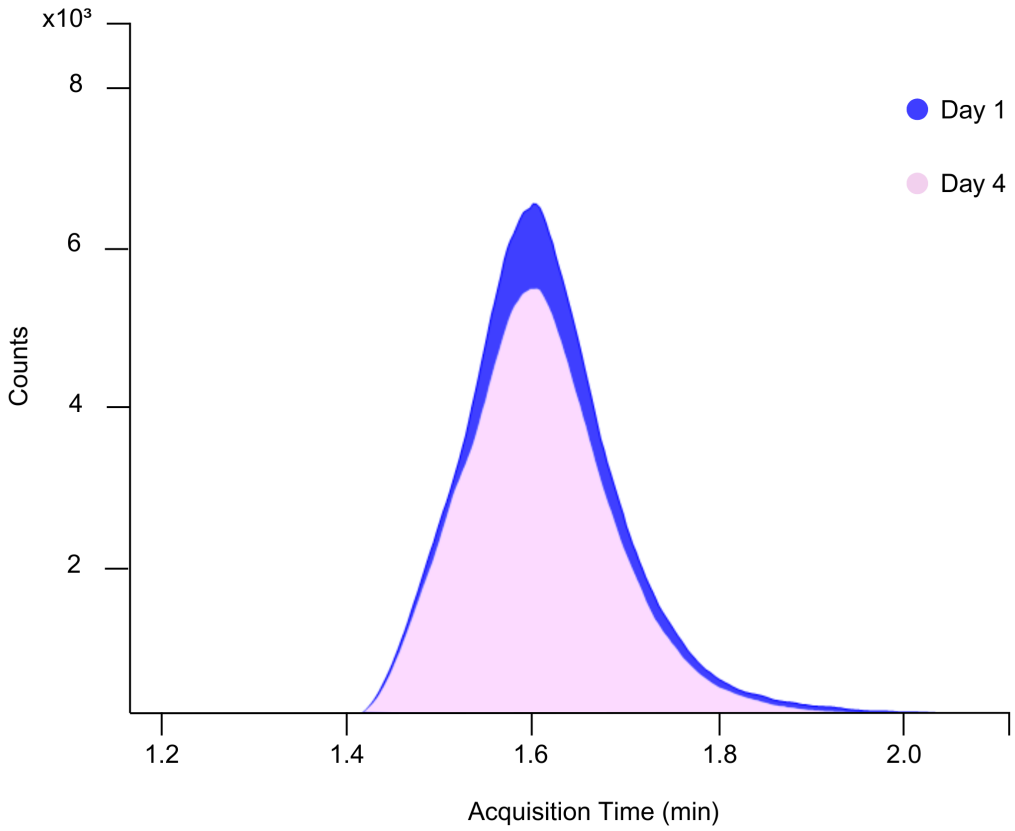


Figure 15. Overlaid MRM chromatograms (72 → 55) of a coffee sample on days 1 and 4.

7.4.7 Summary

The developed methods were validated according to the SANTE/11813/2021 guidelines.[104,106] All validation parameters were intended to be evaluated for both methods. However, recovery was determined only for Method 1, as it reflects the efficiency of sample preparation rather than the chromatographic performance. When recovery was attempted to assess using Method 2, the samples had been stored at room temperature for four days, during which time the acrylamide concentrations had decreased. Therefore, recovery could no longer be determined accurately, but it had already been validated previously, and the results were considered sufficient. The ion ratio was only determined for the final Method 2.

Validation experiments showed that both methods fully met the performance criteria for linearity, sensitivity, precision, accuracy, recovery, and trueness based on SANTE/11813/2021 guidelines and the results are summarized in Table 12.

Table 12. Validation parameters for Method 1 and Method 3 as per SANTE/11813/2021 criteria

Validation parameter	SANTE/11813/2021 criteria	Method 1	Method 2
LOD	S/N = 3	1 µg/L	3 µg/L
LOQ	S/N = 10	3 µg/L	10 µg/L
Linearity (r^2)	≥ 0.99	0.99991	0.99996
Recovery (%)	70–120%	72–99%	N/A
Repeatability (%RSD)	≤ 20%	2.75%	2.95%
Precision	≤ 20%	1.75–6.34%	1.52–3.83%
Ion ratio	±30%	N/A	16%
Retention time	± 0.1 min	0.014 min	0.012 min

In conclusion, no significant differences were observed between these two methods, and both were validated. Method 2 was ultimately selected for sample analysis due to better retention factor, which is highlighted in literature.[77] Method 1 would have required 53% more time than Method 2 when considering the entire run set. The shorter analysis time also supports the principles of green analytical chemistry from an energy-efficiency perspective.[109]

8 Results and Discussion

The quality, food safety and shelf life of coffee are directly related to its physicochemical properties, such as moisture content, pH, water content and color. These factors are linked to sensory properties and chemical stability, as well as the formation of thermal products such as acrylamide. Therefore, the work focused on measuring the changes in acrylamide levels, moisture content, color, and evaluate the chemical stability using thermogravimetric analysis.

8.1 Physicochemical Differences between Specialty and Commercial Coffee

Degree of roast is determined based on the L* value, where light roast: L* ≥ 30.00; medium roast: 27.00 ≤ L* < 30.00; and dark roast L* < 27.00. Table 13 shows the moisture content of specialty coffee samples, the measured L* value, labeled roast levels, roasting times of the coffees, and the final internal temperature of the beans.

Table 13. Moisture and colorimetric analysis of specialty coffee samples (replicates n=3), labeled roast levels, and corresponding times and final temperature of the bean.

ID	Moisture (%)	Color (L*)	Labeled roast levels	Final temperature of the bean (°C)	Roasting time (s)
S1	2.97 ± 0.04	32.57 ± 0.11	Light	209	504
S2	2.48 ± 0.06	30.03 ± 0.63	Light	208	604
S3	2.38 ± 0.03	31.41 ± 0.24	Medium	202	594
S4	3.08 ± 0.03	32.49 ± 0.31	Light	N/A	N/A
S5	2.39 ± 0.03	30.49 ± 0.35	Medium	214	620
S6	2.05 ± 0.04	27.48 ± 0.12	Medium	226	690
S7	1.93 ± 0.06	28.73 ± 0.16	Dark	205	667
S8	1.77 ± 0.01	29.99 ± 0.77	Dark	212	740
S9	1.52 ± 0.06	25.71 ± 0.43	Dark	206	870
S10	1.47 ± 0.05	26.00 ± 0.37	Dark	219	774

Specialty coffees are mainly roasted in small batches (10–30 kg) and the coffee is cooled with air. Results show that the labeled roast degree of the coffees differs from the one classified based on the L* value within a few coffees. What is interesting is that sample 8 is labeled as

a dark roasted and its roasting time and finishing temperature were 740 seconds and 212 °C, while sample 7 was roasted for 667 seconds and the finishing temperature was only 205 °C. Still, sample 7 has darker L* value. Both coffees are grown at the same altitude, have undergone natural processing and contain the same coffee variety (Red Catuai). Different roasting styles may explain the difference. All coffees were roasted by the same brand of roaster, which divides the roasting phases according to temperature as follows: the drying phase 0–160 °C, the Maillard phase 160.01–189.9 °C, and the development phase > 190 °C. Table 14 shows the roasting phase distribution within sample 7, 8 and 10.

Table 14. Roasting phase distribution (%) and total time (s) in samples 7, 8 and 10

ID	Drying (%)	Maillard (%)	Development (%)	Total (s)
S7	53.82	22.34	23.84	667
S8	57.29	20.69	22.02	740
S10	47.93	19.12	32.95	774

Sample 10 is classified as a dark roast coffee, and therefore it is taken into comparison. The drying stage in sample 8 is quite long compared to samples 7 and 10. The drying stage determines the moisture content of the bean during the Maillard reaction, which also affects the formation of acrylamide. The mean moisture content was 2.84 ± 0.32% for light-roasted, 2.27 ± 0.19% for medium-roasted, and 1.67 ± 0.22% for dark-roasted coffees. Values were grouped based on the labeled roast level, rather than by their L* values, since color is not an absolute indicator of roast degree, as demonstrated by sample 8.

Moisture content followed a predictable pattern: lighter roasts retained higher moisture levels than medium and dark roasts. Roasting time had a very strong inverse correlation with moisture ($r = -0.95$) and color ($r = -0.89$), while bean finishing temperature did not show a significant correlation with other characteristics. The unexpected moisture and color of sample 8 slightly weaken the correlations for the dataset.

However, commercial coffee samples did not show similar correlations. The results of moisture and colorimetric analysis of commercial coffee samples are presented in Table 15 with the labeled roasting degree.

Table 15. Moisture and colorimetric analysis of commercial coffee samples (replicates n=3)

ID	Moisture (%)	L*	Labeled roast level
C11	3.30 ± 0.05	29.72 ± 0.48	Light
C12	3.71 ± 0.02	29.01 ± 0.83	Medium
C13	3.93 ± 0.03	30.57 ± 0.54	Light
C14	3.54 ± 0.17	29.27 ± 0.79	Medium
C15	2.55 ± 0.12	26.57 ± 0.14	Dark
C16	3.56 ± 0.10	25.97 ± 0.73	Dark
C17	3.55 ± 0.03	30.04 ± 0.54	Light
C18	3.75 ± 0.04	27.42 ± 0.30	Medium
C19	3.26 ± 0.20	22.59 ± 0.37	Dark

The table shows that the moisture content is significantly higher in commercial coffee than in specialty coffees. The average moisture contents were calculated accordingly for commercial coffees as well. The mean moisture content was 3.59 ± 0.31% for light-roasted, 3.67 ± 0.11% for medium-roasted, and 3.13 ± 0.52% for dark-roasted coffees. Compared to specialty coffee, the moisture content was 23% higher for light-roasted coffees, 61% higher for medium-roasted, and up to 87% higher for dark-roasted coffees.

Specialty coffees are usually roasted using the LTLT style, while commercial coffees are roasted using the HTST style, which leaves the coffee with a lower moisture content and more acrylamide is formed.[58,61] The differences in moisture content of commercial coffees may be explained by the cooling method of the coffee. Specialty coffees are cooled using air, while large-scale industrial production uses the quenching method, which has been shown to leave a higher residual moisture content in the coffee.[9] Therefore, even though the coffee might be drier right after roasting, it can absorb some moisture during cooling process. This, in turn, affects the shelf life and, for example, the degradation rate of molecules including acrylamide and flavor compounds.

8.2 Acrylamide Analysis

The difference in moisture content was observed between commercial and specialty coffees, which directly affects the color. Figure 16 illustrates the relationship between acrylamide measured by HPLC-MS/MS and roast degree for both coffee groups. Quench cooling of commercial coffees increases the moisture content of the coffee after roasting and is therefore

a confounding variable.[9] This is well illustrated by the strong correlation between color (L*) and acrylamide concentrations of specialty coffees (r = 0.79), while there is no correlation for commercial coffees (r = 0.08). The moisture content of specialty coffees is at the original level and this present even stronger correlation (r = 0.90) between color (L*) and moisture (%), while the same correlation within commercial coffee samples is (r = 0.42).

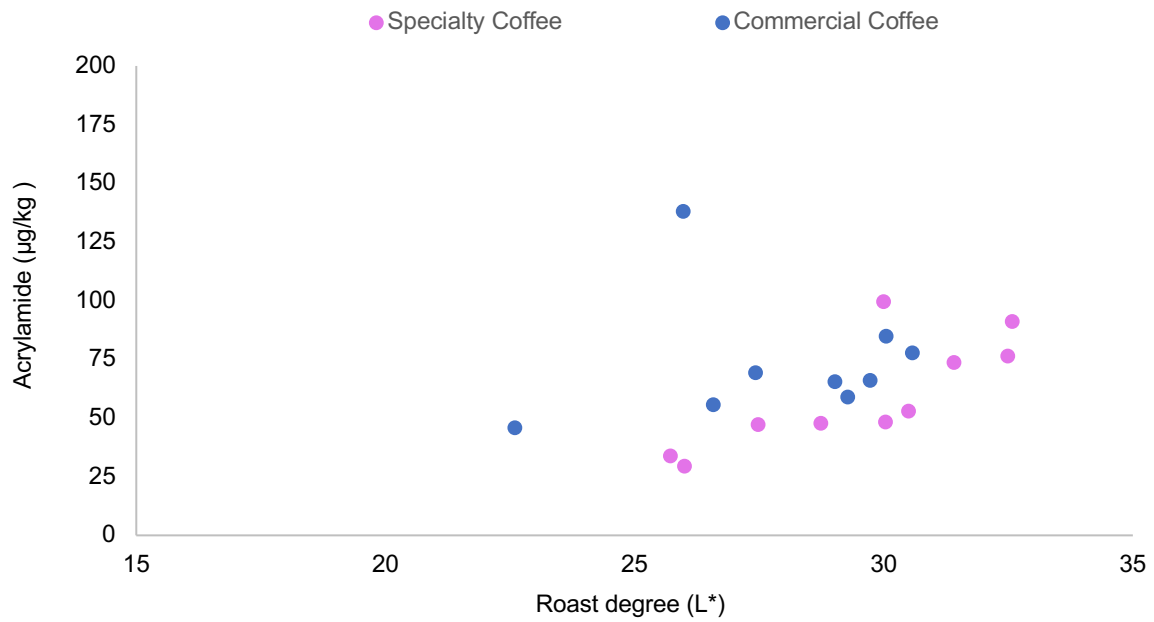


Figure 16. The illustrated relationship between acrylamide and color (L*) for commercial (blue) and specialty (purple) coffee samples.

Senyuva and Gökmen[81] (2007) found that water extraction was not suitable for the analysis of acrylamide in coffee due to matrix interference, and therefore they used methanol in extraction. We faced similar issue and therefore, we used their sample preparation protocol. They reported acrylamide levels of roasted ground coffees ranging from 14 to 29 µg/kg (n = 11). Comparing to their findings, the obtained results in this study are of the same order of magnitude.

Overall, the acrylamide levels appear lower than expected. In general, levels are likely to range from 100 to 600 µg/kg.[19,49,58] This raises questions about the extraction efficiency of methanol as solvent. Extraction efficiency has been studied in cereal and potato crisp matrices using methanol and water as solvents. The study showed that the extraction efficiency depends on both the matrix and the solvent. A single-step water extraction could lead to a 14–30% underestimation of acrylamide, while a single-step methanol extraction could result in 38–48% underestimation.[113]

In order to clearly compare the differences in acrylamide content between specialty coffee and commercial coffee, the effect of roast level must be considered. The roast level (color, L*) affects the measured acrylamide, and comparisons between groups would be biased without taking color into account. Covariate correction (ANCOVA) was applied, where AA was modeled as a nonlinear function of L* (L and L^2).[111]

The predicted reference was modeled using quadratic regression:

$$\widehat{AA}(L) = \beta_2 L^2 + \beta_1 L + \beta_0$$

where β_2 is the second-order coefficient of the polynomial, β_1 is the first-order coefficient, β_0 is a constant term and L is L*-value. For further calculation the median of L* (29.27) was chosen to be reference (L_{ref}). The color adjusted acrylamide was calculated as follow:

$$AA_i^{adj} = AA_i - \beta_1(L_i - L_{ref}) - \beta_2((L_i)^2 - (L_{ref})^2)$$

The success of the correction was assessed by checking that the correlation between adjusted AA and L* was negligible. Pearson's correlation $r(AA_{adj}$, L*) returned a value of 0.00026, indicating that the correction was successful, and the sample groups are comparable to each other regardless of color.

Table 16 show the acrylamide levels before (AA) and after (AA_{adj}) adjustment. Values are reported as mean ± standard deviation (SD), n=3. The standard deviation of the color-corrected AA is not reported because the adjustment was made based on the samples' means, resulting in one value per sample.

Table 16. Concentration of AA before and after color adjustment: measured values (mean ± SD, n=3)

ID	AA (µg/kg)	AA_{adj} (µg/kg)
S1	91.14 ± 1.79	76.15
S2	48.18 ± 1.16	45.17
S3	73.60 ± 0.33	64.44
S4	76.33 ± 0.58	61.79
S5	52.89 ± 0.82	47.89
S6	47.07 ± 0.16	53.22
S7	47.55 ± 1.20	49.56
S8	99.56 ± 1.00	96.69
S9	33.61 ± 0.25	44.47
S10	29.25 ± 0.08	39.44
C11	65.87 ± 2.98	64.09
C12	65.31 ± 3.70	66.27
C13	77.78 ± 1.77	72.46
C14	58.97 ± 0.35	58.97
C15	55.45 ± 2.59	64.20
C16	137.92 ± 3.07	148.18
C17	84.63 ± 2.23	81.56
C18	69.13 ± 3.14	75.47
C19	45.75 ± 3.59	61.62

The original acrylamide levels ranged between 29.25–99.56 µg/kg for specialty coffee samples and 45.75–137.92 µg/kg for commercial coffee samples. The mean and standard deviation for color-adjusted acrylamide levels was for specialty coffee samples 57.88 ± 17.55 µg/kg and for commercial coffee samples 76.98 ± 27.65 µg/kg. Overall, the acrylamide levels are higher in commercial coffee group but so is the standard deviation.

Figure 17 shows how acrylamide levels compare across samples. Each bar shows the percent deviation from the overall mean, where 0% is mean and baseline. Bars above baseline indicate higher than average levels, while bars below baseline indicate lower than average levels.

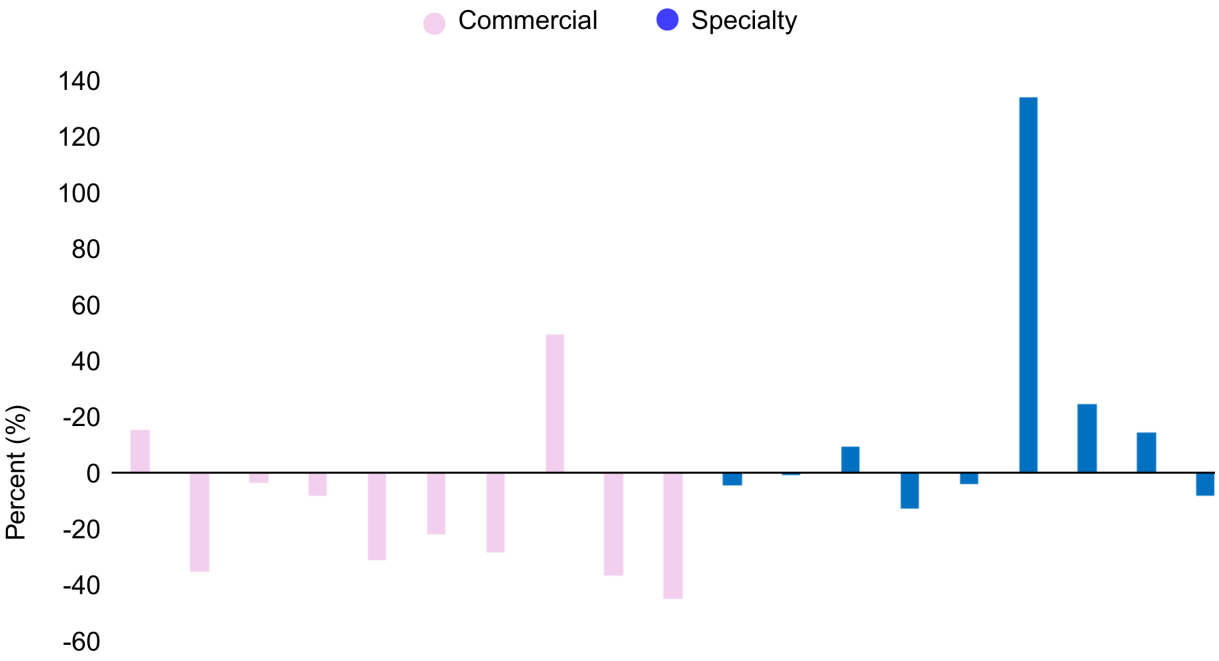


Figure 17. Deviation from mean of all samples (%) of color-corrected acrylamide.

Figure 17 shows that sample S8 has significantly higher compared to other specialty coffee samples, while sample C16 is a similar outlier for the commercial coffee samples. If these two outliers are removed, the mean and standard deviation for color corrected acrylamide levels are 53.57 ± 11.71 µg/kg for specialty coffee and 68.08 ± 7.70 µg/kg for commercial coffee, meaning commercial coffee samples have 27% more acrylamide. Although, the subset of samples is low (n=17, if considering outliers), the previous research supports the phenomenon, since commercial coffees are often roasted with HTST profile and the cherries go through processes leading to higher levels of asparagine.[19,42,51,62]

HTST leads to lower moisture content at the beginning, which may affect acrylamide behavior during roasting.[19] Amrein *et al.*[112] (2006) found that moisture content has a significant effect on acrylamide formation in potato matrix. At higher moisture contents, the reaction proceeds more slowly due to a dilution effect, while low moisture limits the mobility of the reactants, which affects the reaction rate. Thus, the long drying phase of sample S8 before the start of the Maillard reaction may explain such a high acrylamide content compared to the other samples. Lower moisture level in development phase may be the reason why S8 and HTST profiled coffees have higher acrylamide levels, if the mobility of reactants is slower, the acrylamide won't degrade as efficient.

8.3 Acrylamide Behavior During Roasting

Acrylamide behavior was studied during roasting in a real roasting environment. 17 samples were collected at 30-second intervals starting from the moment the internal temperature of the bean reached 120 °C. This sample is from the same batch of green coffee as the previously reviewed S8 but roasted at a different time. Acrylamide could not be quantified with the validated quantifier transition (72 → 55) from early-stage samples. During the early and mid-roasting stages, an isobaric interference peak was observed in the total MRM chromatogram next to the expected retention time. It was overlapping the acrylamide peak and preventing reliable quantification. In Figure 18 is MRM chromatograms from roasting-phase samples showing the progression of an interfering co-eluting compound: A) early sample with separate interference peak, B) mid-roast sample where acrylamide signal is partially overlapped, and C) late sample where only acrylamide remains, which co-elute with the internal standard.

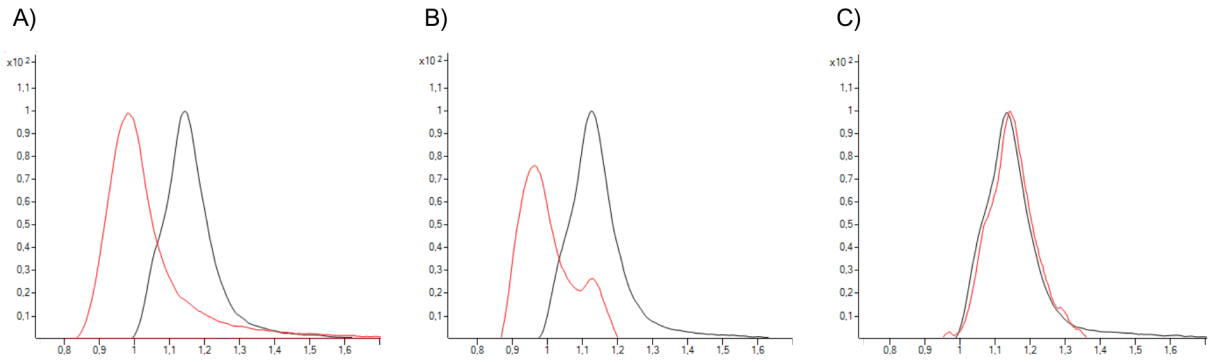


Figure 18. MRM chromatograms (72 → 55, red; 75 → 58, black) showing interference during roasting, where A is early-sample, B is mid-sample and C is the final sample.

To exclude the possibility that the acrylamide peak had shifted due to matrix-related factors such as pH changes during roasting, the same samples were examined with secondary shifts (27/30). These confirmed that the retention time of acrylamide was consistent with the internal standard (AA-d_3) and that the interference peak was not related to acrylamide. The chromatography method was developed and validated for roasted coffee and the isobaric interferences known to occur in it.

Due to interference, absolute concentrations could not be calculated for early-stage samples. However, the acrylamide transition peak heights at the expected retention time, normalized to the internal standard, were used as a semiquantitative indicator of acrylamide levels. This model

allowed to estimate AA levels and follow the temporal trend, which showed a characteristic increase in acrylamide levels up to a maximum and a subsequent decrease with longer roasting times.

In the later stage samples (10–17), interference was no longer as dominative, and acrylamide was successfully quantified in eight sample. Comparison of the peak heights and integrated areas of these samples showed a relative error of 5.0 ± 3.0%, supporting the use of peak heights as a qualitative indicator of acrylamide trends when quantification was not possible. Data is shown in Table 17.

Table 17. Time, sample temperature, quantified acrylamide (µg/kg), model estimate, and relative error (%) across the roast profile (N.Q. = not quantifiable)

ID	Time (s)	Sample T (°C)	Quantified AA (µg/kg)	Estimated AA (µg/kg)	Relative error (%)
R1	257	120	N.Q.	200.16	N/A
R2	287	126	N.Q.	211.43	N/A
R3	317	132	N.Q.	179.59	N/A
R4	347	139	N.Q.	200.95	N/A
R5	377	145	N.Q.	213.45	N/A
R6	407	151	N.Q.	195.82	N/A
R7	437	157	N.Q.	225.44	N/A
R8	467	163	N.Q.	233.17	N/A
R9	497	170	N.Q.	412.48	N/A
R10	527	175	334.07 ± 23.84	340.19	-1.83
R11	557	181	503.07 ± 28.17	464.64	7.64
R12	587	186	345.53 ± 11.50	323.26	6.44
R13	617	192	298.53 ± 3.23	280.27	6.12
R14	647	197	209.42 ± 2.12	201.15	3.95
R15	677	203	238.92 ± 10.37	222.43	6.90
R16	707	208	127.33 ± 2.85	120.84	5.10
R17	720	210	134.17 ± 1.15	126.80	5.49

Acrylamide (AA) was quantifiable in the roasting process after the temperature had reached 170 °C. The first quantified level was observed at 175 °C / 527 s (334.07 ± 23.84 µg/kg), after which the AA concentration reached its peak at 181 °C / 557 s (503.07 ± 28.17µg/kg). As the

temperature and time continued to increase, the concentration decreased, reaching 134.17 ± 1.15 µg/kg at 210 °C / 720 s.

The model-generated estimates followed the measured trend but were on average slightly lower than the measured values. The model predicted the lower AA peak 464.64 µg/kg than the measured maximum 503.07 µg/kg), suggesting a systematic underestimation around the peak. Overall, the model describes the general course of formation and decomposition well, but the timing and height of the peak are incomplete. These results align with the previous research, which report that AA levels peak (468 µg/kg) within the first 5–10 minutes.[25]

The Maillard reaction has been reported to start at 120 °C, but at this point acrylamide had already been formed. This suggests that acrylamide may also be formed by other pathways, as mentioned in Chapter 2. It is also possible that the Maillard reaction starts when the atmosphere reaches the temperature, even if heat has not yet been transferred to the sample.

The decomposition of acrylamide during roasting has been suggested to result from reactions with the formed melanoidins.[113] Acrylamide reacts with amino groups of melanoidins by the Michael addition mechanism, which is presented in Chapter 4.5. [114-117]

8.4 Chemical Stability in Commercial and Specialty Coffee

The differences in shelf life of specialty coffee and commercial coffee were assessed both by measuring acrylamide at different shelf-life time points and using thermogravimetric analysis (TGA) to characterize mass/thermal changes. HTST roasting style expose coffee beans to higher physical stress than LTLT roasting. This creates microcracks, which leads to faster degradation of aromatic compounds and therefore, shorter shelf life.[56]

Acrylamide stability during storage was evaluated by studying four different coffees with two to three different time points: shelf life ranging from 3 to 84 days. Samples C11, S8 and S5 are from earlier discussed coffee group and their moisture (%) and color (L*) is presented in Tables 13 and 15. The fourth coffee (S11) is an Arabica blend specialty coffee with moisture content of 2.32 ± 0.19 % and L* value of 31.06 ± 0.59. Acrylamide levels of these coffees are presented in Figure 19.

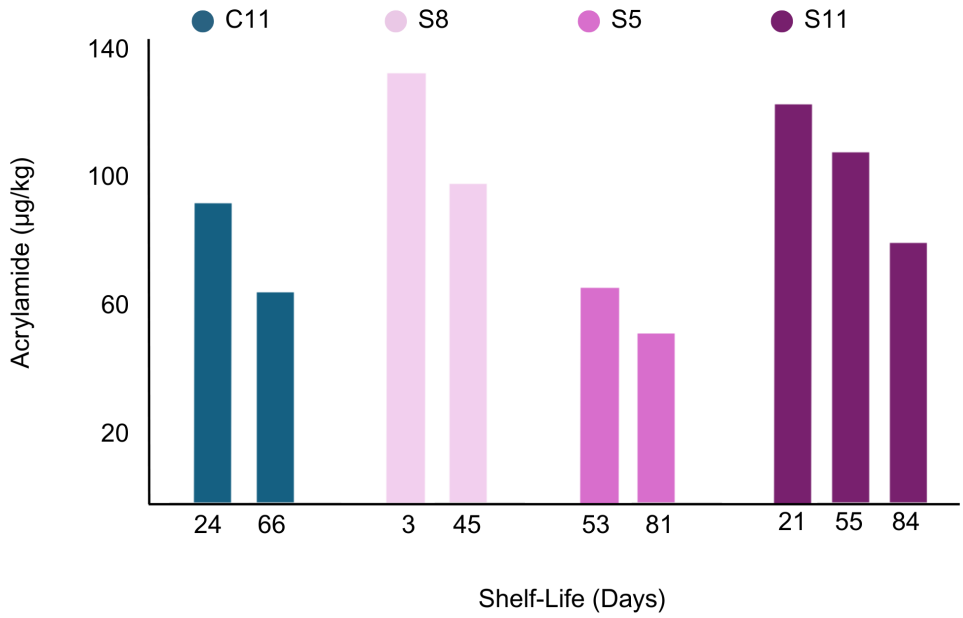


Figure 19. Acrylamide levels in samples C11, S8, S5, and S11 at different time points.

Acrylamide (AA) levels decreased in all samples with increasing shelf-life. The stability of acrylamide during storage has been studied in coffee before, but the rate of reduction varies. Andrzejewski *et al.*[114] (2004) found that acrylamide decreased by 40% in two coffees and 60% in a third after six months of storage at room temperature. Hoenicke & Gatermann[37] (2005) found that acrylamide decreased by 29-30% in coffee after three months of storage at 10°C–12°C. However, the studies did not consider the roast date, roast level, moisture content or measured color of the coffees.

It is possible that the coffee matrix and shelf-life time affect the degradation rate. As observed in this study, in sample S11, acrylamide degraded by only 12.3% over a 34-day period, while over the next 29 days, it degraded more than twice as much, to 26.0%. On the other hand, acrylamide degradation increased by three percentage points in sample S8 compared with C11, with both having 42-day time difference. For sample S5, the degradation was 21.5%, whereas for sample S11, which was nearly the same age between the two data points (53–81 days vs. 55–84 days), degradation reached 26.0%. All samples have been stored under the same conditions. Note that samples S8 and S11 had the highest acrylamide levels out of the specialty coffee samples. This may be due to a roasting style having a particularly long drying phase, 57.29% and 59.43% for S8 and S11, respectively.

However, the sample sets of this study and previous research are too small to draw conclusions about what caused the variation in the reduction rate. The representativeness of the samples

would be increased if the first data point of all samples were the roasting date of coffee and then the reduction would have been systematically monitored at regular intervals. In addition, physicochemical properties such as moisture, color and pH, which have been shown to affect the stability of acrylamide, should be determined for each data point.[63,107]

8.5 Thermogravimetric Analysis

Thermogravimetric analysis (TGA) was included in the study because it provides information on moisture release and indicates thermal and oxidative degradation of coffee, which complements information on chemical stability during storage. TGA provides information on coffee that can help understand the differences in acrylamide between roasting of commercial (HTST) and roasting of specialty coffees (LTLT). The thermal history of coffee affects the coffee matrix, which can be assessed by TGA. Lipid differences between coffee groups may contribute to differences in acrylamide formation and degradation during roasting, as well as degradation during storage.

First, a control sample was analyzed under oxygen and nitrogen atmospheres at two heating rates (2.5 and 10 °C/min). The oxygen atmosphere was selected, because the purpose of the experiment was to evaluate the oxidative stability of roasted coffee under conditions that mimic normal shelf-life. A slower temperature ramping did not provide improved resolution, so a heating rate of 10 °C/min was adopted. Furthermore, the heating rate of 10 °C/min is commonly reported in the literature, so the results are directly comparable to previous studies.[97,98]

Light, medium and dark roast coffee samples were selected from both the specialty coffee and the commercial coffee group. The aim was to minimize effect of thermal history so that it does not cause bias between comparisons. Therefore, the coffees in both groups are roasted by the same roastery, which makes it likely that all coffees have been roasted with the same roaster and cooled with similar method.

Four stages of coffee degradation in oxygen have been reported in the literature: i) 50–200, ii) 200–350, iii) 350–425, iv) 425–700 °C.[97] Instead of reporting individual onset temperatures, mass loss fractions were quantified by integrating TG curves over four temperature windows as follows: i) 20–200, ii) 200–350, iii) 350–425 and iv) 425–600 °C and a residue at 600 °C. This approach is more robust to overlapping events than onset temperatures. At the same time,

it produces directly comparable values for moisture/volatiles, saccharide, lipid/protein degradation, and carbon oxidation. Mass losses and the residue mass are shown in Appendix 3. The data showed consistent differences in TGA profiles between specialty and commercial coffees. In commercial samples, the 20–200 °C loss was systematically higher, indicating a higher load of moisture and other low-temperature volatiles after roasting. In specialty samples, the 350–425 °C window (lipid/protein) and slightly also 425–600 °C (oxidizable carbon mass) were higher. However, the 600 °C residue was almost the same between the groups. This indicates that specialty coffee has an oilier structure and a slightly carbon-rich matrix, while the higher moisture content in commercial samples confirms the quenching as a cooling method.

Figures 20–22 present how thermogravimetric curves differ within same roast but different coffee quality. Thermal losses are systematically higher and occurs in lower temperatures within commercial coffee samples. Roasting coffee with HTST style creates some microcracks, which for one form microlevel tunnels for volatiles to escape.[56] This happens at room temperature, so it is not unlike to see phenomena under thermal stress.

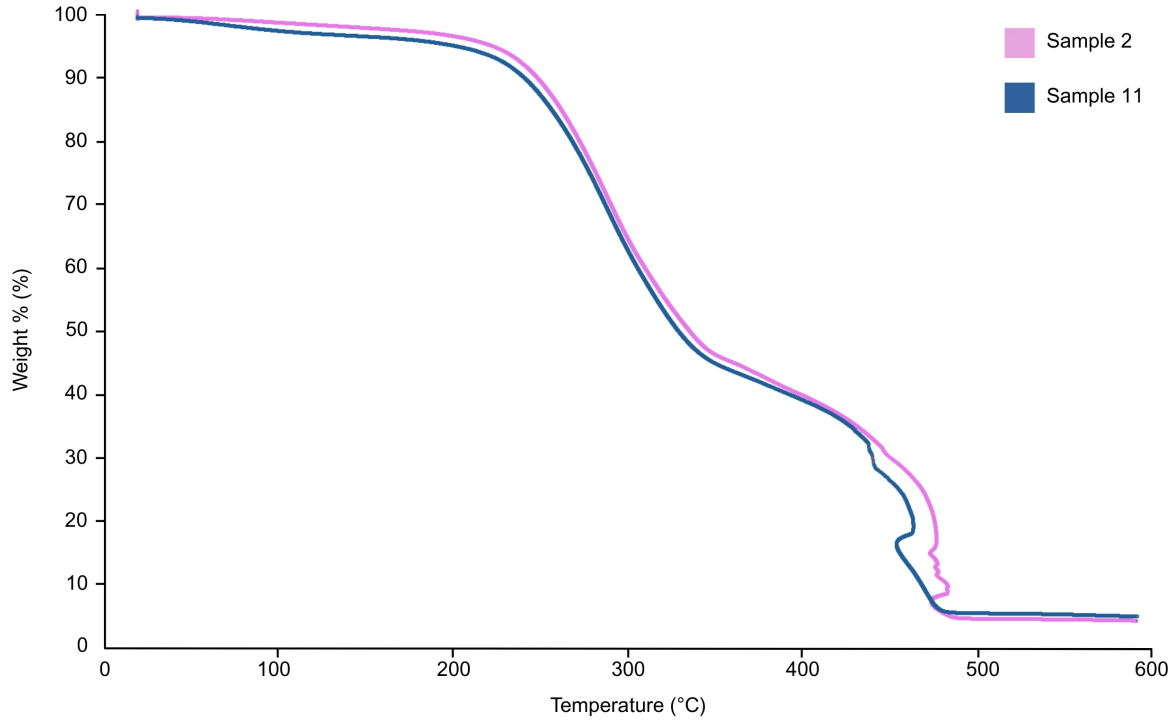


Figure 20. Thermogravimetric curves of light roasted specialty (sample S2) and commercial coffee (sample C11) overlaid.

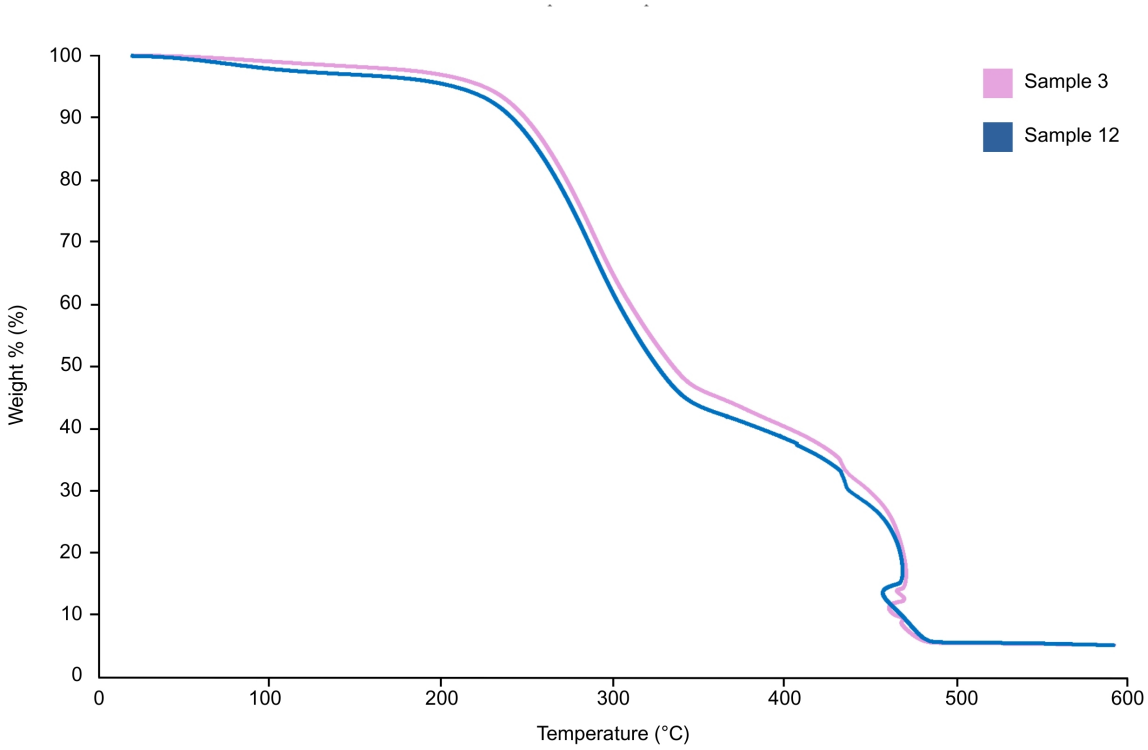


Figure 21. Thermogravimetric curves of medium roasted specialty (sample S3) and commercial coffee (sample C12) overlaid.

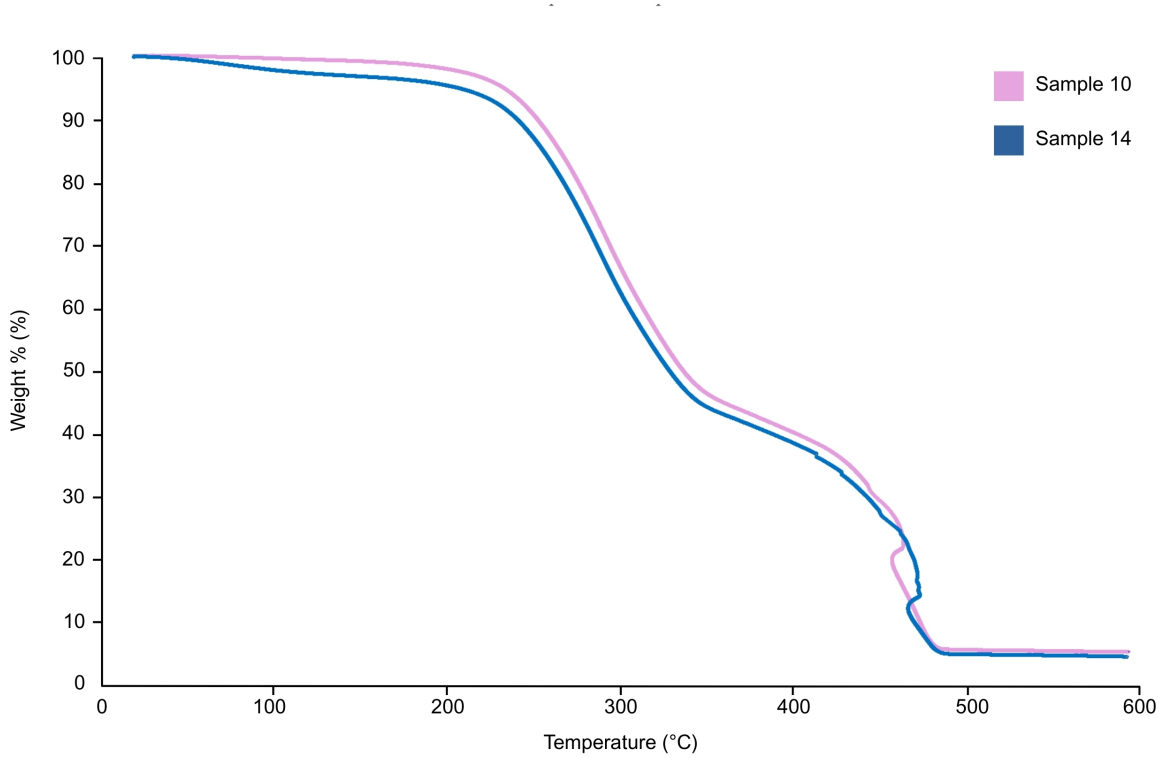


Figure 22. Thermogravimetric curves of dark roasted specialty (sample S10) and commercial coffee (sample C14) overlaid.

In summary, specialty coffees had a slightly lower acrylamide levels on average, lower moisture content, and TGA indicated greater thermal stability: higher lipid-protein fraction and fewer low-temperature volatile compounds.

Green coffee contains 10–17% lipids in a stable form, while roasted coffee has almost the same total amount in a different form. The amount of oxidized lipids and free fatty acids increases, which contributes to changing the chemical reactivity during roasting and storage.[115] During roasting, oxidized lipids can react with asparagine to form acrylamide. On the other hand, as roasting progresses, more acidic functional groups are formed, which can lower the pH of the matrix, so that asparagine no longer reacts with carbonyl compounds to form acrylamide.[115–117] After roasting, oxidation can also affect the degradation of acrylamide.[118]

The HTST profile leads to a higher pH and a stronger diffusion of oils on the bean surface.[119] However, the rapid process does not allow for significant lipid oxidation, resulting in a lower amount of reactive carbonyl compounds, which leads in a shorter acrylamide degradation phase.[115,116] More oxidized lipids and lower pH have been observed in LTLT profiles, suggesting a more extensive formation of lipids and organic acids. In such a matrix, acrylamide is likely to be degraded more efficiently, as the oxidation products of fatty acids can react with acidic functional groups, which even helps the degradation of acrylamide.[118]

Several competing reactions are involved in the formation and degradation of acrylamide during roasting. However, the higher lipid fraction and lower pH of LTLT profiles may be one of many explanatory factors, which is why there is less acrylamide. However, these data alone do not provide explanatory factors for the degradation rate of acrylamide. The subset of samples should be larger, and the pH of the coffee samples should also be measured to assess its effect. In addition, the oil fraction may protect aromas from escaping, which together with lower moisture content supports better shelf life, as oxidation and degassing occur more slowly.[9,65] The advantage of chemical stability is that the desired flavor compounds do not degrade as quickly, and the coffee is better preserved for longer.

9 Conclusion

The two main goals of this thesis were to develop and validate HPLC-MS/MS method to quantify acrylamide (AA) levels in coffee samples and to evaluate whether coffee quality is related to those AA levels. The method was successfully validated based on SANTE/11813/2021 guidelines for linearity, sensitivity, precision, accuracy, recovery and trueness. AA concentrations were measured in specialty coffee and commercial coffee samples having a wide variety of roast levels. In addition, changes in acrylamide content during the roasting process and storage were assessed. Finally, the most relevant physicochemical properties (color and moisture) of the samples were determined, and thermogravimetric analysis (TGA) was performed to support the interpretation of the results.

Coffee has a complex chemical matrix, which made sample preparation and quantification analytically challenging. While the obtained recovery revealed how much acrylamide was retained during sample preparation, it did not account for the performance of extraction efficiency. Measured AA levels ranged between 29.25 and 137.92 µg/kg (n=19) with a median level of 65.31 µg/kg. Previously reported values in literature have been ranging from 100 µg/kg to 600 µg/kg. Methanol extraction was a pragmatical choice based on available resources. It provided systematic and comparable results across samples, but the absolute AA content may be underestimated compared to methods using multi-step or water-based extraction combined with a more extensive clean-up. The purpose of the study was to compare differences between commercial and specialty coffee in different roast levels, not to provide regulatory data for an absolute risk assessment. Although methanol-based extraction was a reasonable compromise, it raised new questions regarding the extraction efficiency of acrylamide in H_2O, MeOH and ACN. Reliable assessment is challenging without a suitable CRM, which is not currently available, as AA is not stable in roasted coffee.

A clear outlier sample was observed in both the specialty coffee and commercial coffee groups. When these were excluded, strong correlations were observed between color and acrylamide: 0.91 for specialty coffee and 0.80 for commercial. Corresponding correlations were observed for moisture and acrylamide: 0.90 and 0.57, respectively. Moisture and color are indicators of roast level, and these correlations describe how evenly acrylamide degrades during roasting. For specialty coffee, a correlation of 0.96 was observed between moisture and L*, while for commercial coffee this was only 0.47. Commercial coffees are often cooled after roasting using a water mist, which may cause moisture to be absorbed into the coffee depending on its porosity.

Dark roast coffees are more porous in structure, which can cause water to be absorbed more strongly. Residual moisture is a confounding factor, which is why the color of the coffee is darker than the roast. Commercial coffee had on average 57% higher moisture content (3.46%) compared to specialty coffee samples (2.20%).

In order to compare the differences in AA levels between specialty coffees and commercial coffee samples without the influence of roast level, covariance color correction was performed. The color-corrected acrylamide concentrations of the specialty coffees were 53.57 ± 11.71 µg/kg, and the commercial coffees were 68.08 ± 7.70 µg/kg. This data indicates a 27% higher acrylamide level in the commercial coffees. Although, the subset of samples is small (n=17 excluding outliers), the previous research supports the findings. Specialty coffee is a high-quality coffee, roasted with LTLT profile. High-quality coffee has less asparagine and their roasting profile has been shown in general to lead to lower levels of acrylamide than coffee roasted with HTST profile. The HTST profile leads to faster moisture evaporation at the beginning of roasting, which may be one explanation for the higher acrylamide levels.

Acrylamide reaches its peak in the middle stages of coffee roasting and generally decreases as roasting progresses further due to thermal decomposition and evaporation. Color (L)* and moisture were practical and easily measurable indicators for AA risk management. Darker color and lower moisture correlated with lower AA content. During storage, AA content decreased by approximately 12–38% over a period of 28–63 days. Findings are consistent with previous literature. An explanation was not found for the variability of acrylamide loss rate across the samples, neither in this study nor in previous research. Future studies should increase the number of samples, determine the physicochemical properties of the coffee more precisely and investigate how the duration of the drying phase affects the formation of acrylamide.

TGA analysis was performed in an oxygen (O_2) atmosphere at a rate of 10 °C/min. The mass loss was integrated over four thermal steps: 1) 20–200 °C, 2) 200–350 °C, 3) 350–425 °C and 4) 425–600 °C, and the residue at 600 °C was reported. The integration provided comparable results for moisture/volatiles, saccharide-derived compounds, lipid/protein content and carbon oxidation. The TGA profiles support that the commercial samples have a higher moisture content, while the specialty samples have a higher lipid/protein content. The TGA results indicate compositional differences between specialty and commercial coffee matrices, which could explain some of the differences in acrylamide levels through variations in lipid oxidations during roasting.

The results obtained address all research questions of the thesis project, and all goals were achieved, even though the sample preparation protocol was not optimal. The validated HPLC-MS/MS method proved to be the most important tool for acrylamide quantification. Therefore, investment in the method development is essential, and potential isobaric interferences should be resolved. This work provides a foundation for future studies and is one of the first to integrate TGA in the assessment of acrylamide formation and degradation. The study focuses on the importance of understanding the coffee matrix and linking compositional variables to acrylamide concentration. Acrylamide is a harmful compound and therefore understanding the roasting parameters that influence its formation is essential for controlling its levels in coffee. This requires close collaboration with roasters.

Future work should move from single factor experiments to systematic, realistic roasting studies. These should include variations in drying stage, airflow rate, and cooling method. Time-resolved samples should be collected from the beginning of the roast and combined with chemical and sensory evaluation to optimize the roast for good taste first and then quantify how the parameters affect the AA levels and other compounds within the boundaries of realistic roasting profiles. It would also be important to monitor how the chemical and sensory profiles of such samples change. A comprehensive understanding of roast style could be applied by comparing HTST and LTLT profiles for specialty coffees, and to map coffee origin-dependent behaviors. TGA-GC-MS coupling helps in quantifying which compounds evolved at each thermal stage using green coffee. In addition, the degradation of such compounds could be monitored in samples of different roast ages, allowing for modeling of chemical stability and shelf life. Finally, using machine learning to integrate process parameters, compositional measurements, TGA properties, and sensory data could produce predictive, multi-objective models that help identify the best trade-off between safety, health-relevant compounds, and taste.

References

1. Somporn, C.; Kamtuo, A.; Theerakulpisut, P.; Siriamornpun, S. Effect of Shading on Yield, Sugar Content, Phenolic Acids and Antioxidant Property of Coffee Beans (Coffea Arabica L. Cv. Catimor) Harvested from North-eastern Thailand. J Sci Food Agric 2012.

2. Usva, K.; Sinkko, T.; Silvenius, F.; Riipi, I.; Heusala, H. Carbon and Water Footprint of Coffee Consumed in Finland—Life Cycle Assessment. *Int J Life Cycle Assess* **2020**, *25* (10), 1976–1990.

3. Bastian, F.; Hutabarat, O. S.; Dirpan, A.; Nainu, F.; Harapan, H.; Emran, T. B.; Simal-Gandara, J. From Plantation to Cup: Changes in Bioactive Compounds during Coffee Processing. *Foods* **2021**, *10* (11), 2827.

4. Hirvonen, T.; Kontto, J.; Jestoi, M.; Valsta, L.; Peltonen, K.; Pietinen, P.; Virtanen, S. M.; Sinkko, H.; Kronberg-Kippilä, C.; Albanes, D.; Virtamo, J. Dietary Acrylamide Intake and the Risk of Cancer among Finnish Male Smokers. *Cancer Causes Control* **2010**, *21* (12), 2223–2229.

5. Lachenmeier, D. W.; Schwarz, S.; Teipel, J.; Hegmanns, M.; Kuballa, T.; Walch, S. G.; Breitling-Utzmann, C. M. Potential Antagonistic Effects of Acrylamide Mitigation during Coffee Roasting on Furfuryl Alcohol, Furan and 5-Hydroxymethylfurfural. *Toxics* **2018**, *7* (1), 1.

6. Svensson, K.; Abramsson, L.; Becker, W.; Glynn, A.; Hellenäs, K.-E.; Lind, Y.; Rosén, J. Dietary Intake of Acrylamide in Sweden. *Food and Chemical Toxicology* **2003**, *41* (11), 1581–1586.

7. Fadai, N. T.; Please, C. P.; Van Gorder, R. A. Modelling Structural Deformations in a Roasting Coffee Bean. *International Journal of Non-Linear Mechanics* **2019**, *110*, 123–130.

8. Toci, A. T.; Azevedo, D. A.; Farah, A. Effect of Roasting Speed on the Volatile Composition of Coffees with Different Cup Quality. *Food Research International* **2020**, *137*, 109546.

9. Baggenstoss, J.; Poisson, L.; Luethi, R.; Perren, R.; Escher, F. Influence of Water Quench Cooling on Degassing and Aroma Stability of Roasted Coffee. *J. Agric. Food Chem.* **2007**, *55* (16), 6685–6691.

10. *ChemSpider: Search and Share Chemistry - Acrylamide*. https://www.chemspider.com/Chemical-Structure.6331.html (2025-02-03).
11. Tareke, E.; Rydberg, P.; Karlsson, P.; Eriksson, S.; Törnqvist, M. Analysis of Acrylamide, a Carcinogen Formed in Heated Foodstuffs. *J. Agric. Food Chem.* **2002**, *50* (17), 4998–5006.

12. Koszucka, A.; Nowak, A.; Nowak, I.; Motyl, I. Acrylamide in Human Diet, its Metabolism, Toxicity, Inactivation and the Associated European Union Legal Regulations in Food Industry. *Crit. Rev. Food Sci. Nutr.* **2020**, *60* (10), 1677-1692.

13. Hirvonen, T.; Jestoi, M.; Tapanainen, H.; Valsta, L.; Virtanen, S. M.; Sinkko, H.; Kronberg-Kippilä, C.; Kontto, J.; Virtamo, J.; Simell, O.; Peltonen, K. Dietary Acrylamide Exposure among Finnish Adults and Children: The Potential Effect of Reduction Measures. *Food Additives & Contaminants: Part A* **2011**, *28* (11), 1483–1491.

14. International Agency for Research on Cancer. *Monographs on the Evaluation of Carcinogenic Risks to Humans: Some Industrial Chemicals*; IARC Monographs on the Evaluation of Carcinogenic Risks to Humans; International Agency for Research on Cancer: Lyon, France; Vol. 1994.

15. European Commission, Commission Regulation (EU) 2017/2158 of 20 November 2017 establishing mitigation measures and benchmark levels for the reduction of the presence of acrylamide in food, Official Journal of the European Union L 304, 21.11.2017,
p. 24–44.

16. Michalak, J.; Gujska, E.; Czarnowska, M.; Klepacka, J.; Nowak, F. Effect of Storage on Acrylamide and 5-Hydroxymethylfurfural Contents in Selected Processed Plant Products with Long Shelf-Life. *Plant Foods Hum Nutr* **2016**, *71* (1), 115–122.

17. Batali, M. E.; Ristenpart, W. D.; Guinard, J.-X. Brew Temperature, at Fixed Brew Strength and Extraction, Has Little Impact on the Sensory Profile of Drip Brew Coffee. *Sci Rep* **2020**, *10* (1), 16450.

18. Specialty Coffee Association, SCA Standard 102-2024: Coffee Value Assessment – Sample Preparation and Tasting Mechanics, Specialty Coffee Association, USA, 2024.

19. Lantz, I.; Ternité, R.; Wilkens, J.; Hoenicke, K.; Guenther, H.; Van Der Stegen, G. H. D. Studies on Acrylamide Levels in Roasting, Storage and Brewing of Coffee. *Molecular Nutrition Food Res* **2006**, *50* (11), 1039–1046.

20. Kocadağlı, T.; Gökmen, V. Formation of Acrylamide in Coffee. *Current Opinion in Food Science* **2022**, *45*, 100842.

21. Stadler, R. H.; Blank, I.; Varga, N.; Robert, F.; Hau, J.; Guy, P. A.; Robert, M.-C.; Riediker, S. Acrylamide from Maillard Reaction Products. *Nature* **2002**, *419* (6906), 449–450.

22. Eerola, S.; Hollebekkers, K.; Hallikainen, A.; Peltonen, K. Acrylamide Levels in Finnish Foodstuffs Analysed with Liquid Chromatography Tandem Mass Spectrometry. *Molecular Nutrition Food Res* **2007**, *51* (2), 239–247.

23. Zyzak, D. V.; Sanders, R. A.; Stojanovic, M.; Tallmadge, D. H.; Eberhart, B. L.; Ewald, D. K.; Gruber, D. C.; Morsch, T. R.; Strothers, M. A.; Rizzi, G. P.; Villagran, M. D. Acrylamide Formation Mechanism in Heated Foods. *J. Agric. Food Chem.* **2003**, *51* (16), 4782–4787.

24. Hamzalıoğlu, A. 5-Hydroxymethylfurfural Accumulation Plays a Critical Role on Acrylamide Formation in Coffee during Roasting as Confirmed by Multiresponse Kinetic Modelling. **2020**.

25. Kocadağlı, T.; Göncüoğlu, N.; Hamzalıoğlu, A.; Gökmen, V. In Depth Study of Acrylamide Formation in Coffee during Roasting: Role of Sucrose Decomposition and Lipid Oxidation. *Food Funct.* **2012**, *3* (9), 970.

26. Nehlig, A.; Cunha, R. A. The Coffee–Acrylamide Apparent Paradox: An Example of Why the Health Impact of a Specific Compound in a Complex Mixture Should Not Be Evaluated in Isolation. *Foods* **2021**, *10* (1310).

27. Liang, N.; Kitts, D. Antioxidant Property of Coffee Components: Assessment of Methods That Define Mechanisms of Action. *Molecules* **2014**, *19* (11), 19180–19208.

28. Asy'ari Hasbullah, U. H.; Rini Umiyati, D. Antioxidant Activity and Total Phenolic Compounds of Arabica and Robusta Coffee at Different Roasting Levels. *J. Phys.: Conf. Ser.* **2021**, *1764* (1), 012033.

29. Nehlig, A.; Cunha, R. A. The Coffee–Acrylamide Apparent Paradox: An Example of Why the Health Impact of a Specific Compound in a Complex Mixture Should Not Be Evaluated in Isolation. Nutrients **2020**, *12*, 3141.

30. Cheng, B.; Furtado, A.; Smyth, H. E.; Henry, R. J. Influence of Genotype and Environment on Coffee Quality. *Trends in Food Science & Technology* **2016**, *57*, 20–30.

31. Fioresi, D. B.; Ramos, A. C.; Bertolazi, A. A.; Pereira, L. L. Adherence and Concordance among Q-GRADERS in the Sensory Analysis of Coffees. *Journal of Sensory Studies* **2023**, *38* (2), e12805.

32. Silva, L. C. F.; Pereira, P. V. R.; Cruz, M. A. D. D.; Costa, G. X. R.; Rocha, R. A. R.; Bertarini, P. L. L.; Amaral, L. R. D.; Gomes, M. S.; Santos, L. D. Enhancing Sensory Quality of Coffee: The Impact of Fermentation Techniques on Coffea Arabica Cv. Catiguá MG2. *Foods* **2024**, *13* (5), 653.

33. Silva, E. A. da; Mazzafera, P.; Brunini, O.; Sakai, E.; Arruda, F. B.; Mattoso, L. H. C.; Carvalho, C. R. L.; Pires, R. C. M. The Influence of Water Management and Environmental Conditions on the Chemical Composition and Beverage Quality of Coffee Beans. *Braz. J. Plant Physiol.* **2005**, *17*, 229–238.

34. Cascos, G.; Montero-Fernández, I.; Marcía-Fuentes, J. A.; Aleman, R. S.; Ruiz-Canales, A.; Martín-Vertedor, D. Electronic Prediction of Chemical Contaminants in Aroma of Brewed Roasted Coffee and Quantification of Acrylamide Levels. *Foods* **2024**,
13 (5), 768.

35. Vezzulli, F.; Triachini, S.; Mulazzi, A.; Lambri, M.; Bertuzzi, T. Acrylamide: Impact of Precursors Concentration, Origin, Post-harvesting Process and Roasting Level in High-quality Arabica and Robusta Coffee. *Int J of Food Sci Tech* **2022**, *57* (12), 7468–7476.

36. Bressanello, D.; Marengo, A.; Cordero, C.; Strocchi, G.; Rubiolo, P.; Pellegrino, G.; Ruosi, M. R.; Bicchi, C.; Liberto, E. Chromatographic Fingerprinting Strategy to Delineate Chemical Patterns Correlated to Coffee Odor and Taste Attributes. *J. Agric. Food Chem.* **2021**, *69* (15), 4550–4560.

37. Hoenicke, K.; Gatermann, R. Studies on the Stability of Acrylamide in Food During Storage. *Journal of AOAC International* **2005**, *88* (1), 268–273.

38. Morales-Ramos, V.; Escamilla-Prado, E.; Ruiz-Carbajal, R. A.; Pérez-Sato, J. A.; Velázquez-Morales, J. A.; Servín-Juárez, R. On the Soil–Bean–Cup Relationships in Coffea Arabica L. *J Sci Food Agric* **2020**, *100* (15), 5434–5441.

39. Hameed, A.; Hussain, S. A.; Ijaz, M. U.; Ullah, S.; Pasha, I.; Suleria, H. A. R. Farm to Consumer: Factors Affecting the Organoleptic Characteristics of Coffee. II: Postharvest Processing Factors. *Comp Rev Food Sci Food Safe* **2018**, *17* (5), 1184–1237.

40. Ságio, S. A.; Lima, A. A.; Barreto, H. G.; De Carvalho, C. H. S.; Paiva, L. V.; Chalfun-Junior, A. Physiological and Molecular Analyses of Early and Late Coffea Arabica Cultivars at Different Stages of Fruit Ripening. *Acta Physiol Plant* **2013**, *35* (11), 3091–3098.

41. Velásquez, S.; Franco, A. P.; Peña, N.; Bohórquez, J. C.; Gutierrez, N. Effect of Coffee Cherry Maturity on the Performance of the Drying Process of the Bean: Sorption Isotherms and Dielectric Spectroscopy. *Food Control* **2021**, *123*, 107692.

42. Dias, E. C.; Borém, F. M.; Pereira, R. G. F. A.; Guerreiro, M. C. Amino Acid Profiles in Unripe Arabica Coffee Fruits Processed Using Wet and Dry Methods. *Eur Food Res Technol* **2012**, *234* (1), 25–32.

43. Orfanou, F.; Dermesonlouoglou, E. K.; Taoukis, P. S. Greek Coffee Quality Loss During Home Storage: Modeling the Effect of Temperature and Water Activity. *Journal of Food Science* **2019**, *84* (10), 2983–2994.

44. Vinecky, F.; Davrieux, F.; Mera, A. C.; Alves, G. S. C.; Lavagnini, G.; Leroy, T.; Bonnot, F.; Rocha, O. C.; Bartholo, G. F.; Guerra, A. F.; Rodrigues, G. C.; Marraccini, P.; Andrade, A. C. Controlled Irrigation and Nitrogen, Phosphorous and Potassium Fertilization Affect the Biochemical Composition and Quality of Arabica Coffee Beans. *J. Agric. Sci.* **2017**, *155* (6), 902–918.

45. Ruan, J.; Wu, X.; Ye, Y.; Hardter, R. Effect of Potassium, Magnesium and Sulphur Applied in Different Forms of Fertilisers on Free Amino Acid Content in Leaves of Tea. *Journal of the science of food and agriculture* **1998**, *76* (3), 389–396.

46. Dos Reis, A. R.; Favarin, J. L.; Gallo, L. A.; Moraes, M. F.; Tezotto, T.; Lavres Junior, J. Inflence of Nitrogen Fertilization on Nickel Accumulation and Chemical Composition of Coffee Plants During Fruit Development.. *Journal of Plant Nutrition* **2011**, *34* (12), 1853–1866.

47. Rydberg, P.; Eriksson, S.; Tareke, E.; Karlsson, P.; Ehrenberg, L.; Törnqvist, M. Factors That Influence the Acrylamide Content of Heated Foods. *Chemistry and Safety of Acrylamide in Food* **2005**, 317–328.

48. Getachew, M.; Tolassa, K.; De Frenne, P.; Verheyen, K.; Tack, A. J. M.; Hylander, K.; Ayalew, B.; Boeckx, P. The Relationship between Elevation, Soil Temperatures, Soil Chemical Characteristics, and Green Coffee Bean Quality and Biochemistry in Southwest Ethiopia. *Agron. Sustain. Dev.* **2022**, *42* (4), 61.

49. Acquaticci, L.; Angeloni, S.; Cela, N.; Galgano, F.; Vittori, S.; Caprioli, G.; Condelli, N. Impact of Coffee Species, Post-Harvesting Treatments and Roasting Conditions on Coffee Quality and Safety Related Compounds. *Food Control* **2023**, *149*, 109714.

50. Kleinwächter, M.; Selmar, D. Influence of Drying on the Content of Sugars in Wet Processed Green Arabica Coffees. *Food Chemistry* **2010**, *119* (2), 500–504.

51. Dias, E. C.; de Lima, R. R.; Fernandes, O.; Casal, S. Biogenic Amine Profile in Unripe Arabica Coffee Beans Processed According to Dry and Wet Methods. *Journal of Agricultural and Food Chemistry* **2012**, No. 60, 4120–4125.

52. Knopp, S.; Bytof, G.; Selmar, D. Influence of Processing on the Content of Sugars in Green Arabica Coffee Beans. *Eur Food Res Technol* **2006**, *223* (2), 195–201.

53. Poisson, L.; Auzanneau, N.; Mestdagh, F.; Blank, I.; Davidek, T. New Insight into the Role of Sucrose in the Generation of α-Diketones upon Coffee Roasting. *Journal of Agricultural and Food Chemistry* **2016**.

54. Schenker, S.; Rothgeb, T., *Chapter 11 – The Roast: Creating the Beans' Signature.* In: Folmer, B. (ed.), *The Craft and Science of Coffee*, Academic Press, London, pp. 245–271, **2017**.

55. Siebald, H.; Möller, M.; Lenz, F.; Kirchner, S.; Hensel, O. Acoustic Condition Monitoring of Coffee Beans, during the Roasting Process. *LWT* **2024**, *199*, 116119.

56. Ganju, E.; Chawla, N. Microstructural Evolution of Ethiopian Coffee Beans by Time-Resolved X-Ray Microcomputed Tomography. *MRS Advances* **2024**, *9* (11), 936–942.

57. Chindapan, N.; Soydok, S.; Devahastin, S. Roasting Kinetics and Chemical Composition Changes of Robusta Coffee Beans During Hot Air and Superheated Steam Roasting. *Journal of Food Science* **2019**, *84* (2), 292–302.

58. Schouten, M. A.; Tappi, S.; Angeloni, S.; Cortese, M.; Caprioli, G.; Vittori, S.; Romani, S. Acrylamide Formation and Antioxidant Activity in Coffee during Roasting – A Systematic Study. *Food Chemistry* **2021**, *343*, 128514.

59. Cascos, G.; Montero-Fernández, I.; Marcía-Fuentes, J. A.; Aleman, R. S.; Ruiz-Canales, A.; Martín-Vertedor, D. Electronic Prediction of Chemical Contaminants in Aroma of Brewed Roasted Coffee and Quantification of Acrylamide Levels. *Foods* **2024**,
13 (5), 768.

60. Grancieri Debona, D.; Catarina Da Silva Oliveira, E.; Schwengber Ten Caten, C.; Carvalho Guarçoni, R.; Rizzo Moreira, T.; Polonini Moreli, A.; Louzada Pereira, L. Sensory Analysis and Mid-Infrared Spectroscopy for Discriminating Roasted Specialty Coffees. *CS* **2021**, *16*, 1–9.

61. Baggenstoss, J.; Poisson, L.; Kaegi, R.; Perren, R.; Escher, F. Coffee Roasting and Aroma Formation: Application of Different Time−Temperature Conditions. *J. Agric. Food Chem.* **2008**, *56* (14), 5836–5846.

62. Baggenstoss, J.; Poisson, L.; Kaegi, R.; Perren, R.; Escher, F. Roasting and Aroma Formation: Effect of Initial Moisture Content and Steam Treatment. *J. Agric. Food Chem.* **2008**, *56* (14), 5847–5851.

63. Budryn, G.; Nebesny, E.; Oracz, J. Correlation Between the Stability of Chlorogenic Acids, Antioxidant Activity and Acrylamide Content in Coffee Beans Roasted in Different Conditions. *International Journal of Food Properties* **2015**, *18* (2), 290–302.

64. Smrke, S.; Adam, J.; Mühlemann, S.; Lantz, I.; Yeretzian, C. Effects of Different Coffee Storage Methods on Coffee Freshness after Opening of Packages. *Food Packaging and Shelf Life* **2022**, *33*, 100893.

65. Anese, M.; Manzocco, L.; Nicoli, M. C. Modeling the Secondary Shelf Life of Ground Roasted Coffee. *J. Agric. Food Chem.* **2006**, *54* (15), 5571–5576.

66. Baggenstoss, J.; Perren, R.; Escher, F. Water Content of Roasted Coffee: Impact on Grinding Behaviour, Extraction, and Aroma Retention. *Eur Food Res Technol* **2008**, *227* (5), 1357–1365.

67. Chau, F.-T.; Liang, Y.-Z.; Gao, J.; Shao, X.-G.; Winefordner, J. D. *Chemometrics: From Basics to Wavelet Transform*; John Wiley & Sons, Incorporated: Newark, UNITED STATES, 2004.

68. Harris, D. C. *Quantitative Chemical Analysis*, 8th ed.; W.H. Freeman and Co: New York, 2010.

69. Cantwell, H. (Ed.); Eurachem Method Validation Working Group. The Fitness for Purpose of Analytical Methods: A Laboratory Guide to Method Validation and Related Topics; **2025**.

70. Brereton, R. G.; Jansen, J.; Lopes, J.; Marini, F.; Pomerantsev, A.; Rodionova, O.; Roger, J. M.; Walczak, B.; Tauler, R. Chemometrics in Analytical Chemistry-Part I: History, Experimental Design and Data Analysis Tools. *Anal Bioanal Chem* **2017**, *409* (25), 5891–5899.

71. Da Costa, D. S.; Albuquerque, T. G.; Costa, H. S.; Bragotto, A. P. A. Thermal Contaminants in Coffee Induced by Roasting: A Review. *IJERPH* **2023**, *20* (8), 5586.

72. Cossu, M.; Sanna, A.; Ledda, G.; Chessa, G.; Mangano, G. Development and Method Validation for Determination of Acrylamide in Food by Modified QuEChERS and UHPLC–MS/MS Analysis. *Journal of Food Composition and Analysis* **2024**, *135*, 106698.

73. Bonilla, L. F.; Sandoval-Aldana, A.; Daza, L. D. Acrylamide: An Approach to Its Knowledge and Importance for Roasted Coffee. *Food Chemistry* **2025**, *466*, 142247.

74. Yoshioka, T.; Izumi, Y.; Nagatomi, Y.; Miyamoto, Y.; Suzuki, K.; Bamba, T. A Highly Sensitive Determination Method for Acrylamide in Beverages, Grains, and Confectioneries by Supercritical Fluid Chromatography Tandem Mass Spectrometry. *Food Chemistry* **2019**, *294*, 486–492.

75. Sebastià, A.; Pallarés, N.; Bridgeman, L.; Juan-García, A.; Castagnini, J. M.; Ferrer, E.; Barba, F. J.; Berrada, H. A Critical Review of Acrylamide Green Extraction and Determination in Food Matrices: Current Insights and Future Perspectives. *TrAC Trends in Analytical Chemistry* **2023**, *167*, 117267.

76. Elbashir, A. A.; Omar, M. M. A.; Ibrahim, W. A. W.; Schmitz, O. J.; Aboul-Enein, H. Y. Acrylamide Analysis in Food by Liquid Chromatographic and Gas Chromatographic Methods. *Critical Reviews in Analytical Chemistry* **2014**, *44* (2), 107–141.

77. Desmarchelier, A.; Bebius, A.; Reding, F.; Griffin, A.; Ahijado Fernandez, M.; Beasley, J.; Clauzier, E.; Delatour, T. Towards a Consensus LC-MS/MS Method for the Determination of Acrylamide in Food That Prevents Overestimation Due to Interferences. *Food Additives & Contaminants: Part A* **2022**, *39* (4), 653–665.

78. Bortolomeazzi, R.; Munari, M.; Anese, M.; Verardo, G. Rapid Mixed Mode Solid Phase Extraction Method for the Determination of Acrylamide in Roasted Coffee by HPLC–MS/MS. *Food Chemistry* **2012**, *135* (4), 2687–2693.

79. Bertuzzi, T.; Rastelli, S.; Mulazzi, A.; Pietri, A. Survey on Acrylamide in Roasted Coffee and Barley and in Potato Crisps Sold in Italy by a LC–MS/MS Method. *Food Additives & Contaminants: Part B* **2017**, 1–8.

80. Stefanović, S.; Đorđevic, V.; Jelušić, V. Rapid and Reliable QuEChERS-Based LC-MS/MS Method for Determination of Acrylamide in Potato Chips and Roasted Coffee. *IOP Conf. Ser.: Earth Environ. Sci.* **2017**, *85*, 012010.

81. Şenyuva, H. Z.; Gökmen, V. Study of Acrylamide in Coffee Using an Improved Liquid Chromatography Mass Spectrometry Method: Investigation of Colour Changes and Acrylamide Formation in Coffee during Roasting. *Food Additives and Contaminants* **2005**, *22* (3), 214–220.

82. Wang, J.; Cai, Z.; Zhang, N.; Hu, Z.; Zhang, J.; Ying, Y.; Zhao, Y.; Feng, L.; Zhang, J.; Wu, P. A Novel Single Step Solid-Phase Extraction Combined with Bromine Derivatization Method for Rapid Determination of Acrylamide in Coffee and Its Products by Stable Isotope Dilution Ultra-Performance Liquid Chromatography Tandem Triple Quadrupole Electrospray Ionization Mass Spectrometry. *Food Chemistry* **2022**, *388*, 132977.

83. Omar, M. M. A.; Elbashir, A. A.; Schmitz, O. J.; Ziyada, A. K.; Osman, A. Validation of High-Performance Liquid Chromatography Coupled with LTQ Orbitrap Mass Spectrometry for Analysis of Acrylamide. *Food Measure* **2024**, *18* (2), 930–939.

84. Liu, J.; Man, Y.; Zhu, Y.; Hu, X.; Chen, F. Simultaneous Analysis of Acrylamide and Its Key Precursors, Intermediates, and Products in Model Systems by Liquid Chromatography–Triple Quadrupole Mass Spectrometry. *Anal. Chem.* **2013**, *85* (19), 9262–9271.

85. Gritti, F.; Gilar, M.; Walter, T. H.; Wyndham, K. Retention Loss of Reversed-Phase Chromatographic Columns Using 100% Aqueous Mobile Phases from Fundamental Insights to Best Practice. *Journal of Chromatography A* **2020**, *1612*, 460662.

86. Gritti, F.; Lawrence, N.; Field, J. Preparation and Investigation of Two-Component Silica-Modified Hydrophobic Layer for Minimizing Retention Loss of Reversed-Phase Chromatographic Columns Using Fully Aqueous Mobile Phases. *Journal of Chromatography A* **2024**, *1719*, 464766.

87. Zhang, Y.; Zhao, H.; Sun, S.; Lu, L.; Xue, X.; Su, S.; Gong, P.; Zheng, W.; Wang, M.; Wang, J.; Zhu, J.; Liu, Y.; Zhang, F. Efficient Optimization and Development of Two Methods for the Determination of Acrylamide in Deep-frying Oil by Liquid Chromatography–Tandem Mass Spectrometry: Application of Multifactor Analysis Assessment Strategy. *J of Separation Science* **2023**, *46* (3), 2200631.

88. Ho, C.; Lam, C.; Chan, M.; Cheung, R.; Law, L.; Lit, L.; Ng, K.; Suen, M.; Tai, H. Electrospray Ionisation Mass Spectrometry: Principles and Clinical Applications. *Clin Biochem Rev* **2003**, *24* (1), 3–12.

89. Desmarchelier, A.; Hamel, J.; Delatour, T. Sources of Overestimation in the Analysis of Acrylamide-in Coffee by Liquid Chromatography Mass Spectrometry. *Journal of Chromatography A* **2020**, *1610*, 460566.

90. *Triple Quadrupole Mass Spectrometry - an overview | ScienceDirect Topics*. https://www.sciencedirect.com/topics/medicine-and-dentistry/triple-quadrupole-mass-spectrometry (2025-07-28).

91. De-Alwis, J.; Ross, E.; Adams, S., *Determination of Acrylamide: Are You Avoiding Isobaric Interferences?* Waters Corporation, Milford, MA, Application Note 720007009, September 2020.

92. Magnusson, B.; Örnemark, U. (eds.), *Eurachem Guide: The Fitness for Purpose of Analytical Methods – A Laboratory Guide to Method Validation and Related Topics*, 2nd ed., Eurachem, 2014. ISBN 978-91-87461-59-0.

93. Wenzl, T.; Szilagyi, S.; Rosén, J.; Karasek, L. Validation by Collaborative Trial of an Isotope Dilution Liquid Chromatographic Tandem Mass Spectrometric Method to Determine the Content of Acrylamide in Roasted Coffee. *Food Additives & Contaminants: Part A* **2009**, *26* (8), 1146–1152.

94. International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), *ICH Harmonised Guideline: Validation of Analytical Procedures Q2(R2)*, Final Version, 1 November 2023.

95. Guenther, H.; Anklam, E.; Wenzl, T.; Stadler, R. H. Acrylamide in Coffee: Review of Progress in Analysis, Formation and Level Reduction. *Food Additives and Contaminants* **2007**, *24* (sup1), 60–70.

96. Vladimir S. Minkina. Thermogravimetry. In *Thermal Analysis and Thermodynamic Properties of Solids*; IntechOpen; Vol. 2020, pp 45–64.

97. Brzezińska, R.; Górska, A.; Wirkowska-Wojdyła, M.; Ostrowska-Ligęza, E.; Bryś, J. Thermal and Kinetic Properties of Brazilian Coffea Arabica Beans. *Applied Sciences* **2021**, *11* (14), 6324.

98. Grzelczyk, J.; Fiurasek, P.; Kakkar, A.; Budryn, G. Evaluation of the Thermal Stability of Bioactive Compounds in Coffee Beans and Their Fractions Modified in the Roasting Process. *Food Chemistry* **2022**, *387*, 132888.

99. Gaisford, S.; Kett, V.; Haines, P. *Principles of Thermal Analysis and Calorimetry*, 2nd edition.; Royal Society of Chemistry: Cambridge, 2016.

100. Kumari, A.; Bhattacharya, B.; Agarwal, T.; Paul, V.; Maurya, V. K.; Chakkaravarthi, S.; Simal-Gandara, J. Method Development and Validation for Acrylamide in Potato Cutlet by UHPLC-MS/MS. *Food Control* **2023**, *151*, 109817.

101. *Agilent Poroshell EC-C18*. https://www.elementlabsolutions.com/uk/agilent-poroshell-ec-c18 (2025-09-05).

102. *Kinetex EVO C18 Column for Polar Analytes | Phenomenex*. https://www.phenomenex.com/products/kinetex-hplc-column/kinetex-evo-c18?srsltid=AfmBOop0zmsQ0hu4_vCgKnsYomawpyL9oNpny7IVMzP4Fv-7x5zwzHeU (2025-09-05).

103. New, L.-S.; Chan, E. C. Y. Evaluation of BEH C18, BEH HILIC, and HSS T3 (C18) Column Chemistries for the UPLC-MS-MS Analysis of Glutathione, Glutathione Disulfide, and Ophthalmic Acid in Mouse Liver and Human Plasma. *J Chromatogr Sci* **2008**, *46* (3), 209–214.

104. Cheng, W.; Wang, X.; Zhang, Z.; Ma, L.; Liu, G.; Wang, Q.; Chen, F.; Cheng, K.-W. Development of an Isotope Dilution UHPLC–QqQ-MS/MS-Based Method for Simultaneous Determination of Typical Advanced Glycation End Products and Acrylamide in Baked and Fried Foods. *J. Agric. Food Chem.* **2021**, *69* (8), 2611–2618.

105. Phenomenex, Inc., *Strata-X Polymeric SPE Phase Overview – A Method Development and Application Guide*, Phenomenex, Torrance, CA, 2023.

106. European Commission, Directorate-General for Health and Food Safety (DG SANTE), Analytical Quality Control and Method Validation Procedures for Pesticide Residues Analysis in Food and Feed (SANTE/11312/2021), Implemented by 01/01/2022, European Commission, Brussels, 2021.

107. Mestdagh, F.; Maertens, J.; Cucu, T.; Delporte, K.; Van Peteghem, C.; De Meulenaer, B. Impact of Additives to Lower the Formation of Acrylamide in a Potato Model System through pH Reduction and Other Mechanisms. *Food Chemistry* **2008**, *107* (1), 26–31.

108. Tang, X.; Bruce, J. E.; Hill, H. H. Characterizing Electrospray Ionization Using Atmospheric Pressure Ion Mobility Spectrometry. *Anal. Chem.* **2006**, *78* (22), 7751–7760.

109. Meher, A. K.; Zarouri, A. Green Analytical Chemistry—Recent Innovations. *Analytica* **2025**, *6* (1), 10.

110. Gökmen, V.; Morales, F. J.; Ataç, B.; Serpen, A.; Arribas-Lorenzo, G. Multiple-Stage Extraction Strategy for the Determination of Acrylamide in Foods. *Journal of Food Composition and Analysis* **2009**, *22* (2), 142–147.

111. Rutherford, A. *Introducing Anova and Ancova: A GLM Approach*; SAGE Publications, Limited: London, UNITED KINGDOM, 2001.

112. Amrein, T. M.; Limacher, A.; Conde-Petit, B.; Amadò, R.; Escher, F. Influence of Thermal Processing Conditions on Acrylamide Generation and Browning in a Potato Model System. *J. Agric. Food Chem.* **2006**, *54* (16), 5910–5916.

113. Pastoriza, S.; Rufián-Henares, J. Á.; Morales, F. J. Reactivity of Acrylamide with Coffee Melanoidins in Model Systems. *LWT - Food Science and Technology* **2012**, *45* (2), 198–203.

114. Andrzejewski, D.; Roach, J. A. G.; Gay, M. L.; Musser, S. M. Analysis of Coffee for the Presence of Acrylamide by LC-MS/MS. *J. Agric. Food Chem.* **2004**, *52* (7), 1996–2002.

115. Aung Moon, S.; Wongsakul, S.; Kitazawa, H.; Saengrayap, R. Lipid Oxidation Changes of Arabica Green Coffee Beans during Accelerated Storage with Different Packaging Types. *Foods* **2022**, *11* (19), 3040.

116. Gertz, C.; Aladedunye, F.; Popp, M.; Matthäus, B. The Impact of Fat Deterioration on Formation of Acrylamide in Fried Foods. *Euro J Lipid Sci & Tech* **2023**, *125* (4), 2200144.

117. Capuano, E.; Oliviero, T.; Açar, Ö. Ç.; Gökmen, V.; Fogliano, V. Lipid Oxidation Promotes Acrylamide Formation in Fat-Rich Model Systems. *Food Research International* **2010**, *43* (4), 1021–1026.

118. Wang, X.; Tang, B.; Bao, L.; Zhang, H.; He, M.; Yuan, S. Degradation Evaluation of Acrylamide in Advanced Oxidation Processes Based on Theoretical Method: Mechanisms, Kinetics, Toxicity Evaluation and the Role of Soil Particles. *Journal of Hazardous Materials* **2022**, *424*, 127592.

119. Wang, N.; Lim, L.-T. Fourier Transform Infrared and Physicochemical Analyses of Roasted Coffee. *J. Agric. Food Chem.* **2012**, *60* (21), 5446–5453.

List of Appendix

Appendix 1

Description of Coffee Sample Group 1

This appendix presents detailed information on coffee sample group 1. It includes the shelf-life (roast) ages of the samples at the time of analysis. In addition, the table shows the labeled roast degrees.

The coffee samples were either received or purchased, and both sample sets (S1–S10 and C11–C19) include coffees from several different roasters. For specialty coffees, some roasters provided roasting data, e.g. the final internal temperature of the coffee bean and roasting time, which are included in the table.

ID	Category	Roast degree (labeled)	DAR*	Final temperature of the bean (°C)	Roasting time (s)
1	Specialty	Light	70	209	504
2	Specialty	Light	52	208	604
3	Specialty	Medium	52	202	594
4	Specialty	Light	59	N/A	N/A
5	Specialty	Medium	53	214	620
6	Specialty	Medium	56	226	690
7	Specialty	Dark	57	205	667
8	Specialty	Dark	48	212	740
9	Specialty	Dark	87	206	870
10	Specialty	Dark	60	219	774
11	Commercial	Light	66	N/A	N/A
12	Commercial	Medium	61	N/A	N/A
13	Commercial	Light	74	N/A	N/A
14	Commercial	Medium	68	N/A	N/A
15	Commercial	Dark	70	N/A	N/A
16	Commercial	Dark	71	N/A	N/A
17	Commercial	Light	41	N/A	N/A
18	Commercial	Medium	81	N/A	N/A
19	Commercial	Dark	77	N/A	N/A

*DAR refers to the number of days after roasting when the sample is extracted.

Appendix 2

Raw Data from Precision Evaluation for Method 2

This appendix includes the raw data obtained from the precision experiment, which is described in Chapter 7.4.3. Precision was evaluated by spiking a coffee sample at three acrylamide levels (10 µg/L, 50 µg/L and 100 µg/L). Each sample contained isotope-labeled acrylamide (AA-d_3, c=200 µg/L). The samples were injected in triplicate, and the resulting peak areas were used to determine the relative standard deviations (%RSD) for each AA levels.

Added AA (µg/L)	Area (AA)	Area (AA-d3)
10	6867	67407
	7045	65437
	7818	64452
50	20698	67869
	20987	68637
	23798	71234
100	37265	66020
	39123	69220
	36338	68606

Appendix 3

Thermogravimetric Analysis (TGA) of Coffee Samples

This appendix summarizes the thermogravimetric analysis (TGA) of selected coffee samples. Mass loss fractions were obtained by integrating TG curves over four temperature ranges 1: 20–200, 2: 200–350, 3: 350–425, and 4: 425–600 °C, and for the residue mass at 600 °C.

Sample ID	m-% loss 20–200 °C	m-% loss 200–350 °C	m-% loss 350–425 °C	m-% loss 425–600 °C	Residual mass (%)
S2	2,48 ± 0,14	49,39 ± 0,64	10,58 ± 0,30	33,10 ± 0,42	4,44 ± 0,21
S3	2,65 ± 0,07	50,39 ± 0,11	10,00 ± 0,19	32,27 ± 0,16	4,69 ± 0,14
S10	1,50 ± 0,16	50,29 ± 1,03	10,39 ± 0,53	32,28 ± 0,94	5,55 ± 0,48
C11	3,97 ± 0,24	49,39 ± 0,38	9,66 ± 0,22	31,75 ± 0,29	5,24 ± 0,13
C12	4,30 ± 0,29	51,31 ± 0,48	9,08 ± 0,13	30,99 ± 0,61	4,32 ± 0,42
C14	4,07 ± 0,14	50,89 ± 0,13	9,45 ± 0,43	31,18 ± 0,57	4,42 ± 0,07